

The Schwarzschild Metric as a Solution to Einstein's Field Equations

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Abstract This manuscript presents a detailed, self-contained exposition of the differential geometry underlying general relativity, with particular emphasis on the derivation of the Schwarzschild solution from its physical and geometric assumptions. Its purpose is pedagogical: to make the subject accessible to graduate students by providing explicit calculations and intermediate arguments often omitted in standard references. Rather than claiming new results, we reconstruct and examine established derivations in detail. We begin by reviewing tensors, differentiable manifolds, vector bundles, tensor fields, and Riemannian and pseudo-Riemannian metrics. We then introduce causal structure and Lorentzian spacetimes and derive the Schwarzschild metric as a spherically symmetric vacuum solution of Einstein's field equations. Finally, we use variational methods and the Euler–Lagrange equations to derive the Schwarzschild geodesic equations, discuss timelike and null motion, and study radial free fall, highlighting the distinction between proper time and Schwarzschild coordinate time.

To the reader: The results of this manuscript were developed as part of my Master's thesis under the supervision of Professor Dr. Rodrigo Castro Marín at the University of Valparaíso in 2021 [6]. A few minor details have been revised, mainly to ensure the accuracy of the translation from Spanish .

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Introduction

Two central themes have shaped the history of differential geometry: the desire to understand and extend the work of F. Gauss, and the continuing influence of physical problems on the direction of research. Between 1854 and 1900, geometers were more preoccupied with the language of differential geometry than with the subject itself. This preoccupation with symbolism culminated in the development of tensor calculus by G. Ricci (1853–1925) and his student T. Levi-Civita (1873–1941); see [5], [9], and [14]. It was on B. Riemann’s abstract conception of space and on tensor calculus that A. Einstein (1879–1955) based his theory of gravitation, commonly known as the general theory of relativity. This theory is regarded as the foremost achievement of human thought in the formulation of a mathematical description of nature; see [2].

Einstein’s special theory of relativity, proposed in 1905, arose from the aim of reconciling two major theories of classical physics: Newtonian mechanics and Maxwell’s electromagnetic theory. In 1908, H. Minkowski, Einstein’s former teacher in Zurich, proposed a geometric formulation of Einstein’s theory by introducing Minkowski spacetime; see [4]. Einstein later concluded that a gravitational field produces curvature in spacetime. This replaced the action at a distance associated with the gravitational field generated by a body in the classical theory of gravitation with the idea that the body distorts the space around it. This distortion is measured by curvature, a concept studied in Riemannian geometry. Einstein further concluded that this should be expressed through a change in the metric, replacing the Minkowski metric η with a Lorentzian metric g ; see [11].

In 1915, Einstein arrived at the complete formulation of the general theory of relativity, presenting the equations that determine the gravitational field g from the physical conditions of the system. These equations relate the presence of matter to the curvature of spacetime. More precisely, the greater the concentration of matter, represented by the energy–momentum tensor, the larger the components of the Ricci curvature tensor; see [11] and [12].

Despite the difficulty of these equations, and to Einstein’s surprise, only a few months after their presentation Karl Schwarzschild (1873–1916) used symmetry to obtain an exact solution describing the gravitational field outside a star with a uniform, time-independent mass distribution. This result is of great importance, among other reasons because it makes it possible to study gravitational phenomena such as the advance of Mercury’s perihelion, which the classical theory of gravitation proposed by I. Newton in the seventeenth century could not explain. In 1923, G. D. Birkhoff (1884–1944) established that the only spherically symmetric solutions of Einstein’s equations in vacuum are given by the Schwarzschild metric, which predicts phenomena such as the existence of

black holes and gravitational waves; see [2] and [10].

The purpose of this manuscript is to describe the gravitational field produced by a static, spherically symmetric body—a star—within the spacetime framework of general relativity. We assume that no physical phenomena of a nongravitational nature occur outside the star, in which case the metric representing the gravitational field must have vanishing Ricci tensor.

The main objective is therefore to present a study of the geometry of the Schwarzschild solution. Since general relativity has developed over several decades, with authors adopting a variety of approaches and notational conventions, our aim is to provide a unified presentation of selected results in this field.

Chapter 1

Differential Geometry

Through a careful review of the literature, we shall examine several results on tensors, differentiable manifolds, and pseudo-Riemannian geometry. This chapter thus provides the theoretical foundations. Further details can be found in [1], [5], [9], [14], and [15].

1.1 Tensors

There are several ways to define a tensor. We shall study tensors in the language of multilinear maps and also as collections of quantities governed by a transformation law under changes of coordinates, establishing the equivalence of these two definitions. We begin by defining tensors on a vector space as multilinear maps from a Cartesian product of a fixed number of copies of the vector space and its dual to the underlying field.

Our motivation for studying tensor calculus is that it will allow us to give formal definitions of Riemannian and pseudo-Riemannian metrics.

Let V be a vector space over $\mathbb{R}$ of dimension $n \in \mathbb{N}$, and let $\{e_1, e_2, \dots, e_n\}$ be a basis of V . Every vector $v \in V$ can be written uniquely as $v = \sum_{i=1}^n v^i e_i$, where the real scalars v^i are the components of v in the basis $\{e_i\}_{i=1}^n$. To simplify the notation, we introduce the following convention

Notation. Einstein summation convention: Once the number n has been fixed, we omit the summation symbol $\sum_{i=1}^n$ whenever "i" appears as both a subscript and a superscript in the same expression. Under this convention,

$$x_1 a^1 + \dots + x_n a^n = \sum_{i=1}^n x_i a^i \quad \text{is written as} \quad x_i a^i.$$

Note that $x_i a^i$, $x_j a^j$, and $x_\alpha a^\alpha$ represent the same sum. Thus we write the vector as $v = v^i e_i$, omitting the summation symbol.

1.1.1 Dual and Bidual Spaces

We first discuss some features of the dual space that will be useful in studying tensors. Throughout what follows, we assume that the vector spaces have finite dimension n .

Definition 1.1.1. Let V be a vector space over a field $\mathbb{K}$. The set

$$L(V, \mathbb{K}) := \{ f : V \rightarrow \mathbb{K} \mid f \text{ is linear} \}$$

is a vector space over $\mathbb{K}$ called the dual space of V , denoted by V^* .

Its vector space structure is given by the usual addition and scalar multiplication:

$$(\lambda_1 f_1 + \lambda_2 f_2)(v) := \lambda_1 f_1(v) + \lambda_2 f_2(v).$$

Since V^* is a vector space, we can define its dual.

Definition 1.1.2. Let V be a vector space over a field $\mathbb{K}$, and let V^* be the dual space of V . The set

$$(V^*)^* = L(V^*, \mathbb{K}) := \{ \phi : V^* \rightarrow \mathbb{K} \mid \phi \text{ is linear} \}$$

is a vector space over $\mathbb{K}$ called the bidual space of V , denoted by V^{**} .

Theorem 1.1.3. Let V be a vector space over $\mathbb{R}$, and let $\mathcal{B} = \{e_\alpha\}_{\alpha=1}^n$ be a basis of V . Then the family $\{e^{*\alpha}\}_{\alpha=1}^n$ defined by

$$\begin{aligned} e^{*\alpha} : V &\longrightarrow \mathbb{R} \\ u &\longmapsto u^\alpha, \quad \text{where } u = u^\alpha e_\alpha \end{aligned}$$

is a basis of V^* . Moreover,

$$e^{*\alpha}(e_\beta) = \delta_\beta^\alpha$$

Proof. We first show that $e^{*\alpha} : V \longrightarrow \mathbb{R}$ is linear. Indeed,

$$\begin{aligned} e^{*\alpha}(\gamma u + \delta v) &= e^{*\alpha}(\gamma(u^\beta e_\beta) + \delta(v^\beta e_\beta)) \\ &= e^{*\alpha}((\gamma u^\beta + \delta v^\beta) e_\beta) \\ &= (\gamma u^\alpha + \delta v^\alpha) \\ &= \gamma e^{*\alpha}(u) + \delta e^{*\alpha}(v) \end{aligned}$$

We now show that $\{e^{*\alpha}\}_{\alpha=1}^n$ is linearly independent and spans the whole space. First, consider the linear combination $x_1 e^{*1} + x_2 e^{*2} + \dots + x_n e^{*n} = 0$, where zero is defined by

$$\begin{aligned} 0 : V &\longrightarrow \mathbb{R} \\ x &\longmapsto 0(x) = 0, \end{aligned}$$

Evaluating at the elements e_α of V , with $\alpha = 1, \dots, n$, gives

$$x_1 e^{*1}(e_\alpha) + x_2 e^{*2}(e_\alpha) + \dots + x_n e^{*n}(e_\alpha) = 0$$

that is, $x_\alpha = 0$ for $\alpha = 1, 2, \dots, n$: when $\alpha = 1$, we have $x_1 e^{*1}(e_1) = x_1$, and similarly up to $\alpha = n$, for which $x_n e^{*n}(e_n) = x_n$. This proves that $\{e^{*\alpha}\}_{\alpha=1}^n$ is linearly independent.

Let $f \in V^*$. We show that f is a linear combination of the elements of $\{e^{*\alpha}\}_{\alpha=1}^n$. More precisely, we prove that

$$f = f(e_\alpha) e^{*\alpha}.$$

Indeed, let $u \in V$ and write $u = u^\beta e_\beta$. Then, for every $u \in V$,

$$\begin{aligned} f(e_\alpha) e^{*\alpha}(u) &= f(e_\alpha) e^{*\alpha}(u^\beta e_\beta) \\ &= f(e_\alpha) u^\alpha \\ &= f(u^\alpha e_\alpha) \\ &= f(u) \end{aligned}$$

Hence $f(e_\alpha) e^{*\alpha} = f$. Thus $\{e^{*\alpha}\}_{\alpha=1}^n$ spans V^* . We conclude that $\{e^{*\alpha}\}_{\alpha=1}^n$ is a basis of the vector space V^* . $\square$

For $f \in V^*$, write $f = f(e_\alpha) e^{*\alpha}$. The quantities $f(e_\alpha)$ are called the covariant components of f in the basis $\{e^{*\alpha}\}_{\alpha=1}^n$.

Remark: Let V be a real vector space of dimension $n \geq 1$, and let $\mathcal{B} = \{e_1, \dots, e_n\}$ be a basis, with dual basis $\mathcal{B}^* = \{e^{*1}, \dots, e^{*n}\}$. There is a unique linear map

$$\varphi_{\mathcal{B}} : V \longrightarrow V^*$$

such that

$$\varphi_{\mathcal{B}}(e_i) = e^{*i}, \quad i = 1, \dots, n.$$

Explicitly,

$$\varphi_{\mathcal{B}} \left(\sum_{i=1}^n v^i e_i \right) = \sum_{i=1}^n v^i e^{*i}.$$

Since this map sends a basis of V to a basis of V^* , it is an isomorphism.

However, this construction depends on the chosen basis. To see this, consider the basis $\mathcal{C} = \{b_1, \dots, b_n\}$ defined by

$$b_1 = 2e_1, \quad b_i = e_i \quad (i = 2, \dots, n).$$

Its dual basis satisfies

$$b^{*1} = \frac{1}{2} e^{*1}, \quad b^{*i} = e^{*i} \quad (i = 2, \dots, n).$$

Consequently,

$$\varphi_{\mathcal{C}}(e_1) = \varphi_{\mathcal{C}}\left(\frac{1}{2}b_1\right) = \frac{1}{2}b^{*1} = \frac{1}{4}e^{*1},$$

whereas

$$\varphi_{\mathcal{B}}(e_1) = e^{*1}.$$

Thus $\varphi_{\mathcal{C}} \neq \varphi_{\mathcal{B}}$.

Nevertheless, distinct bases need not induce distinct isomorphisms. For example, if $b_i = -e_i$ for every i , then $b^{*i} = -e^{*i}$, and hence

$$\varphi_{\mathcal{C}}(e_i) = \varphi_{\mathcal{C}}(-b_i) = -b^{*i} = e^{*i} = \varphi_{\mathcal{B}}(e_i).$$

In this case, $\varphi_{\mathcal{C}} = \varphi_{\mathcal{B}}$.

Therefore, although V and V^* are isomorphic in finite dimension, the construction above does not provide a basis-independent identification between them.

Example 1.1.4. Let $\{e_{\alpha}\}_{\alpha=1}^3$ be the standard basis of $\mathbb{R}^3$. That is,

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Then

$$\begin{aligned} e^{*1}(e_1) &= 1, & e^{*1}(e_2) &= 0, & e^{*1}(e_3) &= 0 \\ e^{*2}(e_1) &= 0, & e^{*2}(e_2) &= 1, & e^{*2}(e_3) &= 0 \\ e^{*3}(e_1) &= 0, & e^{*3}(e_2) &= 0, & e^{*3}(e_3) &= 1 \end{aligned}$$

that is, $(e^{*\alpha}(e_{\beta}))_{3 \times 3} = I_3$. Furthermore,

$$\begin{aligned} e^{*1}\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) &= e^{*1}(xe_1 + ye_2 + ze_3) \\ &= xe^{*1}(e_1) + ye^{*1}(e_2) + ze^{*1}(e_3) \\ &= x. \end{aligned}$$

Similarly, $e^{*2}\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = y$ and $e^{*3}\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = z$.

Thus $e^{*i} = \pi_i$, where $\pi_i(x_1, x_2, x_3) = x_i$ is the projection onto the i th coordinate.

Example 1.1.5. Let $U \subset \mathbb{R}^n$ be open, and let $f : U \rightarrow \mathbb{R}$ be differentiable at $a \in U$. By definition, there exists a linear map

$$df(a) : \mathbb{R}^n \rightarrow \mathbb{R}$$

such that

$$f(a+h) = f(a) + df(a)(h) + r(h), \quad \lim_{\substack{h \rightarrow 0 \\ h \neq 0}} \frac{|r(h)|}{\|h\|} = 0.$$

Here h is sufficiently small that $a + h \in U$. Thus

$$df(a) \in L(\mathbb{R}^n, \mathbb{R}) = (\mathbb{R}^n)^*.$$

Let $\{e_i\}_{i=1}^n$ be the standard basis of $\mathbb{R}^n$, and let $\{e^{*i}\}_{i=1}^n$ be its dual basis. For each i , the definition of differentiability gives

$$\begin{aligned} \frac{\partial f}{\partial x^i}(a) &= \lim_{t \rightarrow 0} \frac{f(a + te_i) - f(a)}{t} \\ &= df(a)(e_i). \end{aligned}$$

Indeed, the remainder satisfies

$$\left| \frac{r(te_i)}{t} \right| = \frac{|r(te_i)|}{\|te_i\|} \rightarrow 0.$$

Consequently, for $v = \sum_{i=1}^n v^i e_i$, linearity yields

$$\begin{aligned} df(a)(v) &= \sum_{i=1}^n v^i df(a)(e_i) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) v^i \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) e^{*i}(v). \end{aligned}$$

Since this holds for every $v \in \mathbb{R}^n$, we obtain

$$df(a) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) e^{*i}.$$

To relate this expression to the usual differential notation, let $x^i : U \rightarrow \mathbb{R}$ be the coordinate function $x^i(p) = p^i$. Since

$$x^i(a + h) - x^i(a) = h^i,$$

its differential satisfies

$$dx^i(a)(v) = v^i = e^{*i}(v).$$

Therefore $dx^i(a) = e^{*i}$, and

$$df(a) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) dx^i(a).$$

With respect to the standard Euclidean inner product, the gradient is

$$\nabla f(a) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) e_i,$$

so

$$df(a)(v) = \langle \nabla f(a), v \rangle \quad \text{for every } v \in \mathbb{R}^n.$$

The differential is a covector, whereas the gradient is the vector corresponding to it under the identification induced by the Euclidean inner product.

Theorem 1.1.6. *Let V be a $\mathbb{K}$ -vector space, with $x \in V$. The map given by*

$$\begin{aligned} \phi: V &\longrightarrow V^{**} \\ x &\mapsto \phi(x): V^* \longrightarrow \mathbb{K} \\ f &\mapsto \phi(x)(f) = f(x). \end{aligned}$$

is an isomorphism independent of the choice of bases, called the natural isomorphism.

Proof. We show that ϕ is an isomorphism. First, we check that the map is well defined. Let $f \in V^*$ be arbitrary. Then

$$\begin{aligned} \phi(\alpha x + \beta y)(f) &= f(\alpha x + \beta y) \\ &= \alpha f(x) + \beta f(y) \\ &= \alpha \phi(x)(f) + \beta \phi(y)(f) \end{aligned}$$

Thus $\phi(\alpha x + \beta y) = \alpha \phi(x) + \beta \phi(y)$, and hence $\phi(x) \in V^{**}$, where $\{e_\alpha\}_{\alpha=1}^n$ is a basis of V .

To prove that ϕ is injective, suppose that $x \in \ker \phi$. Then $\phi(x) = 0$, so $f(x) = 0, \forall f \in V^*$. Since $e^{*\alpha} \in V^*$, we have $e^{*\alpha}(x) = 0$, with $x = x^\beta e_\beta \in V$. Thus

$$\begin{aligned} e^{*\alpha}(x^\beta e_\beta) &= 0 \\ x^\alpha &= 0, \end{aligned}$$

that is, $x^\alpha = 0$ for $\alpha = 1, 2, \dots, n$, and hence $x = 0$. Therefore $\ker \phi = \{0\}$, so ϕ is injective.

On the other hand, the rank-nullity theorem gives $\dim(V) - \dim(\ker \phi) = \dim(\text{Im } \phi)$,

so $\dim(V) = \dim(\text{Im } \phi) = \dim(V^{**})$. Hence ϕ is bijective. Note that this isomorphism is independent of the choice of bases. $\square$

Remark: Let $\{e_\alpha\}$ be a basis of V and $\{e^{*\alpha}\}$ the dual basis of V^* . Since V^{**} is the dual space of V^* , we can consider the dual basis $\{e_\alpha\}$ in V^{**} . It is easy to show that $\phi(e_\alpha) = e_\alpha$ for the map in Theorem 1.1.6, with $\phi(e_\alpha)(e^{*\beta}) = e^{*\beta}(e_\alpha) = \delta_\alpha^\beta$, which implies that

$$\begin{aligned} e_\alpha: V^* &\longrightarrow \mathbb{K} \\ e^{*\beta} &\mapsto e_\alpha(e^{*\beta}) = \delta_\alpha^\beta, \end{aligned}$$

Moreover, $e_\alpha(u_\beta e^{*\beta}) = u^\alpha$. Note that we can write

$$\begin{aligned} x &= x^1 e_1 + x^2 e_2 + \dots + x^n e_n \\ \chi &= \chi^1 e_1 + \chi^2 e_2 + \dots + \chi^n e_n \end{aligned}$$

where $\phi(x) = \chi$. Thus the image of x under the isomorphism ϕ is χ ; furthermore, $\chi(e^{*\beta}) = \chi^\alpha e_\alpha(e^{*\beta}) = \delta_\alpha^\beta$.

We now generalize linearity to define multilinear maps on a Cartesian product of vector spaces.

Definition 1.1.7. Let $V_1, V_2, \dots, V_p, V$ be vector spaces over a field $\mathbb{K}$. The map

$$f : V_1 \times V_2 \times \dots \times V_p \longrightarrow V$$

is p -linear if it satisfies

1. $f(x_1, \dots, x_i + x'_i, \dots, x_p) = f(x_1, \dots, x_i, \dots, x_p) + f(x_1, \dots, x'_i, \dots, x_p)$
2. $f(x_1, \dots, \lambda x_i, \dots, x_p) = \lambda f(x_1, \dots, x_i, \dots, x_p)$

for every $x_j \in V_j$ and every $x'_i \in V_i$, with $i, j \in \{1, 2, \dots, p\}$, and every $\lambda \in \mathbb{K}$.

The vector space of all multilinear maps from $V_1 \times V_2 \times \dots \times V_p$ to V is denoted by $L(V_1, \dots, V_p; V)$. In particular, if $V_i = V$ for every $i \in I$, then $L(\underbrace{V, V, \dots, V}_{p\text{-times}}; V)$ is denoted by $L^p(V, V)$. Otherwise, if $f \in L^p(V, \mathbb{R})$, then f is called a multilinear form on V .

Remark: $L^1(V, \mathbb{R}) = L(V, \mathbb{R}) = V^*$.

Example 1.1.8. Define

$$\begin{aligned} f : V \times V^* &\longrightarrow \mathbb{R} \\ (x, g) &\mapsto g(x) \end{aligned}$$

Then f is linear in the first argument by the linearity of $g \in V^*$, and linear in the second argument by the definition of addition of functions.

Definition 1.1.9. Let V be a vector space over a field $\mathbb{K}$.

1. The set given by

$$T_p(V) = \{f : \underbrace{V \times V \times \dots \times V}_{p\text{-times}} \longrightarrow \mathbb{K} \mid f \text{ is } p\text{-linear} \}$$

is a vector space over $\mathbb{K}$ called the space of p -fold covariant tensors over $\mathbb{K}$. The elements of $T_p(V)$ are called tensors of type $\begin{pmatrix} 0 \\ p \end{pmatrix}$.

2. The set given by

$$T^q(V) = \{ \phi : \underbrace{V^* \times V^* \times \dots \times V^*}_{q\text{-times}} \longrightarrow \mathbb{K} \mid \phi \text{ is } q\text{-linear} \}$$

is a vector space over $\mathbb{K}$ called the space of q -fold contravariant tensors over $\mathbb{K}$. The elements of $T^q(V)$ are called tensors of type $\binom{q}{0}$.

3. The set

$$T_p^q(V) = \{ \eta : \underbrace{V^* \times V^* \times \dots \times V^*}_{q\text{-times}} \times \underbrace{V \times V \times \dots \times V}_{p\text{-times}} \longrightarrow \mathbb{K} \mid \eta \text{ is } (p+q)\text{-linear} \}$$

is a vector space over $\mathbb{K}$ called the space of p -fold covariant and q -fold contravariant tensors over $\mathbb{K}$. The elements of $T_p^q(V)$ are called tensors of type $\binom{q}{p}$.

Example 1.1.10. Let V be a vector space, and let $x_1, x_2, \dots, x_p \in V$. If we define

$$\begin{aligned} f : \underbrace{V^* \times V^* \times \dots \times V^*}_{p\text{-times}} \times \underbrace{V \times V \times \dots \times V}_{p\text{-times}} &\longrightarrow \mathbb{R} \\ (g_1, \dots, g_p)(x_1, \dots, x_p) &\mapsto g_1(x_1)g_2(x_2)\dots g_p(x_p) \end{aligned}$$

then $f \in T_p^p(V)$.

Remark:

1. Let $f_j \in V^*$, with $j = 1, 2, \dots, q$. Let $\mathcal{B} = \{e_\alpha\}$ be a basis of V and $\mathcal{B}^* = \{e^{*\alpha}\}$ a basis of V^* , and write $f_j = f_j(e_\alpha)e^{*\alpha}$. Then

$$\phi(f_1(e_{\alpha_1})e^{*\alpha_1}, \dots, f_q(e_{\alpha_q})e^{*\alpha_q}) = f_1(e_{\alpha_1}) \dots f_q(e_{\alpha_q})\phi(e^{*\alpha_1}, \dots, e^{*\alpha_q}).$$

2. Let $G \in T_p^q(V)$ and $f_j \in V^*$, with $j = 1, 2, \dots, q$, and let $x_1, x_2, \dots, x_p \in V$. Write $f_j = f_j(e_{\alpha_j})e^{*\alpha_j}$ and $x_i = x_i^{\beta_i}e_{\beta_i}$. Then

$$\begin{aligned} G(x_1, x_2, \dots, x_p, f_1, \dots, f_q) &= G(x_1^{\beta_1}e_{\beta_1}, \dots, x_p^{\beta_p}e_{\beta_p}, f_1(e_{\alpha_1})e^{*\alpha_1}, \dots, f_q(e_{\alpha_q})e^{*\alpha_q}) \\ &= x_1^{\beta_1} \dots x_p^{\beta_p} f_1(e_{\alpha_1}) \dots f_q(e_{\alpha_q}) G(e_{\beta_1}, \dots, e_{\beta_p}, e^{*\alpha_1}, \dots, e^{*\alpha_q}). \end{aligned}$$

1.1.2 The Tensor Product

In various areas of mathematics, the tensor product is constructed on modules using generators and relations. Here, it is defined as a bilinear form.

Definition 1.1.11. 1. Let $f \in T_p(V)$ and $g \in T_q(V)$. The tensor product of f and g is defined

as the $p + q$ -linear map

$$\begin{aligned} f \otimes g : \underbrace{V \times V \times \dots \times V}_{p\text{-times}} \times \underbrace{V \times V \times \dots \times V}_{q\text{-times}} &\longrightarrow \mathbb{K} \\ (x_1, \dots, x_p, y_1, \dots, y_q) &\mapsto f(x_1, \dots, x_p)g(y_1, \dots, y_q) \end{aligned}$$

2. Let $\phi \in T^p(V)$ and $\varphi \in T^q(V)$. The tensor product of ϕ and φ is defined as the $p + q$ -linear map

$$\begin{aligned} \phi \otimes \varphi : \underbrace{V^* \times V^* \times \dots \times V^*}_{p\text{-times}} \times \underbrace{V^* \times V^* \times \dots \times V^*}_{q\text{-times}} &\longrightarrow \mathbb{K} \\ (f_1, \dots, f_p, g_1, \dots, g_q) &\mapsto \phi(f_1, \dots, f_p)\varphi(g_1, \dots, g_q) \end{aligned}$$

3. Let $f \in T_p(V)$ and $\varphi \in T^q(V)$. The tensor product of f and φ is defined as the $p + q$ -linear map

$$\begin{aligned} f \otimes \varphi : \underbrace{V \times V \times \dots \times V}_{p\text{-times}} \times \underbrace{V^* \times V^* \times \dots \times V^*}_{q\text{-times}} &\longrightarrow \mathbb{K} \\ (x_1, \dots, x_p, h_1, \dots, h_q) &\mapsto f(x_1, \dots, x_p)\varphi(h_1, \dots, h_q) \end{aligned}$$

Generalizing this definition, we obtain the following

Definition 1.1.12. If $F \in T_l^k(V)$ and $G \in T_q^p(V)$, then $F \otimes G \in T_{l+q}^{k+p}(V)$ is given by

$$\begin{aligned} F \otimes G(\omega_1, \dots, \omega_k, \omega_{k+1}, \dots, \omega_{k+p}, v_1, \dots, v_l, v_{l+1}, \dots, v_{l+q}) \\ := F(\omega_1, \dots, \omega_k, v_1, \dots, v_l)G(\omega_{k+1}, \dots, \omega_{k+p}, v_{l+1}, \dots, v_{l+q}) \end{aligned}$$

with $\omega_i \in V^*$ and $v_i \in V$.

The following properties follow directly from the definition

Theorem 1.1.13. 1. If $f \in T_p(V)$ and $g \in T_q(V)$, then $f \otimes g \in T_{p+q}(V)$.

2. If $\phi \in T^p(V)$ and $\varphi \in T^q(V)$, then $\phi \otimes \varphi \in T^{p+q}(V)$.

3. If $f \in T_p(V)$ and $\varphi \in T^q(V)$, then $f \otimes \varphi \in T_p^q(V)$.

Remark: In general, $f \otimes g \neq g \otimes f$, as the next example shows.

Example 1.1.14. Let $f = e^{*1} - e^{*2}$ and $g = 2e^{*1}$. Then

$$\begin{aligned} f \otimes g(e_1, e_2) &= ((e^{*1} - e^{*2}) \otimes 2e^{*1})(e_1, e_2) \\ &= ((e^{*1} - e^{*2})(e_1)2e^{*1}(e_2)) \\ &= 2(e^{*1} - e^{*2})(e_1)e^{*1}(e_2) \\ &= 0. \end{aligned}$$

Similarly, it is easy to see that $g \otimes f(e_1, e_2) = -2$.

Proposition 1.1.15. *Let $f, f' \in T_p(V)$, $g, g' \in T_r(V)$, and $h \in T_s(V)$. Then*

$$\begin{aligned}(f + f') \otimes g &= f \otimes g + f' \otimes g \\ (\lambda f) \otimes g &= f \otimes \lambda g = \lambda(f \otimes g) \\ f \otimes (g + g') &= f \otimes g + f \otimes g' \\ f \otimes (g \otimes h) &= (f \otimes g) \otimes h.\end{aligned}$$

These properties also hold for contravariant and mixed tensors.

Theorem 1.1.16. *Let $\{e_\alpha\}_{\alpha=1}^n$ be a basis of V and $\{e^{*\alpha}\}_{\alpha=1}^n$ a basis of V^* . Then $\{e^{*\alpha} \otimes e^{*\beta}\}_{\alpha,\beta=1}^n$ is a basis of $T_2(V)$, and therefore $\dim T_2(V) = n^2$.*

Proof. We know that $e^{*\alpha} \otimes e^{*\beta} \in T_2(V)$, since $e^{*\alpha} \in T_1(V)$ and $e^{*\beta} \in T_1(V)$. It suffices to prove that $\{e^{*\alpha} \otimes e^{*\beta}\}$ is linearly independent and spans $T_2(V)$.

Consider the linear combination

$$\lambda_{\alpha\beta} e^{*\alpha} \otimes e^{*\beta} = 0_{\mathbb{K}} \in T_1(V),$$

For fixed i, j with $1 \leq i, j < n$, we have

$$\lambda_{\alpha\beta} e^{*\alpha} \otimes e^{*\beta}(e_i, e_j) = 0_{\mathbb{K}}(e_i, e_j) \in T_1(V),$$

that is, $\lambda_{ij} = 0$ for $i, j \in \{1, 2, \dots, n\}$. Hence $\{e^{*\alpha} \otimes e^{*\beta}\}$ is linearly independent. We now show that $\{e^{*\alpha} \otimes e^{*\beta}\}$ spans $T_2(V)$. Let $f \in T_2(V)$. We claim that

$$f = f(e_\alpha, e_\beta) e^{*\alpha} \otimes e^{*\beta}.$$

Let $(u_1, u_2) \in V \times V$, with $u_i = u_i^\beta e_\beta$. Then

$$\begin{aligned}f(e_\alpha, e_\beta) e^{*\alpha} \otimes e^{*\beta}(u_1, u_2) &= f(e_\alpha, e_\beta) e^{*\alpha}(u_1) e^{*\beta}(u_2) \\ &= f(e_\alpha, e_\beta) u_1^\alpha u_2^\beta \\ &= f(u_1^\alpha e_\alpha, u_2^\beta e_\beta) \\ &= f(u_1, u_2).\end{aligned}$$

Remark: More generally, one can prove that □

1. If $\{e_\alpha\}$ is a basis of V and $\{e^{*\alpha}\}$ a basis of the dual V^* , then

(a) $\{e^{*i_1} \otimes \dots \otimes e^{*i_p}\}_{i_1, \dots, i_p=1, \dots, n}$ is a basis of $T_p(V)$.

(b) $\dim T_p(V) = n^p$

- (c) If $f \in T_p(V)$, then $f = f(e_{i_1}, \dots, e_{i_p})e^{*i_1} \otimes \dots \otimes e^{*i_p}$
2. If $\{e^{*\alpha}\}$ is a basis of the dual V^* and $\{e_\alpha\}$ a basis of the bidual V^{**} , then
- (a) $\{e_{i_1} \otimes \dots \otimes e_{i_q}\}_{i_1, \dots, i_q=1, \dots, n}$ is a basis of $T^q(V)$.
- (b) $\dim T^q(V) = n^q$
- (c) If $\phi \in T^q(V)$, then $\phi = \phi(e^{*i_1}, \dots, e^{*i_q})e_{i_1} \otimes \dots \otimes e_{i_q}$
3. If $\{e_\alpha\}$ is a basis of V , $\{e^{*\alpha}\}$ a basis of the dual V^* , and $\{e_\alpha\}$ a basis of the bidual V^{**} , then
- (a) $\{e^{*i_1} \otimes \dots \otimes e^{*i_p} \otimes e_{j_1} \otimes \dots \otimes e_{j_q}\}$ is a basis of $T_p^q(V)$.
- (b) $\dim T_p^q(V) = n^{p+q}$
- (c) If $G \in T_p^q(V)$, then

$$G = G(e_{i_1}, \dots, e_{i_p}, e^{*j_1}, \dots, e^{*j_q})e^{*i_1} \otimes \dots \otimes e^{*i_p} \otimes e_{j_1} \otimes \dots \otimes e_{j_q}$$

1.1.3 Change of Basis

To study coordinate transformations (changes of basis) for elements expressed in different bases, consider the following.

Let $\{e_\alpha\}$ and $\{b_\beta\}$ be bases of the vector space V , with dual bases $\{e^{*\alpha}\}$ and $\{b^{*\beta}\}$, respectively. If we know how $\{e_\alpha\}$ and $\{b_\beta\}$ are related, it is natural to ask how $\{e^{*\alpha}\}$ and $\{b^{*\beta}\}$ are related.

Let $f \in V^*$, and write $f = f(e_\alpha)e^{*\alpha}$ and $f = f(b_\beta)b^{*\beta}$ in the respective bases. Since $e^{*\alpha} \in V^*$ and $\{b^{*\beta}\}$ is a basis of V^* , we have

$$e^{*\alpha} = e^{*\alpha}(b_\beta)b^{*\beta}, \quad (1.1)$$

Moreover, since $b_\beta \in V$ and $\{e_\alpha\}$ is a basis of V , there exist $\eta_\beta^1, \eta_\beta^2, \dots, \eta_\beta^n \in \mathbb{K}$ such that the elements of $\{b_\beta\}$ can be expressed in the basis $\{e_\alpha\}$ as

$$\begin{aligned} b_\beta &= \eta_\beta^1 e_1 + \eta_\beta^2 e_2 + \dots + \eta_\beta^n e_n \\ &= \eta_\beta^\sigma e_\sigma. \end{aligned}$$

Substituting this expression into equation 1.1 gives

$$\begin{aligned} e^{*\alpha} &= e^{*\alpha}(\eta_\beta^\sigma e_\sigma)b^{*\beta} \\ &= \eta_\beta^\alpha b^{*\beta}, \end{aligned}$$

Hence $f = f(e_\alpha)\eta_\beta^\alpha b^{*\beta}$, and therefore $f(b_\beta)b^{*\beta} = f(e_\alpha)\eta_\beta^\alpha b^{*\beta}$. By uniqueness of the representation in a basis,

$$f(b_\beta) = f(e_\alpha)\eta_\beta^\alpha,$$

Thus, knowing the relations between the bases allows us to express the same element in either basis. This is particularly useful for avoiding lengthy computations in a specific coordinate system.

Example 1.1.17. Let $f \in T_2(V)$ and $x_1, x_2 \in V$, with $x_1 = x_1^\alpha e_\alpha$ and $x_2 = x_2^\alpha e_\alpha$, respectively. Then

$$\begin{aligned} f(x_1, x_2) &= f(x_1^\alpha e_\alpha, x_2^\beta e_\beta) \\ &= x_1^\alpha x_2^\beta f(e_\alpha, e_\beta) \end{aligned}$$

This expression can be written in matrix form by expanding the summations:

$$\begin{aligned} f(x_1, x_2) &= x_1^1 x_2^1 f(e_1, e_1) + x_1^1 x_2^2 f(e_1, e_2) + \dots + x_1^1 x_2^n f(e_1, e_n) + \\ &\quad x_1^2 x_2^1 f(e_2, e_1) + x_1^2 x_2^2 f(e_2, e_2) + \dots + x_1^2 x_2^n f(e_2, e_n) + \\ &\quad \vdots \quad \quad \quad \vdots \quad \quad \quad \ddots \quad \quad \quad \vdots \\ &\quad x_1^n x_2^1 f(e_n, e_1) + x_1^n x_2^2 f(e_n, e_2) + \dots + x_1^n x_2^n f(e_n, e_n) \end{aligned}$$

whence we can write

$$f(x_1, x_2) = x_1^1 (x_2^{i1} f(e_1, e_{i1})) + x_2^2 (x_2^{i2} f(e_2, e_{i2})) + \dots + x_1^n (x_2^{in} f(e_n, e_{in}))$$

Since the sums are independent, we can replace the indices $i1, i2, \dots, in$ with a common summation index σ . Then

$$f(x_1, x_2) = (x_1^1, x_1^2, \dots, x_1^n) \begin{pmatrix} x_2^\sigma f(e_1, e_\sigma) \\ x_2^\sigma f(e_2, e_\sigma) \\ \vdots \\ x_2^\sigma f(e_n, e_\sigma) \end{pmatrix}$$

Expanding the sums yields the matrix equation

$$f(x_1, x_2) = (x_1^1, x_1^2, \dots, x_1^n) \begin{pmatrix} f(e_1, e_1) & f(e_1, e_2) & \dots & f(e_1, e_n) \\ f(e_2, e_1) & f(e_2, e_2) & \dots & f(e_2, e_n) \\ \vdots & \vdots & \ddots & \vdots \\ f(e_n, e_1) & f(e_n, e_2) & \dots & f(e_n, e_n) \end{pmatrix} \begin{pmatrix} x_2^1 \\ x_2^2 \\ \vdots \\ x_2^n \end{pmatrix},$$

equivalently,

$$f(x_1, x_2) = x_1^\alpha f(e_\alpha, e_\beta) x_2^\beta.$$

Remark: In the preceding example, if $\{b_\alpha\}$ is another basis of V and $b_i = \eta_i^\tau e_\tau$ and $b_j = \eta_j^\sigma e_\sigma$, then the preceding equation becomes

$$f(b_i, b_j) = \eta_i^\tau \eta_j^\sigma f(e_\tau, e_\sigma).$$

In general, let $\{e_\alpha\}$ and $\{b_\beta\}$ be bases of V , and let $\{e^{*\alpha}\}$ and $\{b^{*\beta}\}$ be bases of V^* .

If

$$\begin{aligned} b_{j1} &= \eta_{j1}^{i1} e_{i1}, \\ b_{j2} &= \eta_{j2}^{i2} e_{i2}, \\ &\vdots \\ b_{jp} &= \eta_{jp}^{ip} e_{ip}, \end{aligned}$$

and $f \in T_p(V)$, then

$$\begin{aligned} f(b_{j1}, b_{j2}, \dots, b_{jp}) &= f(\eta_{j1}^{i1} e_{i1}, \eta_{j2}^{i2} e_{i2}, \dots, \eta_{jp}^{ip} e_{ip}) \\ &= \eta_{j1}^{i1} \eta_{j2}^{i2} \dots \eta_{jp}^{ip} f(e_{i1}, e_{i2}, \dots, e_{ip}). \end{aligned}$$

On the other hand, if we take

$$\begin{aligned} b^{*k1} &= \rho_{\alpha 1}^{k1} e^{*\alpha 1}, \\ b^{*k2} &= \rho_{\alpha 2}^{k2} e^{*\alpha 2}, \\ &\vdots \\ b^{*kq} &= \rho_{\alpha q}^{kq} e^{*\alpha q}, \end{aligned}$$

and $\psi \in T^q(V)$, then

$$\psi(b^{*k1}, b^{*k2}, \dots, b^{*kq}) = \rho_{\alpha 1}^{k1} \rho_{\alpha 2}^{k2} \dots \rho_{\alpha q}^{kq} \psi(e^{*\alpha 1}, e^{*\alpha 2}, \dots, e^{*\alpha q}).$$

For $G \in T_p^q(V)$, we obtain

$$G(b_{j1}, b_{j2}, \dots, b_{jp}, b^{*k1}, b^{*k2}, \dots, b^{*kq}) = \eta_{j1}^{i1} \eta_{j2}^{i2} \dots \eta_{jp}^{ip} \rho_{\alpha 1}^{k1} \rho_{\alpha 2}^{k2} \dots \rho_{\alpha q}^{kq} G(e_{i1}, \dots, e_{ip}, e^{*\alpha 1}, \dots, e^{*\alpha q}).$$

1.1.4 Tensors from a Physical and Geometric Perspective

Definition 1.1.18. Let $(x^1, \dots, x^n)$ and $(\bar{x}^1, \dots, \bar{x}^n)$ be the coordinates of the same point in two coordinate systems. Suppose that, for $k = 1, 2, \dots, n$, there are independent relations between these coordinates:

$$\bar{x}^k = \bar{x}^k(x^1, \dots, x^n).$$

All the functions are assumed to be continuous with continuous derivatives. Each coordinate tuple $(\bar{x}^1, \dots, \bar{x}^n)$ in one system corresponds to a unique coordinate tuple $(x^1, \dots, x^n)$ in the other, so that

$$x^k = x^k(\bar{x}^1, \dots, \bar{x}^n)$$

for $k = 1, 2, \dots, n$.

These relations give the coordinate transformation formulas from one-coordinate system into other.

Definition 1.1.19. 1. Suppose that we have n scalar functions $A^\alpha = A^\alpha(x^1, \dots, x^n)$ in one coordinate system and n scalar functions $\bar{A}^i = \bar{A}^i(\bar{x}^1, \dots, \bar{x}^n)$ in another, for $i = 1, 2, \dots, n$. If they satisfy the transformation law

$$\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^\alpha} A^\alpha \quad i = 1, 2, \dots, n,$$

we say that A^α are the components of a once-contravariant tensor, or a contravariant tensor of order one.

2. Similarly, suppose that we have n functions $A_\alpha = A_\alpha(x^1, \dots, x^n)$ in one coordinate system and n functions $\bar{A}_i = \bar{A}_i(\bar{x}^1, \dots, \bar{x}^n)$ in another, with $i = 1, 2, \dots, n$. If they satisfy the transformation law

$$\bar{A}_i = \frac{\partial x^\alpha}{\partial \bar{x}^i} A_\alpha \quad i = 1, 2, \dots, n,$$

we say that A_α are the components of a once-covariant tensor, or a covariant tensor of order one.

Definition 1.1.20. 1. Suppose that we have n^2 scalar functions $A^{\alpha\beta} = A^{\alpha\beta}(x^1, \dots, x^n)$, where $\alpha, \beta \in \{1, 2, \dots, n\}$. If another collection of n^2 scalar functions $\bar{A}^{ij} = \bar{A}^{ij}(\bar{x}^1, \dots, \bar{x}^n)$ is obtained by the transformation

$$\bar{A}^{ij} = \frac{\partial \bar{x}^i}{\partial x^\alpha} \frac{\partial \bar{x}^j}{\partial x^\beta} A^{\alpha\beta}, \quad 1 \leq i, j \leq n$$

we say that $A^{\alpha\beta}$ are the components of a twice-contravariant tensor, also called a contravariant tensor of order two.

2. Suppose that we have n^2 scalar functions $A_{\alpha\beta} = A_{\alpha\beta}(x^1, \dots, x^n)$, where $\alpha, \beta \in \{1, 2, \dots, n\}$. If another collection of n^2 scalar functions $\bar{A}_{ij} = \bar{A}_{ij}(\bar{x}^1, \dots, \bar{x}^n)$ is obtained by the transformation

$$\bar{A}_{ij} = \frac{\partial x^\alpha}{\partial \bar{x}^i} \frac{\partial x^\beta}{\partial \bar{x}^j} A_{\alpha\beta}, \quad 1 \leq i, j \leq n$$

we say that $A_{\alpha\beta}$ are the components of a twice-covariant tensor, also called a covariant tensor of order two.

Definition 1.1.21. Suppose that we have n^{p+q} scalar functions

$A_{\alpha_1 \alpha_2 \dots \alpha_p}^{\beta_1 \beta_2 \dots \beta_q} = A_{\alpha_1 \alpha_2 \dots \alpha_p}^{\beta_1 \beta_2 \dots \beta_q}(x^1, \dots, x^n)$, where $1 \leq \beta_i, \alpha_j \leq n$, for $1 \leq i \leq q$ and $1 \leq j \leq p$. If another collection of n^{p+q} scalar functions $\bar{A}_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q} = \bar{A}_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q}(\bar{x}^1, \dots, \bar{x}^n)$ is obtained by the transformation

$$\bar{A}_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q} = \frac{\partial x^{\alpha_1}}{\partial \bar{x}^{i_1}} \frac{\partial x^{\alpha_2}}{\partial \bar{x}^{i_2}} \dots \frac{\partial x^{\alpha_p}}{\partial \bar{x}^{i_p}} \frac{\partial \bar{x}^{j_1}}{\partial x^{\beta_1}} \frac{\partial \bar{x}^{j_2}}{\partial x^{\beta_2}} \dots \frac{\partial \bar{x}^{j_q}}{\partial x^{\beta_q}} A_{\alpha_1 \alpha_2 \dots \alpha_p}^{\beta_1 \beta_2 \dots \beta_q}.$$

we say that $A_{\alpha_1 \alpha_2 \dots \alpha_p}^{\beta_1 \beta_2 \dots \beta_q}$ are the components of a q -fold contravariant and p -fold covariant tensor.

1.1.5 Equivalence of the Two Definitions of Tensors

Suppose that $T \in T_1^1(\mathbb{R}^n) = \{T : \mathbb{R}^n \times (\mathbb{R}^n)^* \longrightarrow \mathbb{R} \mid T \text{ is bilinear} \}$, where $(\mathbb{R}^n)^* = \{f : \mathbb{R}^n \longrightarrow \mathbb{R} \mid f \text{ is linear} \}$, and consider a coordinate transformation

$$\begin{aligned} \phi : \quad U \subset \mathbb{R}^n &\longrightarrow V \subset \mathbb{R}^n \\ (\bar{x}^1, \bar{x}^2, \dots, \bar{x}^n) &\mapsto (x^1, x^2, \dots, x^n) \end{aligned}$$

where U and V are open subsets of $\mathbb{R}^n$. Thus $x^i = x^i(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^n)$, where $i = 1, 2, \dots, n$. Suppose also that $\{e_i\}_{i=1}^n$ and $\{b_\alpha\}_{\alpha=1}^n$ are bases of $\mathbb{R}^n$ in the coordinate spaces $(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^n)$ and $(x^1, x^2, \dots, x^n)$, respectively. Write

$$T_i^j = T(e_i, e^{*j}) \quad \text{and} \quad T_\alpha^\beta = T(b_\alpha, b^{*\beta}). \quad (1.2)$$

We know that

$$\begin{aligned} T &= T(e_i, e^{*j}) e^{*i} \otimes e_j \\ &= T_i^j e^{*j} \otimes e_j. \end{aligned} \quad (1.3)$$

On the other hand, since

$$e^{*i} = d\bar{x}^i, \quad b^{*\beta} = dx^\beta, \quad d\bar{x}^i = \frac{\partial \bar{x}^i}{\partial x^\alpha} dx^\alpha \quad \text{and} \quad dx^\beta = \frac{\partial x^\beta}{\partial \bar{x}^j} d\bar{x}^j$$

equations (1.2) and (1.3) give

$$\begin{aligned} T_\alpha^\beta &= T(b_\alpha, b^{*\beta}) \\ &= T_i^j e^{*i} \otimes e_j(b_\alpha, b^{*\beta}) \\ &= T_i^j d\bar{x}^i \otimes e_j(b_\alpha, dx^\beta) \\ &= T_i^j \frac{\partial \bar{x}^i}{\partial x^\alpha} dx^\alpha \otimes e_j \left(b_\alpha, \frac{\partial x^\beta}{\partial \bar{x}^j} d\bar{x}^j \right) \\ &= T_i^j \frac{\partial \bar{x}^i}{\partial x^\alpha} b^{*\alpha} \otimes e_j \left(b_\alpha, \frac{\partial x^\beta}{\partial \bar{x}^j} e^{*j} \right) \\ &= T_i^j \frac{\partial \bar{x}^i}{\partial x^\alpha} \frac{\partial x^\beta}{\partial \bar{x}^j} b^{*\alpha}(b_\alpha) e_j(e^{*j}) \\ &= T_i^j \frac{\partial \bar{x}^i}{\partial x^\alpha} \frac{\partial x^\beta}{\partial \bar{x}^j}, \end{aligned}$$

that is,

$$T_\alpha^\beta = T_i^j \frac{\partial \bar{x}^i}{\partial x^\alpha} \frac{\partial x^\beta}{\partial \bar{x}^j}.$$

Thus every tensor of type $(1, 1)$ in the algebraic sense gives rise to a tensor in the physical and geometric sense, and conversely. The general case is analogous.

1.1.6 Tensor Spaces

We define $T_0^k(V) = T^k(V)$ and $T_l^0(V) = T_l(V)$. We also have $T^0(V) = \mathbb{R}$, $T_1(V) = V^*$, and $T^1(V) = V^{**} \cong V$. Since V is finite-dimensional, we also have $T_1^1(V) \cong \text{End}(V)$, as we shall prove in the next proposition, where $T_1^1(V) = \{f : V^* \times V \rightarrow \mathbb{R} \mid f \text{ is bilinear}\}$.

Proposition 1.1.22. *Let $\phi : V \rightarrow V$ be linear. Then $f : \text{End}(V) \rightarrow T_1^1(V)$, where*

$$\begin{aligned} f(\phi) : V^* \times V &\longrightarrow \mathbb{R} \\ (w, v) &\longmapsto w(\phi(v)) \end{aligned}$$

is an isomorphism.

Proof. Let $\{e_\alpha\}$ be a basis of V and $\{e^{*\alpha}\}$ the dual basis of V^* . Note that f is linear by the linearity of ω .

To show that f is injective, suppose that $f(\phi) = f(\psi)$ for some $\phi, \psi \in \text{End}(V)$. For arbitrary $x \in V$ and $\omega \in V^*$, we have

$$f(\phi)(\omega, x) = f(\psi)(\omega, x) \iff \omega(\phi(x) - \psi(x)) = 0$$

By the linearity of f , it follows that $\phi x = \psi x$, and hence $\phi = \psi$.

For surjectivity, let $T \in T_1^1(V)$ and write $T = T_\beta^\alpha e_\alpha \otimes e^{*\beta}$. We seek $\phi \in \text{End}(V)$ such that $f(\phi) = T$. Choose $\phi \in \text{End}(V)$ so that ϕ has matrix T_β^α . For $x = x^\alpha e_\alpha$ and $\omega = \omega(e_\alpha) e^{*\alpha}$, we obtain

$$\begin{aligned} f(\phi)(\omega, x) &= \omega(\phi(x)) \\ &= \omega(e_k) e^{*k} (T_j^i x^j e_i) \\ &= T_j^i x^j \omega(e_k) e^{*k}(e_i) \\ &= T_j^i x^j \omega(e_k) \delta_i^k \\ &= T_j^k x^j \omega(e_k). \end{aligned}$$

On the other hand,

$$\begin{aligned} T(\omega, x) &= T(w(e_\alpha) e^{*\alpha}, x^i e_i) \\ &= \omega(e_\alpha) x^i T(e^{*\alpha}, e_i) \\ &= T_i^\alpha x^i \omega(e_\alpha). \end{aligned}$$

Since x and ω were arbitrary, taking $i = j$ and $\alpha = k$ gives $f(\phi) = T$. Thus f is linear and bijective, hence an isomorphism. $\square$

The following statement generalizes this case.

Lemma 1.1.23. *There is a natural isomorphism between $T_{l+1}^k(V)$ and the space of maps defined by*

$$\{\phi : \underbrace{V^* \times \dots \times V^*}_{l\text{-times}} \times \underbrace{V \times \dots \times V}_{k\text{-times}} \longrightarrow V \mid \phi \text{ is multilinear} \}$$

Proof. Let $\mathcal{L} := \{\phi : \underbrace{V^* \times \dots \times V^*}_{l\text{-times}} \times \underbrace{V \times \dots \times V}_{k\text{-times}} \longrightarrow V \mid \phi \text{ is multilinear} \}$.

Define

$$\begin{aligned} f : \mathcal{L} &\longrightarrow T_k^{l+1}(V) \\ \phi &\mapsto f(\phi) : \underbrace{V^* \times \dots \times V^*}_{l+1\text{-times}} \times \underbrace{V \times \dots \times V}_{k\text{-times}} \longrightarrow \mathbb{R} \\ &\quad (\omega_1, \dots, \omega_{l+1}, v_1, \dots, v_k) \mapsto \omega_{l+1}(\phi(\omega_1, \dots, \omega_l, v_1, \dots, v_k)). \end{aligned}$$

with $\phi(\omega_1, \dots, \omega_l, v_1, \dots, v_k) \in V$. Linearity follows immediately from ϕ . We also define

$$\begin{aligned} g : T_k^{l+1}(V) &\longrightarrow \mathcal{L} \\ T &\mapsto g(T) : \underbrace{V^* \times \dots \times V^*}_{l\text{-times}} \times \underbrace{V \times \dots \times V}_{k\text{-times}} \longrightarrow V \\ &\quad (\omega_1, \dots, \omega_l, v_1, \dots, v_k) \mapsto T(\omega_1, \dots, \omega_l, \cdot, v_1, \dots, v_k), \end{aligned}$$

where

$$(f \circ g)(T) = T$$

and

$$(g \circ f)(\phi) = \phi$$

giving the required isomorphism. $\square$

Remark: If $\{e_i\}$ is a basis of V , we denote the dual basis by $\{e^{*i}\}$ and the basis of $V^{**} \cong V$ by $\{e_i\}$.

A basis of $T_l^k(V)$ is therefore

$$\{e_{i_1} \otimes e_{i_2} \otimes \dots \otimes e_{i_k} \otimes e^{*j_1} \otimes \dots \otimes e^{*j_l}\}$$

with $(j_1, \dots, j_l) \in S_n$ and $(i_1, \dots, i_k) \in S_n$.

A tensor $F \in T_l^k(V)$ can be written as

$$F = F_{j_1 j_2 \dots j_l}^{i_1 i_2 \dots i_k} e_{i_1} \otimes e_{i_2} \otimes \dots \otimes e_{i_k} \otimes e^{*j_1} \otimes \dots \otimes e^{*j_l}$$

where

$$F_{j_1 j_2 \dots j_l}^{i_1 i_2 \dots i_k} = F(e^{*i_1}, \dots, e^{*i_k}, e_{j_1}, \dots, e_{j_l})$$

so that

$$F(\omega_1, \dots, \omega_k, v_1, \dots, v_l) = F_{j_1 j_2 \dots j_l}^{i_1 i_2 \dots i_k} e_{i_1}(\omega_1) \dots e_{i_k}(\omega_k) e^{*j_1}(v_1) \dots e^{*j_l}(v_l)$$

where $e_{j_r}(\omega_r) = \omega_r(e_{j_r}) = e_{j_r}(\omega_r)$.

Definition 1.1.24 (Contraction or trace). *The linear map $Tr : T_1^1(V) \longrightarrow T_0^0(V) \cong \mathbb{R}$ defined by $Tr(T) = T_i^i$, where $T = T_j^i e_i \otimes e^{*j}$, is called the trace of the tensor T .*

More generally, if $F \in T_l^k(V)$, write

$$F = F_{j_1 \dots j_l}^{i_1 \dots i_k} e^{*j_1} \otimes \dots \otimes e^{*j_l} \otimes e_{i_1} \otimes \dots \otimes e_{i_k},$$

Contracting the l -th covariant index $j_l = m$ with the first contravariant index $i_1 = n$ gives

$$\begin{aligned} Tr(F) &= F_{j_1 \dots j_{l-1} m}^{n i_2 \dots i_k} e^{*n}(e_n) e^{*j_1} \otimes \dots \otimes e^{*j_{l-1}} \otimes e_{i_2} \otimes \dots \otimes e_{i_k} \\ &= F_{j_1 \dots m}^{n \dots i_k} \delta_n^m e^{*j_1} \otimes \dots \otimes e^{*j_{l-1}} \otimes e_{i_2} \otimes \dots \otimes e_{i_k} \\ &= F_{j_1 \dots m}^{m \dots i_k} e^{*j_1} \otimes \dots \otimes e^{*j_{l-1}} \otimes e_{i_2} \otimes \dots \otimes e_{i_k}, \end{aligned}$$

with $Tr(F) \in T_{l-1}^{k-1}(V)$.

In general, if T has components $T_{j_1 \dots j_l}^{i_1 \dots i_k}$, then $Tr(T)$ has components $T_{j_1 \dots j_{l-1} m}^{m i_2 \dots i_k}$.

Definition 1.1.25. *Let $\phi \in T_l(V)$. The tensor ϕ is called symmetric if, for every $v_1, \dots, v_l \in V$ and every $\sigma \in S_n$, we have*

$$\phi(v_1, \dots, v_l) = \phi(v_{\sigma(1)}, \dots, v_{\sigma(l)}).$$

The tensor ϕ is called antisymmetric if, for every $v_1, \dots, v_l \in V$ and every $\sigma \in S_n$, we have

$$\phi(v_1, \dots, v_l) = \text{sgn}(\sigma) \phi(v_{\sigma(1)}, \dots, v_{\sigma(l)}),$$

where $\text{sgn}(\sigma) = 1$ for an even permutation and $\text{sgn}(\sigma) = -1$ for an odd permutation.

For example, if the tensor T_{ab} is symmetric, then $T_{ab} = T_{ba}$; if T_{ab} is antisymmetric, then $T_{ab} = -T_{ba}$.

1.2 Differentiable Manifolds and Tangent Spaces

Manifold theory allows us to study non-Euclidean objects that are locally homeomorphic to $\mathbb{R}^n$, making it possible to apply the tools of analysis locally.

Definition 1.2.1. *A chart, or m -dimensional coordinate system, on a topological space M is a pair (ϕ, U) , where $U \subseteq M$ and $\phi : U \longrightarrow \mathbb{R}^m$ is a homeomorphism onto its image.*

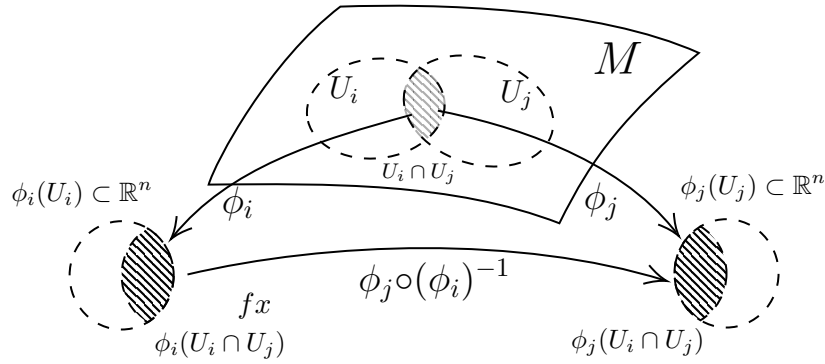
Definition 1.2.2. An m -dimensional atlas of class C^k , with $k \geq 1$, on a topological space M is a collection $\mathcal{A} = \{(U_i, \phi_i)\}_{i \in I}$ of charts on M such that:

1. $M = \bigcup_{i \in I} U_i$
2. If $U_i \cap U_j \neq \emptyset$, with $i, j \in I$, then the change of coordinates

$$\phi_j \circ (\phi_i)^{-1} : \phi_i(U_i \cap U_j) \subseteq \mathbb{R}^m \longrightarrow \phi_j(U_i \cap U_j) \subseteq \mathbb{R}^m$$

is a diffeomorphism between the open subsets $\phi_i(U_i \cap U_j)$ and $\phi_j(U_i \cap U_j)$ of $\mathbb{R}^m$.

Definition 1.2.3. M is called an m -dimensional differentiable manifold if the atlas $\mathcal{A} = \{(U_i, \phi_i)\}$ is maximal with respect to conditions 1. and 2. of the preceding definition.



Remark: Under condition 2 of Definition 1.2.2, the charts (U_i, ϕ_i) and (U_j, ϕ_j) are said to be compatible.

An atlas is maximal in the sense that it contains all compatible charts. Thus, if (U_i, ϕ_i) and (U_j, ϕ_j) are compatible, they belong to the same maximal atlas, which equips the manifold with a differentiable structure.

Remark: Manifolds may have different differentiable structures. Namely, two coordinate charts that define differentiable structures on M but are not compatible give rise to distinct differentiable structures on M .

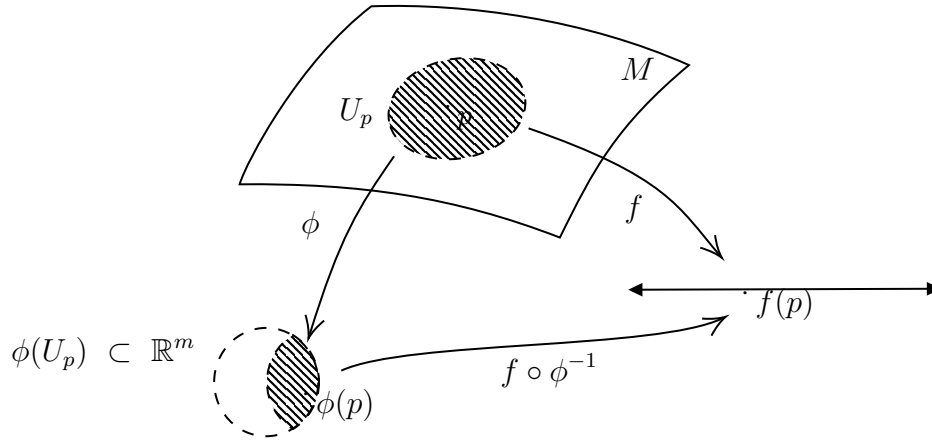
Definition 1.2.4. An atlas on M is called oriented if, for any pair of charts (U_i, ϕ_i) and (U_j, ϕ_j) , either $U_i \cup U_j = \emptyset$ or, if $U_i \cup U_j \neq \emptyset$, then

$$\det \left[J \left(\phi_j \circ (\phi_i)^{-1} \right) (\phi_i(x)) \right] > 0, \quad \text{for all } x \in U_i \cap U_j.$$

Definition 1.2.5. A manifold M is called orientable if M admits an oriented atlas.

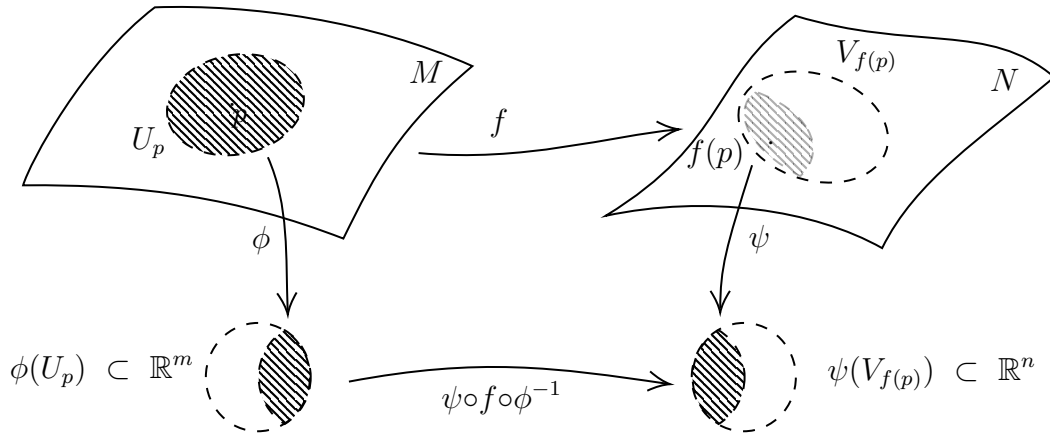
Definition 1.2.6. Let $f : M \longrightarrow \mathbb{R}$, where M is an m -dimensional manifold of class C^k , with $k \leq \infty$. We say that f is differentiable at $p \in M$ if there is a chart (U_p, ϕ) in the atlas, with $p \in U_p$, such that $f \circ \phi^{-1} : \phi(U) \subset \mathbb{R}^m \longrightarrow \mathbb{R}$ is differentiable at $\phi(p) \in \mathbb{R}^m$.

We say that f is differentiable if it is differentiable at every point of M .



Definition 1.2.7. Let $f : M \rightarrow N$ be a map between two manifolds of dimensions m and n , respectively.

We say that f is differentiable at $p \in M$ if there are charts (U_p, ϕ) and $(V_{f(p)}, \psi)$ in the atlases of M and N , respectively, such that $f(U_p) \subseteq V_{f(p)}$ and $\psi \circ f \circ \phi^{-1} : \phi(U_p) \rightarrow \psi(V_{f(p)})$ is differentiable at $\phi(p)$.



We say that f is differentiable if it is differentiable at every point of M .

Definition 1.2.8. The family of smooth scalar-valued functions on the manifold M is defined by $\mathfrak{T}(M) = \{f : M \rightarrow \mathbb{R} \mid f \text{ is of class } C^\infty\}$

Definition 1.2.9. A tangent vector to M at p is a map $X_p : \mathfrak{T}(M) \rightarrow \mathbb{R}$ that satisfies the following for every $f, g \in \mathfrak{T}(M)$ and every $r \in \mathbb{R}$:

- i. $X_p(f + g) = X_p(f) + X_p(g)$
- ii. $X_p(rf) = rX_p(f)$ and
- iii. $X_p(fg) = f(p)X_p(g) + g(p)X_p(f)$.

Here fg denotes the product of functions, and $f(p)X_p(g)$ is the product of the real numbers $f(p)$ and $X_p(g)$.

Remark: We shall show that this definition generalizes the tangent planes of surfaces in $\mathbb{R}^n$, by establishing a bijective correspondence with tangent vectors in the classical definition in $\mathbb{R}^n$. We shall also show that the set of vectors X_p is a vector space.

There are several equivalent ways to define tangent spaces.

Proposition 1.2.10. *Let X_p be a tangent vector at $p \in M$. If $\gamma : M \rightarrow \mathbb{R}$ is the constant function $\gamma(m) \equiv r$, with $r \in \mathbb{R}$, then $X_p(\gamma) = 0$.*

Proof. Let $\mathbf{1} : M \rightarrow \mathbb{R}$ be the constant function $\mathbf{1}(x) \equiv 1$. Then

$$X_p(\mathbf{1}) = X_p(\mathbf{1} \cdot \mathbf{1}) = X_p(\mathbf{1}) + X_p(\mathbf{1}) = 2X_p(\mathbf{1}).$$

Hence $X_p(\mathbf{1}) = 0$, and consequently $X_p(\gamma) = X_p(r\mathbf{1}) = rX_p(\mathbf{1}) = 0$. $\square$

Example 1.2.11. *Let $\alpha : (-\epsilon, \epsilon) \rightarrow M$ be a differentiable curve in M , with $\alpha(0) = p$. Define $X_p^\alpha : \mathfrak{T}(M) \rightarrow \mathbb{R}$ by $X_p^\alpha(f) = \left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0}$. We show that X_p^α is a tangent vector to M at p .*

Indeed, let $f, g \in \mathfrak{T}(M)$ and $r \in \mathbb{R}$. Then

$$\begin{aligned} X_p^\alpha(f + g) &= \left. \frac{d((f + g) \circ \alpha)}{dt} \right|_{t=0} \\ &= \left. \frac{d((f \circ \alpha) + (g \circ \alpha))}{dt} \right|_{t=0} \\ &= \left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0} + \left. \frac{d(g \circ \alpha)}{dt} \right|_{t=0} \\ &= X_p^\alpha(f) + X_p^\alpha(g). \end{aligned}$$

Moreover,

$$\begin{aligned} X_p^\alpha(rf) &= \left. \frac{d((rf) \circ \alpha)}{dt} \right|_{t=0} \\ &= \left. \frac{d(r(f \circ \alpha))}{dt} \right|_{t=0} \\ &= r \left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0} \\ &= rX_p^\alpha(f). \end{aligned}$$

On the other hand,

$$\begin{aligned}
 X_p^\alpha(fg) &= \left. \frac{d((fg) \circ \alpha)}{dt} \right|_{t=0} \\
 &= \left. \frac{d((f \circ \alpha)(g \circ \alpha))}{dt} \right|_{t=0} \\
 &= \left(\left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0} \right) g(\alpha(0)) + f(\alpha(0)) \left(\left. \frac{d(g \circ \alpha)}{dt} \right|_{t=0} \right) \\
 &= X_p^\alpha(f)g(p) + X_p^\alpha(g)f(p).
 \end{aligned}$$

Therefore X_p^α is a tangent vector to M at p .

Similarly, one can show that $\left(\frac{\partial}{\partial x^i}\right)_p$ is also a tangent vector. This leads to the following definition.

Definition 1.2.12. Let (U, ϕ) be a chart around $p \in M$, and let $u^1, \dots, u^n$ be coordinates on $\mathbb{R}^n$. Then $\left(\frac{\partial}{\partial x^i}\right)_p$ is the tangent vector to M at p given by

$$\left(\frac{\partial}{\partial x^i}\right)_p(f) = \frac{\partial f \circ \phi^{-1}}{\partial u^i}(\phi(p)).$$

This vector depends on the choice of chart (U, ϕ) .

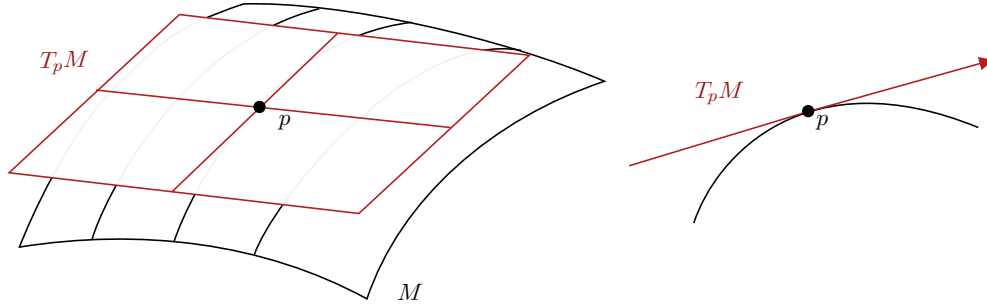
Definition 1.2.13. The i th local coordinate function on M with respect to the chart (U, ϕ) is the function

$$\begin{aligned}
 x^i : U &\longrightarrow \mathbb{R} \\
 p &\longmapsto x^i(p) = u^i(\phi(p)),
 \end{aligned}$$

In particular, note that

$$\begin{aligned}
 \left(\frac{\partial}{\partial x^i}\right)_p(x^j) &= \left(\frac{\partial}{\partial x^i}\right)_p(u^j \circ \phi)(p) \\
 &= \left(\frac{\partial}{\partial u^i}\right)(u^j \circ \phi \circ \phi^{-1})(\phi(p)) \\
 &= \left(\frac{\partial u^j}{\partial u^i}\right)(\phi(p)) \\
 &= \delta_j^i.
 \end{aligned}$$

Definition 1.2.14. The tangent space to M at p , denoted by $T_p M$, is the set of all tangent vectors to M at p .



Theorem 1.2.15. $T_p M$ is a vector space.

Proof. For $X_p, Y_p \in T_p M$, and $r \in \mathbb{R}$, define $(X_p + Y_p)$ and (rX_p) by $(X_p + Y_p)(f) = X_p(f) + Y_p(f)$ and $(rX_p)(f) = r \cdot X_p(f)$.

$$\begin{aligned}
 (X_p + Y_p)(f + g) &= X_p(f + g) + Y_p(f + g) \\
 &= X_p(f) + X_p(g) + Y_p(f) + Y_p(g) \\
 &= X_p(f) + Y_p(f) + X_p(g) + Y_p(g) \\
 &= (X_p + Y_p)(f) + (X_p + Y_p)(g),
 \end{aligned}$$

$$\begin{aligned}
 (X_p + Y_p)(rf) &= X_p(rf) + Y_p(rf) \\
 &= rX_p(f) + rY_p(f) \\
 &= r(X_p(f) + Y_p(f)) \\
 &= r(X_p + Y_p)(f),
 \end{aligned}$$

$$\begin{aligned}
 (X_p + Y_p)(fg) &= X_p(fg) + Y_p(fg) \\
 &= f(p)X_p(g) + g(p)X_p(f) + f(p)Y_p(g) + g(p)Y_p(f) \\
 &= f(p)X_p(g) + f(p)Y_p(g) + g(p)X_p(f) + g(p)Y_p(f) \\
 &= f(p)(X_p + Y_p)(g) + g(p)(X_p + Y_p)(f).
 \end{aligned}$$

On the other hand,

$$\begin{aligned}(rX_p)(f + g) &= rX_p(f + g) \\ &= rX_p(f) + rX_p(g),\end{aligned}$$

$$\begin{aligned}(rX_p)(kf) &= rX_p(kf) \\ &= rkX_p(f),\end{aligned}$$

$$\begin{aligned}(rX_p)(fg) &= rX_p(fg) \\ &= f(p)rX_p(g) + g(p)rX_p(f).\end{aligned}$$

Hence $(X_p + Y_p)$ and (rX_p) belong to T_pM and clearly satisfy the vector space axioms. The zero element $\mathbf{0}_p$ is the zero operator, satisfying $\mathbf{0}_p(f) = 0 \in \mathbb{R}, \forall f \in \mathfrak{T}(M)$. $\square$

Lemma 1.2.16. *If $F : \mathbb{R}^n \rightarrow \mathbb{R}$ is of class C^k and $(a^1, \dots, a^n) \in \mathbb{R}^n$, then there exist functions h_{ij} of class C^{k-2} such that*

$$\begin{aligned}F(u^1, \dots, u^n) &= F(a^1, \dots, a^n) + \sum \frac{\partial F}{\partial u^i}(a^1, \dots, a^n)(u^i - a^i) \\ &\quad + \sum h_{ij}(u^1, \dots, u^n)(u^i - a^i)(u^j - a^j).\end{aligned}$$

This follows from Taylor's theorem with remainder.

Theorem 1.2.17. *Let (U, ϕ) be a chart with $p \in U$. Then $\mathcal{B} = \left\{ \left(\frac{\partial}{\partial x^i} \right)_p \right\}_{i=1}^n$ is a basis of T_pM ,*

$$\text{and } X_p = \sum_{i=1}^n X_p(x^i) \left(\frac{\partial}{\partial x^i} \right)_p.$$

Proof. To show that $\mathcal{B}$ is linearly independent, suppose that $c^i \left(\frac{\partial}{\partial x^i} \right)_p = \mathbf{0}_p$. Then $c^i \left(\frac{\partial}{\partial x^i} \right)_p(f) = \mathbf{0}_p(f) = 0 \in \mathbb{R}, \forall f \in \mathfrak{T}(M)$. In particular, $c^i \left(\frac{\partial}{\partial x^i} \right)_p(x^j) = 0, j = 1, \dots, n$.

By Definition 1.2.13, we have $c^i \delta_i^j = 0$, and hence $c^j = 0$ for $j = 1, \dots, n$.

We now show that $\mathcal{B}$ spans T_pM . Let $f \in \mathfrak{T}(M)$, $a = \phi(p)$ with $a \in \mathbb{R}^n$, and $F = f \circ \phi^{-1} : \phi(U) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. For $m \in U$, we have $\phi(m) = (x^1(m), \dots, x^n(m))$ and $\phi(p) = (a^1, \dots, a^n)$, where $x^i(p) = a^i$. By Lemma 1.2.16, there exist $h_{ij} : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$F(\phi(m)) = F(\phi(p)) + \sum_{i=1}^n \frac{\partial F}{\partial u^i}(\phi(p))(x^i(m) - a^i) + \sum_{i=1}^n \sum_{j=1}^n h_{ij}(\phi(m))(x^i(m) - a^i)(x^j(m) - a^j),$$

Thus

$$f(m) = f(p) + \sum_{i=1}^n \frac{\partial f \circ \phi^{-1}}{\partial u^i}(\phi(p))(x^i(m) - a^i) + \sum_{i=1}^n \sum_{j=1}^n h_{ij}(\phi(m))(x^i(m) - a^i)(x^j(m) - a^j),$$

whence

$$f = f(p) + \sum_{i=1}^n \left(\frac{\partial}{\partial x^i} \right)_p (f)(x^i - a^i) + \sum_{i=1}^n \sum_{j=1}^n h_{ij} \circ \phi(x^i - a^i)(x^j - a^j).$$

Therefore,

$$X_p(f) = X_p(f(p)) + \sum_{i=1}^n \left(\frac{\partial}{\partial x^i} \right)_p (f) (X_p(x^i) - X_p(a^i)) + \sum_{i=1}^n \sum_{j=1}^n X_p(h_{ij} \circ \phi(x^i - a^i)(x^j - a^j)),$$

Since $f(p)$ is constant, Proposition 1.2.10 gives $X_p(f(p)) = 0$. Thus

$$X_p(f) = \sum_{i=1}^n \left(\frac{\partial}{\partial x^i} \right)_p (f) X_p(x^i) + \sum_{i=1}^n \sum_{j=1}^n X_p(h_{ij} \circ \phi(x^i - a^i)(x^j - a^j))$$

Since $x^i(p) = a^i$, both $(h_{ij} \circ \phi)(x^i - a^i)$ and $(x^j - a^j)$ vanish at $p = \phi^{-1}(a^1, \dots, a^n)$. We conclude that $X_p(f) = \sum_{i=1}^n \left(\frac{\partial}{\partial x^i} \right)_p (f) X_p(x^i)$, $\forall f \in \mathfrak{T}(M)$, that is,

$$X_p = \sum_{i=1}^n X_p(x^i) \left(\frac{\partial}{\partial x^i} \right)_p$$

Hence $\mathcal{B}$ spans $T_p M$. □

Since $T_p M$ is a vector space, we can define its dual.

Definition 1.2.18. *The cotangent space to M at p is $T_p^* M = (T_p M)^*$.*

Let $\{x^1, \dots, x^n\}$ be coordinates on a manifold M . If $\left\{ \left(\frac{\partial}{\partial x^i} \right)_p \right\}$ is a basis of $T_p M$, we denote its dual basis by $\{(dx^i)_p\}$. Thus

$$(dx^i)_p \left(\left(\frac{\partial}{\partial x^j} \right)_p \right) = \delta_j^i.$$

1.3 Vector Bundles

To define concepts such as vector fields, Riemannian metrics, and pseudo-Riemannian metrics formally, we first introduce vector bundles, fibres, the tangent bundle, sections, and tensor bundles.

Definition 1.3.1. *A vector bundle is a triple (E, M, π) , where E and M are manifolds and π is a surjective map $\pi : E \rightarrow M$ such that:*

1. $E_p = \pi^{-1}(p)$ is a vector space for every $p \in M$.

The space E_p is called the fibre of the bundle over p .

2. *For every $p \in M$, there is a neighbourhood U of p and a diffeomorphism $\phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$, called a local trivialization of the bundle, such that $\phi|_{\pi^{-1}(p)} : \pi^{-1}(p) \rightarrow \{p\} \times \mathbb{R}^k$ is an isomorphism, where π_1 is the projection onto the first factor.*

That is, $\pi = \pi_1 \circ \phi$, so the following diagram commutes:

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\phi} & U \times \mathbb{R}^k \\ & \searrow \pi & \downarrow \pi_1 \\ & & U \end{array}$$

We generally say that E is a vector bundle with base M and projection π .

In a vector bundle, the inverse image of a point under π is a vector space, locally modelled on $\mathbb{R}^k$.

Example 1.3.2. The tangent bundle is defined by $TM := \{(p, V_p) \mid p \in M, V_p \in T_p M\}$. Note that (TM, M, π) is a vector bundle, where π is the projection

$$\begin{aligned} \pi : TM &\longrightarrow M \\ (p, V_p) &\mapsto \pi(p, V_p) = p. \end{aligned}$$

Indeed, $\pi^{-1}(p) = \{p\} \times T_p M$: for $(x, y) \in \pi^{-1}(p)$, we have $\pi(x, y) = p$, hence $x = p$ and $y \in T_p M$, so $(x, y) \in \{p\} \times T_p M$.

Let $(x^1, \dots, x^n)$ be coordinates on M . A point $p \in U$ has coordinates $(x^1(p), \dots, x^n(p))$ in M , and $V_p \in T_p M$ has coordinates $x^1, \dots, x^n$ such that $V_p = v^i(p) \frac{\partial}{\partial x^i} \Big|_p$.

The map

$$\begin{aligned} \phi : \pi^{-1}(U) \subset TM &\longrightarrow U \times \mathbb{R}^n \\ (p, V_p) &\mapsto \phi \left(p, v^1(p) \frac{\partial}{\partial x^1} + \dots + v^n(p) \frac{\partial}{\partial x^n} \right) = (p, v^1(p), \dots, v^n(p)). \end{aligned}$$

is called a local trivialization of the bundle. Here $\phi|_{\pi^{-1}(p)}$ is a vector space isomorphism, since for $V = v^i \frac{\partial}{\partial x^i}$ and $W = w^j \frac{\partial}{\partial x^j}$ we have

$$\begin{aligned} \phi(p, V) + \phi(p, W) &= (p, v^1 + w^1, \dots, v^n + w^n) \\ &= \phi(p, V + W). \end{aligned}$$

Similarly, for the cotangent bundle defined by $T^*M = (TM)^* = \{(p, \omega_p) \mid p \in M, \omega_p \in T_p^* M\}$, one can show that (T^*M, M, π) is a vector bundle, with π the projection onto p .

Example 1.3.3. $(\mathbb{R}^k \times \mathbb{R}^n, \mathbb{R}^k, \pi_2)$ is a vector bundle with projection

$$\begin{aligned} \pi_2 : \mathbb{R}^k \times \mathbb{R}^n &\longrightarrow \mathbb{R}^n \\ (x, y) &\mapsto \pi_2(x, y) = y \end{aligned}$$

The fibre is $\pi^{-1}(y_0) = \mathbb{R}^k \times \{y_0\}$, $\phi : \pi^{-1}U \longrightarrow \mathbb{R}^k \times U$ is the identity, and $\phi|_{\pi^{-1}(p)}$ is a vector space isomorphism.

For details of the following lemma, see [7].

Lemma 1.3.4. *Let M be a manifold, E a set, and $\pi : E \rightarrow M$ a surjective map. Suppose that there is an open cover $\{U_\alpha\}_{\alpha \in I}$ of M and bijective maps $\phi_\alpha : \pi^{-1}(U_\alpha) \subset E \rightarrow U_\alpha \times \mathbb{R}^k$ such that*

1. $\pi_1 \circ \phi_\alpha = \pi$
2. *If U_α and U_β are nonempty, the map $\phi_\alpha \circ \phi_\beta^{-1} : U_\alpha \cap U_\beta \times \mathbb{R}^k \rightarrow U_\alpha \cap U_\beta \times \mathbb{R}^k$ has the form $\phi_\alpha \circ \phi_\beta^{-1}(p, v) = (p, \mathcal{T}(p)(v))$, where $\mathcal{T} : U_\alpha \cap U_\beta \rightarrow GL(k, \mathbb{R})$,*

Then E has a unique vector bundle structure over M .

In other words, $\mathcal{T}(p) : \mathbb{R}^k \rightarrow \mathbb{R}^k$ is an invertible linear transformation for every $p \in U_\alpha \cap U_\beta$. The maps ϕ_α are called local trivializations, and $\mathcal{T}$ is a transition function.

We can apply this construction to the tangent bundle $TM = \bigcup_{p \in M} T_p M$, with $p \in M$.

$$\begin{aligned} \pi : TM &\longrightarrow M \\ (p, V_p) &\mapsto p \end{aligned}$$

Take a chart $(U, (x^i))$ on M and express $V_p \in T_p M$ in coordinates as $V_p = v^i(p) \frac{\partial}{\partial x^i} \Big|_p$. This representation always gives a bijection

$$\begin{aligned} \phi : \pi^{-1}(U) &\longrightarrow U \times \mathbb{R}^k \\ V_p &\mapsto (p, v^1, v^2, \dots, v^k). \end{aligned}$$

If $(U, (x^i))$ and $(\tilde{U}, (\tilde{x}^i))$ are charts with $U \cap \tilde{U} \neq \emptyset$, then locally we have

$$\frac{\partial}{\partial x^i} = \frac{\partial \tilde{x}^j}{\partial x^i} \frac{\partial}{\partial \tilde{x}^j},$$

Thus, for V_p , we obtain

$$\begin{aligned} V_p &= \tilde{v}^j \frac{\partial}{\partial \tilde{x}^j} \\ &= v_i \frac{\partial}{\partial x^i} \\ &= v^i \frac{\partial \tilde{x}^j}{\partial x^i} \frac{\partial}{\partial \tilde{x}^j}, \end{aligned}$$

so

$$\tilde{v}^j = v^i \frac{\partial \tilde{x}^j}{\partial x^i}.$$

Hence the new coordinates $\tilde{v}^j$ are related to the previous ones by the transition function $\mathcal{T}(p) = \left(\frac{\partial \tilde{x}^j}{\partial x^i} \Big|_p \right)$.

Definition 1.3.5. *If $\pi : E \rightarrow M$ is a vector bundle, a map $S : M \rightarrow E$ is a section of E if $\pi \circ S(p) = p$, that is, $S(p) \in E_p \cong \pi^{-1}(p)$.*

Example 1.3.6. *Let $\pi : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the canonical projection. In this case, sections have the form $S(p) = (p, f(p))$, where f is a differentiable map.*

Remark: These constructions are used to define tensor bundles and tensor fields on M . For each $p \in M$, we can form $T_l^k(T_p M)$.

The bundle of tensors of type $\binom{k}{l}$ on M is defined by $T_l^k(M) = \bigcup_{p \in M} T_l^k(T_p M)$. As we shall see, a tensor field is a section of $T_l^k(M)$. To show that a tensor bundle is a vector bundle, consider the projection

$$\begin{aligned} \pi : T_l^k(M) &\longrightarrow M \\ F &\longmapsto \pi(F) = p, \quad \text{for all } F \in T_l^k(T_p M) \text{ and } p \in M, \end{aligned}$$

and define

$$\begin{aligned} \phi : \pi^{-1}(U) \subset T_l^k(M) &\longrightarrow U \times \mathbb{R}^{n(k+l)} \\ F &\longmapsto \phi(F) = (p, F_{j_1 \dots j_l}^{i_1 \dots i_k}), \quad \text{for all } F \in T_l^k(T_p M) \text{ and } p \in M, \end{aligned}$$

Thus, if (x^i) is a coordinate system on $U \subset M$, then $\left\{ \frac{\partial}{\partial x^i} \Big|_p \right\}$ is a basis of $T_p M$ and $\{dx^i|_p\}$ is a basis of $T_p^* M$. A tensor of type $\binom{k}{l}$ can be written as

$$F(p) = F_{j_1 \dots j_l}^{i_1 \dots i_k}(p) \frac{\partial}{\partial x^{i_1}} \Big|_p \otimes \dots \otimes \frac{\partial}{\partial x^{i_k}} \Big|_p \otimes dx^{j_1} \Big|_p \otimes \dots \otimes dx^{j_l} \Big|_p.$$

1.4 Vector Fields and Connections

Whenever M denotes a manifold, we mean an n -dimensional differentiable manifold.

Having defined the tangent bundle, we can now define vector fields.

Definition 1.4.1. Let M be a manifold and $\pi : TM \longrightarrow M$ a projection such that (TM, M, π) is a vector bundle. A vector field on M is a section of the tangent bundle TM ; that is, a differentiable map $S : M \longrightarrow TM$ given by $S(p) = (p, X_p)$.

Informally, we can describe a vector field as follows:

Definition 1.4.2. If X is a field of vectors and $f \in \mathfrak{T}(M)$, we can define the function Xf by

$$\begin{aligned} Xf : M &\longrightarrow \mathbb{R} \\ p &\longmapsto X(f)(p) = X_p(f). \end{aligned}$$

If $Xf \in \mathfrak{T}(M)$ for every $f \in \mathfrak{T}(M)$, we say that X is a vector field.

Remark: If (U, ϕ) is a chart with $p \in U \subset M$, we can construct a basis $\mathcal{B} = \left\{ \left(\frac{\partial}{\partial x^i} \right)_p \right\}_{i=1}^n$ of $T_p M$, in which

$$X_p = x^i(p) \left(\frac{\partial}{\partial x^i} \right)_p.$$

Thus $X = x^i \left(\frac{\partial}{\partial x^i} \right)$ only makes sense on an open subset $U \subset M$ for which (U, ϕ) is a chart. This gives a vector field defined on U .

Definition 1.4.3. We denote the set of all vector fields on M by $\mathfrak{X}(M)$. Thus

$$\mathfrak{X}(M) = \{ S : M \longrightarrow TM \mid S \text{ is differentiable and } \pi \circ S = id_M \}$$

Remark: Let $f \in \mathfrak{T}(M)$ and $X \in \mathfrak{X}(M)$. We can define the vector field

$$\begin{aligned} fX : M &\longrightarrow TM, \\ p &\longmapsto f(p)X_p. \end{aligned}$$

whereas

$$\begin{aligned} Xf : M &\longrightarrow \mathbb{R} \\ (p) &\longmapsto X_p f, \end{aligned}$$

where $X_p : \mathfrak{T}(M) \longrightarrow \mathbb{R}$.

If $X, Y \in \mathfrak{X}(M)$, $r \in \mathbb{R}$, and $f \in \mathfrak{T}(M)$, then

$$\begin{aligned} (X + Y)_p &= X_p + Y_p, \\ (rX)_p &= rX_p, \\ (fX)_p &= f(p)X_p, \end{aligned}$$

Thus $fX \in \mathfrak{X}(M)$, whereas $Xf \in \mathfrak{T}(M)$. With pointwise addition and multiplication by real constants, $\mathfrak{X}(M)$ is a real vector space. Multiplication by smooth functions also makes it a module over $\mathfrak{T}(M)$. Its zero vector field O is the section mapping every $p \in M$ to the zero tangent vector in $T_p M$; that is, $O_p(f) = 0$, $\forall p \in U$, $f \in \mathfrak{T}(M)$.

Definition 1.4.4. Let $X, Y \in \mathfrak{X}(M)$. The Lie bracket of X and Y is defined by

$$[X, Y]_p f = X_p(Yf) - Y_p(Xf), \quad f \in \mathfrak{T}(M), \quad p \in M.$$

Lemma 1.4.5. $[X, Y]$ is a vector field on M .

Example 1.4.6. Let $M = \mathbb{R}^2$ and consider $X = x^1 x^2 \left(\frac{\partial}{\partial x^1} \right)$ and $Y = x^2 \left(\frac{\partial}{\partial x^2} \right)$. We compute $[X, Y]$. If $f \in \mathfrak{T}(M)$, then

$$\begin{aligned} [X, Y]f &= x^1 x^2 \frac{\partial}{\partial x^1} \left(x^2 \frac{\partial f}{\partial x^2} \right) - x^2 \frac{\partial}{\partial x^2} \left(x^1 x^2 \frac{\partial f}{\partial x^1} \right) \\ &= x^1 (x^2)^2 \frac{\partial^2 f}{\partial x^1 \partial x^2} - x^2 x^1 \frac{\partial f}{\partial x^1} - x^1 (x^2)^2 \frac{\partial^2 f}{\partial x^1 \partial x^2} \\ &= -x^2 x^1 \frac{\partial f}{\partial x^1}, \end{aligned}$$

Therefore $[X, Y] = -x^2 x^1 \frac{\partial}{\partial x^1}$.

The following properties follow directly from the definition:

Lemma 1.4.7. *Let $X, Y, Z \in \mathfrak{X}(M)$ and $r \in \mathbb{R}$. Then*

1. $[X, Y] = -[Y, X]$ and $[rX, Y] = r[X, Y]$,
2. $[X + Y, Z] = [X, Z] + [Y, Z]$ and $[Z, X + Y] = [Z, X] + [Z, Y]$,
3. $[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$.

1.4.1 The Differential and Submanifolds

In this section, we introduce the differential of a map between manifolds and apply it to the notion of a submanifold. Let M and N be manifolds. For every $X_p \in T_p M$, we wish to define an element $(\Phi_*)_p(X_p)$ of $T_{\Phi(p)} N$ such that $(\Phi_*)_p(X_p)$ is a linear transformation for each $p \in M$.

Definition 1.4.8. *If $\Phi : M \rightarrow N$ is differentiable, the differential of Φ at p is defined by*

$$\begin{aligned} (\Phi_*)_p : T_p M &\longrightarrow T_{\Phi(p)} N \\ X_p &\mapsto (\Phi_*)_p X_p : \mathfrak{T}(N) \longrightarrow \mathbb{R} \\ &\quad f \mapsto (\Phi_*)_p(X_p)f \end{aligned}$$

where $(\Phi_*)_p(X_p)f := X_p(f \circ \Phi) \in \mathbb{R}$, for $X_p \in T_p M$ and $f \in \mathfrak{T}(N)$.

Lemma 1.4.9. *$(\Phi_*)_p$ is well defined. That is, if $\Phi : M \rightarrow N$ and $X_p \in T_p M$, then $(\Phi_*)_p(X_p) \in T_{\Phi(p)} N$.*

Proof. We must prove that $(\Phi_*)_p X_p \in T_{\Phi(p)} N$; namely, for every $f, g \in \mathfrak{T}(N)$ and $r \in \mathbb{R}$,

1. $(\Phi_*)_p(X_p)(f + g) = (\Phi_*)_p(X_p)f + (\Phi_*)_p(X_p)g$,
2. $(\Phi_*)_p(X_p)(rf) = r(\Phi_*)_p(X_p)f$,
3. $(\Phi_*)_p(X_p)(fg) = f(\Phi(p))(\Phi_*)_p(X_p)(g) + g(\Phi(p))(\Phi_*)_p(X_p)(f)$.

First,

$$\begin{aligned} (\Phi_*)_p(X_p)(f + g) &= X_p((f + g) \circ \Phi) \\ &= X_p(f \circ \Phi) + X_p(g \circ \Phi) \\ &= (\Phi_*)_p(X_p)(f) + (\Phi_*)_p(X_p)(g), \end{aligned}$$

Property 2 also holds, since

$$\begin{aligned} (\Phi_*)_p(X_p)(rf) &= X_p(rf \circ \Phi) \\ &= rX_p(f \circ \Phi) \\ &= r(\Phi_*)_p(X_p)f, \end{aligned}$$

For property 3, note that $((fg) \circ \Phi)(q) = (fg)(\Phi(q)) = f(\Phi(q))g(\Phi(q))$ for every $q \in M$, so

$$(fg) \circ \Phi = (f \circ \Phi)(g \circ \Phi).$$

Therefore

$$\begin{aligned} (\Phi_*)_p(X_p)(fg) &= X_p((fg) \circ \Phi) \\ &= X_p((f \circ \Phi)(g \circ \Phi)) \\ &= (f \circ \Phi)(p)X_p(g \circ \Phi) + (g \circ \Phi)(p)X_p(f \circ \Phi) \\ &= f(\Phi(p))\Phi_*(X_p)(g) + g(\Phi(p))\Phi_*(X_p)(f). \end{aligned}$$

□

Lemma 1.4.10. *Let $\Phi : M \rightarrow N$ and $p \in M$. Then $(\Phi_*)_p : T_p M \rightarrow T_{\Phi(p)} N$ is a linear transformation.*

Proof. Let $r \in \mathbb{R}$ and $X_p, Y_p \in T_p M$. We wish to show that

$$(\Phi_*)_p(rX_p + Y_p) = r(\Phi_*)_p X_p + (\Phi_*)_p Y_p.$$

Indeed, let $f \in \mathfrak{F}(M)$ be arbitrary.

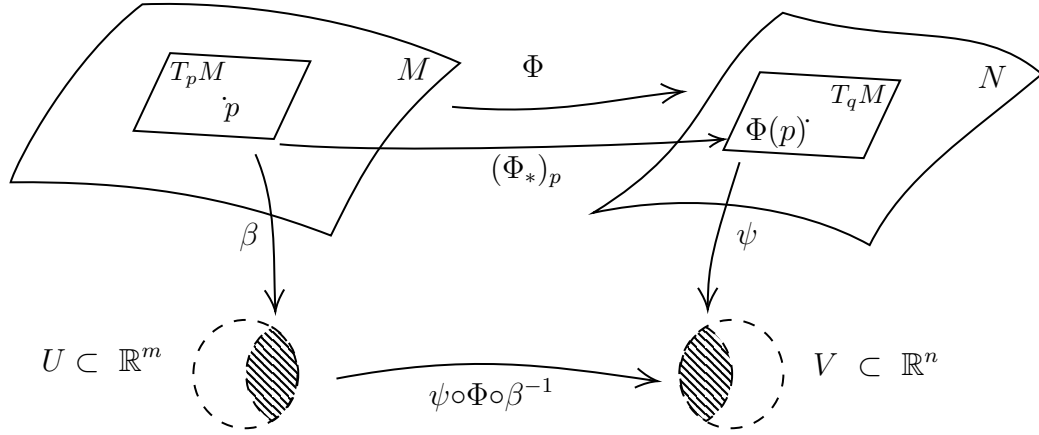
$$\begin{aligned} (\Phi_*)_p(rX_p + Y_p)(f) &= (rX_p + Y_p)(f \circ \Phi) \\ &= rX_p(f \circ \Phi) + Y_p(f \circ \Phi) \\ &= r(\Phi_*)_p(X_p)f + (\Phi_*)_p(Y_p)f. \end{aligned}$$

□

Theorem 1.4.11. *Let $\Phi : M \rightarrow N$ be differentiable, (U, β) a chart with $p \in U$, and (V, ψ) a chart with $q = \Phi(p) \in V$ such that $\Phi(U) \subset V$. If $\left\{\left(\frac{\partial}{\partial x^i}\right)_p\right\}_{i=1}^m$ is the basis of $T_p M$ associated with (U, β) and $\left\{\left(\frac{\partial}{\partial y^j}\right)_q\right\}_{j=1}^n$ is the basis of $T_{\Phi(p)} M = T_q M$ associated with (V, ψ) , then the matrix of $(\Phi_*)_p$ in these bases is the Jacobian matrix of $\psi \circ \Phi \circ \beta^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^n$. Thus*

$$(\Phi_*)_p \left(\left(\frac{\partial}{\partial x^i} \right)_p \right) = \sum_{j=1}^n \frac{\partial(y^j \circ \Phi \circ \beta^{-1})}{\partial u^i}(s) \left(\frac{\partial}{\partial y^j} \right)_q.$$

where $s = \beta(p)$.



Proof. Let $q = \Phi(p)$. If $X_p \in T_p M$ and $Y_q \in T_q N$, then

$$X_p = X_p(x^i) \left(\frac{\partial}{\partial x^i} \right)_p, \quad Y_q = Y_q(y^j) \left(\frac{\partial}{\partial y^j} \right)_q,$$

Since $(\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p \in T_q N$, we have

$$\begin{aligned} (\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p &= (\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p (y^j) \left(\frac{\partial}{\partial y^j} \right)_q \\ &= \left(\frac{\partial}{\partial x^i} \right)_p (y^j \circ \Phi) \left(\frac{\partial}{\partial y^j} \right)_q \\ &= \left(\frac{\partial}{\partial u^i} \right)_p (y^j \circ \Phi \circ \beta^{-1}) \beta(p) \left(\frac{\partial}{\partial y^j} \right)_q. \end{aligned}$$

Thus the matrix of the transformation is

$$\begin{aligned} [(\Phi_*)_p]_{\beta_1}^{\beta_2} &= \left[(\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p \right]_{\beta_1} \cdots \left[(\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p \right]_{\beta_1} \\ &= \left(\frac{\partial}{\partial u^i} (y^j \circ \Phi \circ \beta^{-1})_{\beta(p)} \right)_{m \times n} \\ &= J(\psi \circ \Phi \circ \beta^{-1})_{\beta(p)}. \end{aligned}$$

□

Definition 1.4.12. Let M and N be manifolds of dimensions m and n , respectively. We say that M is a submanifold of N if there is an injective differentiable map $\Phi : M \rightarrow N$ whose differential $(\Phi_*)_p$ is injective for every $p \in M$. If this map is also a homeomorphism onto its image, then Φ is called an embedding of M into N .

Remark: Since $(\Phi_*)_p$ is injective, $(\Phi_*)_p(T_p M) \subset T_q N$ is a subspace of $T_q N$ of dimension $m \leq n$. Thus $T_p M$ may be identified with a subspace of $T_q N$, where $q = \Phi(p)$. If $(\Phi_*)_p$ is injective but Φ is not necessarily injective, then Φ is called an immersion.

Example 1.4.13. Let $M = M_f$ be the hypersurface defined by $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, namely $M_f = \{x \in \mathbb{R}^{n+1} \mid f(x) = 0\}$. We know that M_f is an n -dimensional manifold if $\text{grad}f \neq 0, \forall x \in M_f$.

Without loss of generality, suppose that $\frac{\partial f}{\partial u^{n+1}}(p) \neq 0$. Recall that, locally, $w = (w^1, \dots, w^{n+1}) \in M_f$ if and only if $w^{n+1} = g(w^1, \dots, w^n)$, where g exists by the implicit function theorem. In this case, a chart on M_f is (W, β) , where $\beta^{-1}(u^1, \dots, u^n) = (u^1, \dots, u^n, g(u^1, \dots, u^n))$.

Let $\phi : M_f \rightarrow \mathbb{R}^{n+1}$ be the inclusion and ψ the identity. Then

$$\begin{aligned} \psi \circ \phi \circ \beta^{-1}(u^1, \dots, u^n) &= \psi(\phi(u^1, \dots, u^n, g(u^1, \dots, u^n))) \\ &= \psi(u^1, \dots, u^n, g(u^1, \dots, u^n)) \\ &= (u^1, \dots, u^n, g(u^1, \dots, u^n)), \end{aligned}$$

$$\text{Thus } \frac{\partial}{\partial u^i} (\psi \circ \phi \circ \beta^{-1})(u^1, \dots, u^n) = \begin{cases} 1; & i = j; \quad j = n+1 \\ 0; & i \neq j; \quad j = n+1 \\ \frac{\partial g}{\partial u^i}; & j = n+1 \end{cases} \quad \text{Hence}$$

$$(\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p = \left(\frac{\partial}{\partial y^i} \right)_p + \frac{\partial g}{\partial u^i}(p) \left(\frac{\partial}{\partial y^n} \right)_p.$$

For surfaces of this type, the tangent plane is isomorphic to the Euclidean plane.

Theorem 1.4.14. Let $M = M_f$ be the hypersurface defined by $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$. Then $T_p M$ is isomorphic as a vector space to

$$(\Phi_*)_p(T_p M) = \{X_p \in T_p \mathbb{R}^{n+1} \mid \langle (\text{grad}f)_p, X_p \rangle = 0\}$$

Proof. Clearly, $T_p M \cong (\Phi_*)_p(T_p M)$, since $(\Phi_*)_p$ is injective and its image has the same dimension. We show that $(\Phi_*)_p(T_p M) = \{X_p \in T_p \mathbb{R}^{n+1} \mid \langle (\text{grad}f)_p, X_p \rangle = 0\}$.

Write $E = \{X_p \in T_p \mathbb{R}^{n+1} \mid \langle (\text{grad}f)_p, X_p \rangle = 0\}$. Since $\dim E = \dim(\Phi_*)_p(T_p M)$, it suffices to prove that $(\Phi_*)_p(T_p M) \subset \{X_p \in T_p \mathbb{R}^{n+1} \mid \langle (\text{grad}f)_p, X_p \rangle = 0\}$. For this, it is enough to show that the basis elements of $T_p M$ lie in E . Thus it suffices to verify that

$$(\Phi_*)_p \left(\left(\frac{\partial}{\partial x^i} \right)_p \right) \in \{X_p \in T_p \mathbb{R}^{n+1} \mid \langle (\text{grad}f)_p, X_p \rangle = 0\}, \quad \text{for } i = 1, 2, \dots, n.$$

Assume, without loss of generality, that

$$\frac{\partial f(p)}{\partial u^{n+1}} \neq 0.$$

From the preceding example,

$$\begin{aligned} \left\langle (\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p, (gradf)_p \right\rangle &= \left\langle e_i(p) + \frac{\partial g}{\partial u^i}(p) e_{n+1}(p), \sum_{j=1}^{n+1} \frac{\partial f}{\partial u^j}(p) e_j(p) \right\rangle \\ &= \frac{\partial f}{\partial u^j}(p) + \frac{\partial g}{\partial u^i}(p) \frac{\partial f}{\partial u^{n+1}}(p) \end{aligned}$$

where $g = g(u^1, \dots, u^n)$ is defined by $f(u^1, u^2, \dots, u^n), g(u^1, \dots, u^n) = 0$, by the implicit function theorem. Differentiating gives

$$\frac{\partial f}{\partial u^j} + \frac{\partial g}{\partial u^i} \frac{\partial f}{\partial u^{n+1}} = 0, \quad \text{for } i = 1, 2, \dots, n.$$

Hence

$$\left\langle (\Phi_*)_p \left(\frac{\partial}{\partial x^i} \right)_p, (gradf)_p \right\rangle = 0,$$

so

$$(\Phi_*)_p \left(\left(\frac{\partial}{\partial x^i} \right)_p \right) \in \left\{ X_p \in T_p \mathbb{R}^{n+1} \mid \langle (gradf)_p, X_p \rangle = 0 \right\}, \quad \text{for } i = 1, 2, \dots, n,$$

Therefore

$$(\Phi_*)_p(T_p M) \subset \left\{ X_p \in T_p \mathbb{R}^{n+1} \mid \langle (gradf)_p, X_p \rangle = 0 \right\}.$$

since $T_p M$ is a vector space. □

We now introduce some notions concerning 1-forms on manifolds.

Definition 1.4.15. A differential 0-form of class C^k , with $k \geq 1$, on M is a map $f : M \rightarrow \mathbb{R}$ of class C^k .

Definition 1.4.16. A differential 1-form of class C^k on M is a map $\omega : TM \rightarrow \mathbb{R}$ of class C^k such that, for every $p \in M$, the map $\omega_p : T_p M \rightarrow \mathbb{R}$ given by $\omega_p(v) = \omega(p, v)$ is linear. Thus, for every $p \in M$, we have $\omega_p \in T_p^* M$.

In terms of the cotangent bundle, a 1-form on M is a section of T^*M ; that is, a map $\omega : M \rightarrow T^*M$ such that $\pi \circ \omega = Id_M$.

We have seen that, for a differentiable map $\Phi : M \rightarrow N$, the differential $(\Phi_*)_p : T_p M \rightarrow T_{\Phi(p)} N$ is linear.

Definition 1.4.17. Let $\Phi : M \rightarrow N$ be a smooth map, and let ω be a smooth 1-form on N . The pullback of ω by Φ is the 1-form $\Phi^* \omega$ on M defined by

$$(\Phi^* \omega)_p(v) = \omega_{\Phi(p)}((\Phi_*)_p(v)), \quad p \in M, \quad v \in T_p M.$$

Equivalently,

$$(\Phi^*\omega)_p = \omega_{\Phi(p)} \circ (\Phi_*)_p.$$

This is a linear functional on T_pM , since both $(\Phi_*)_p$ and $\omega_{\Phi(p)}$ are linear.

A smooth function $f : N \rightarrow \mathbb{R}$ is a differential 0-form. Its pullback is the smooth function

$$\Phi^*f = f \circ \Phi.$$

The chain rule relates these two notions of pullback:

$$\Phi^*(df) = d(f \circ \Phi).$$

Indeed, for every $p \in M$ and $v \in T_pM$,

$$(\Phi^*(df))_p(v) = df_{\Phi(p)}((\Phi_*)_p(v)) = d(f \circ \Phi)_p(v).$$

1.5 Riemannian and Pseudo-Riemannian Metrics

Recall that a pseudo-inner product $\langle \cdot, \cdot \rangle$, which need not be positive definite, is a nondegenerate symmetric bilinear form on V . The Gram–Schmidt process gives a basis in which the matrix of $\langle \cdot, \cdot \rangle$ has the form

$$\begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 0 \\ 0 & 0 & \cdots & 0 & -1 \end{pmatrix} \quad \text{with } p \text{ entries equal to } 1 \text{ and } q \text{ entries equal to } -1 \text{ on the diagonal.} \quad (1.4)$$

The numbers of entries 1 and -1 are independent of the basis. We say that this inner product has signature (p, q) .

A nondegenerate symmetric bilinear form $g = \langle \cdot, \cdot \rangle$ induces a linear map

$$\flat_g : V \rightarrow V^*, \quad v \mapsto v^\flat := g(v, \cdot).$$

Thus

$$v^\flat(w) = g(v, w) \quad \text{for every } w \in V.$$

If $v^\flat = 0$, then $g(v, w) = 0$ for every $w \in V$. By nondegeneracy, this implies $v = 0$, so $\flat_g$ is injective. Since V is finite-dimensional and $\dim V = \dim V^*$, it is an isomorphism.

Its inverse is denoted by

$$\sharp_g : V^* \rightarrow V, \quad \omega \mapsto \omega^\sharp,$$

and is characterized by

$$g(\omega^\sharp, w) = \omega(w) \quad \text{for every } w \in V.$$

This identification is independent of the choice of basis, but depends on the chosen bilinear form g . Positive definiteness is not required.

In a basis $\{e_i\}$ with dual basis $\{e^{*i}\}$, write $g_{ij} = g(e_i, e_j)$ and let (g^{ij}) be the inverse matrix. Then

$$v = v^i e_i \implies v^\flat = v_i e^{*i}, \quad v_i = g_{ij} v^j,$$

and

$$\omega = \omega_i e^{*i} \implies \omega^\sharp = g^{ij} \omega_j e_i.$$

These operations are called lowering and raising indices, respectively.

Remark: The inner products of interest in physics have signature $(p, 1)$ on a vector space of dimension $n + 1$ and are called Lorentzian inner products.

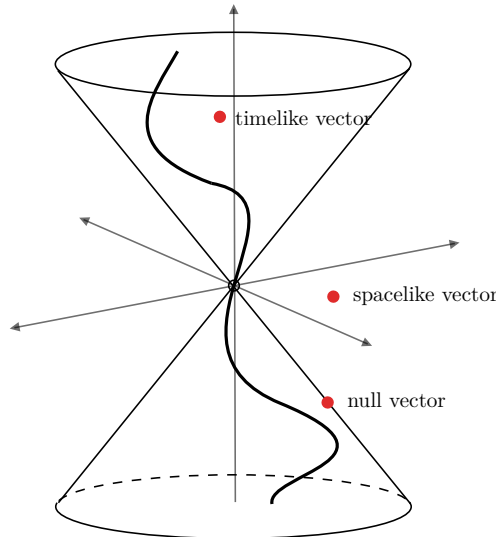
Example 1.5.1. Consider the inner product on $\mathbb{R}^4$ defined by

$$\langle (x, y, z, w), (x^1, y^1, z^1, w^1) \rangle = xx^1 + yy^1 + zz^1 - ww^1.$$

The space $(\mathbb{R}^4, \langle \cdot, \cdot \rangle)$ is called Minkowski space.

Note that $\langle v, v \rangle = x^2 + y^2 + z^2 - w^2$ if $v = (x, y, z, w) \in \mathbb{R}^4$. Thus $\langle v, v \rangle$ can be positive, negative, or zero for more than one vector. This allows us to divide vectors into three classes:

1. *Spacelike:* if $\langle v, v \rangle > 0 \iff x^2 + y^2 + z^2 > w^2$,
2. *Timelike:* if $\langle v, v \rangle < 0 \iff x^2 + y^2 + z^2 < w^2$,
3. *Lightlike or null:* if $\langle v, v \rangle = 0 \iff x^2 + y^2 + z^2 = w^2$.



This is called the causal classification of Minkowski space. According to the postulates of general relativity; light travels on the surface of the cone, while particles travel inside it.

Definition 1.5.2. *If M is a manifold, a metric g on M is a section of the tensor bundle $T_2^0(M) = \bigcup_{p \in M} T_2^0(T_p M)$, where $T_2^0(T_p M) = \{f : T_p M \times T_p M \longrightarrow \mathbb{R} \mid f \text{ is bilinear}\}$. Thus $g(p) \in T_2^0(T_p M)$ is a nondegenerate symmetric bilinear form.*

Hence at each point $p \in M$ we have an inner product on $T_p M$ satisfying the following properties for all $W_p, X_p, Y_p, Z_p \in T_p M$ and $a, b \in \mathbb{R}$:

1. Bilinearity: $g_p(aW_p + bX_p, Y_p) = ag_p(W_p, Y_p) + bg_p(X_p, Y_p)$,
2. Symmetry: $g_p(X_p, Y_p) = g_p(Y_p, X_p)$,
3. Nondegeneracy: $g_p(X_p, Y_p) = 0, \forall Y_p \in T_p M$ if and only if $X_p = 0$.

Remark: A Riemannian metric corresponds to the case in which $g(p)$ has signature $(p, 0)$, meaning that it is positive definite: $g_p(X_p, X_p) \geq 0$, with $g_p(X_p, X_p) = 0$ if and only if $X_p = 0$. If $g(p)$ has signature (p, q) , the metric is called pseudo-Riemannian. A Lorentzian metric is the particular case of a pseudo-Riemannian metric in which $g(p)$ has signature $(n - 1, 1)$ in dimension n .

Definition 1.5.3. *A manifold M equipped with a fixed Riemannian (pseudo-Riemannian) metric is called a Riemannian (pseudo-Riemannian) manifold.*

Remark: Every manifold can be equipped with a Riemannian metric (*Hicks, 1965*).

Definition 1.5.4. *On a Riemannian manifold (M, g) , the norm of $v \in T_p M$ is*

$$\|v\|_g = \sqrt{g_p(v, v)}.$$

For a pseudo-Riemannian metric, the quantity $g_p(v, v)$ may be negative. We may instead define the magnitude

$$|v|_g = \sqrt{|g_p(v, v)|}.$$

In the indefinite case, this is not a norm: a nonzero null vector has zero magnitude.

In particular, for a Lorentzian metric with signature $(-, +, \dots, +)$, a unit timelike vector satisfies

$$g_p(v, v) = -1,$$

whereas a unit spacelike vector satisfies

$$g_p(v, v) = 1.$$

Definition 1.5.5. *Two vectors $V_p, W_p \in T_p M$ are orthogonal if and only if $g_p(V_p, W_p) = 0$.*

Definition 1.5.6. Let (M, g) and (M', g') be pseudo-Riemannian manifolds. A diffeomorphism $\varphi : M \rightarrow M'$ is an isometry if

$$g'_{\varphi(p)}(d\varphi_p(v), d\varphi_p(w)) = g_p(v, w)$$

for every $p \in M$ and every $v, w \in T_p M$, where

$$d\varphi_p : T_p M \rightarrow T_{\varphi(p)} M'$$

is the differential of φ at p . Equivalently, $\varphi^* g' = g$.

The manifolds (M, g) and (M', g') are called isometric if such a diffeomorphism exists.

Remark: Many properties of interest are preserved by isometries and are often called intrinsic properties.

A more sophisticated approach to relativity is to study the space of Lorentzian metrics up to isometry. This is called a moduli space:

$$M/[g] \quad ; \quad [g] = \{g' \mid g \text{ is isometric to } g'\}$$

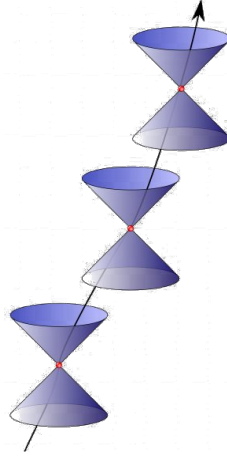
Remark: Consider an open neighbourhood of $p \in M$ with local coordinates $(x^1, x^2, \dots, x^n)$. If $V_p, W_p \in T_p M$, with $V_p = v^i \frac{\partial}{\partial x^i} \Big|_p$ and $W_p = w^j \frac{\partial}{\partial x^j} \Big|_p$, we can write $g_p(V_p, W_p) = g_{ij}(p)v^i w^j$, where the functions $g_{ij}(p) = g_p\left(\left(\frac{\partial}{\partial x^i}\right)_p, \left(\frac{\partial}{\partial x^j}\right)_p\right)$ are the coefficients of the metric g in local coordinates.

If g is a metric on M , then in local coordinates $(x^1, x^2, \dots, x^n)$ on M , g can be written as $g_{ij}dx^i \otimes dx^j$, where $g_{ij} = g_{ji}$ and $\det(g_{ij}) \neq 0$ (nondegeneracy).

Recall that g is a pseudo-Riemannian metric if, for every $p \in M$, $g(p)$ is an inner product on $T_p M$. Thus, at every point $p \in M$ there is a basis of $T_p M$ in which the matrix of g has signature (p, q) .

The metrics of interest in relativity are Riemannian metrics, with $p = n$ and $q = 0$, and Lorentzian metrics, with $p = n - 1$ and $q = 1$ (some texts use the opposite convention). An inner product of signature $(n - 1, 1)$ thus divides vectors into three classes: timelike, $g_p(V, V) < 0$; spacelike, $g_p(V, V) > 0$; and lightlike, $g_p(V, V) = 0$.

General relativity takes $(3, 1)$ as the appropriate signature. On a Lorentzian manifold, there is a cone based at every point of M . Each cone lies in the tangent space, since it consists of all lightlike vectors in $T_p M$. These cones determine the “causal structure” of the manifold M .



A spacetime is a manifold M of dimension 4 equipped with a Lorentzian metric. However, not every manifold M is a suitable model of spacetime. For this purpose, M must be orientable, connected, and paracompact.

The Gram–Schmidt process shows that, near any point, there is a local frame of vector fields $x_1, x_2, \dots, x_n$ such that the matrix of $g(p)$ in the basis $x_1(p), \dots, x_n(p)$ has the form in equation 1.4.

This local frame need not have the form $\left\{ \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right\}$. Such a frame occurs when the manifold is flat, that is, locally isometric to $\mathbb{R}^n$.

In local coordinates $\{x^1, \dots, x^n\}$ on $U \subset M$, the metric g is written as $g_{ij}dx^i \otimes dx^j$, and the inverse of the matrix (g_{ij}) is (g^{ij}) .

A metric allows us to pass from covariant to contravariant tensors and conversely. This is traditionally called raising and lowering indices.

Example 1.5.7. If $X = x^i \frac{\partial}{\partial x^i}$, then $x_i = g_{ij}x^j$.

Example 1.5.8. Let $B \in T_2^1(M)$ have components B_{ik}^l , that is, $B = B_{ik}^l dx^i \otimes \frac{\partial}{\partial x^l} \otimes dx^k$. We can then write $B_{ilk} = g_{lj}B_{ik}^j$, namely $B_{ilk} = g_{l1}B_{ik}^1 + g_{l2}B_{ik}^2 + \dots + g_{ln}B_{ik}^n$.

Remark: Every paracompact manifold admits a Riemannian metric.

Every noncompact manifold admits a Lorentzian metric. For compact manifolds, the existence of a pseudo-Riemannian metric is subject to topological restrictions.

1.5.1 Covariant Differentiation and Curvature

On a Lorentzian manifold M , there is a unique connection $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$, given by $\nabla(X, Y) = \nabla_X Y$, satisfying the following properties for $X, Y, Z \in \mathfrak{X}(M)$, $a, b \in \mathbb{R}$, and $f, h \in \mathfrak{T}(M)$:

1. $\nabla_X Y$ is $\mathbb{R}$ -linear in Y : $\nabla_X(aY + bZ) = a\nabla_X Y + b\nabla_X Z$,
2. $\nabla_X Y$ is M -linear in X : $\nabla_{fX+hY} Z = f\nabla_X Z + h\nabla_Y Z$,

3. $\nabla_X fY = (Xf)Y + f\nabla_X Y$,
4. ∇ is symmetric, or torsion-free: $\nabla_X Y - \nabla_Y X = [X, Y]$,
5. ∇ is compatible with the metric: $Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

Here $fX_p = f(p)X_p \in T_p M$, with $f \in \mathfrak{F}(M)$, whereas $Xf(p) := X_p(f) \in \mathbb{R}$.

We call $\nabla_X Y$ the covariant derivative of Y in the direction of the vector field X , and ∇ a linear or affine connection on M .

This definition generalizes the classical notion of differentiation in $\mathbb{R}^n$.

Definition 1.5.9. Let (U, ϕ) be a chart on M , with $p \in U$, and let $\left\{ \left(\frac{\partial}{\partial x^i} \right)_p \right\}_{i=1}^m$ be a basis of $T_p M$. Then $\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} \in \mathfrak{X}(M)$. Since $T_p M$ is a vector space for every $p \in M$, there are scalar functions $\Gamma_{ijk} : M \rightarrow \mathbb{R}$ such that

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \Gamma_{ij}^k \left(\frac{\partial}{\partial x^k} \right).$$

Example 1.5.10. Let $M = \mathbb{R}^n$ be flat Euclidean space. If $Y \in \mathfrak{X}(\mathbb{R}^n)$, then $Y = y^i e_i$ for some $y^i \in \mathfrak{F}(M)$. Define the flat connection on $\mathbb{R}^n$ by

$$\begin{aligned} \nabla_X Y &= (Xy^i)e_i \\ &= (Xy^i) \frac{\partial}{\partial x^i}. \end{aligned}$$

Furthermore, $\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = 0$ for every i, j , so $\Gamma_{ij}^k = 0$ for $\forall i, j$ and k . In this case, $\nabla_X Y$ is the usual directional derivative of a real-valued function.

In general, many covariant derivatives satisfy properties 1. through 3., but exactly one satisfies properties 1. through 5.. It is called the Levi-Civita connection.

Theorem 1.5.11. (Fundamental Theorem of Riemannian Geometry). Let M be a manifold with metric g . There is a unique symmetric affine connection compatible with the metric.

Proof. Since ∇ is compatible with the metric, for every $X, Y, Z \in \mathfrak{X}(M)$ we have

$$\begin{aligned} Xg(Y, Z) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z), \\ Yg(Z, X) &= g(\nabla_Y Z, X) + g(Z, \nabla_Y X), \\ Zg(X, Y) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y), \end{aligned}$$

Since $g(X, Y) = g(Y, X)$ for all $X, Y \in \mathfrak{X}(M)$ and g is bilinear, we obtain

$$\begin{aligned} Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) &= g(\nabla_X Y + \nabla_Y X, Z) \\ &\quad + g(\nabla_X Z - \nabla_Z X, Y) + g(\nabla_Y Z - \nabla_Z Y, X). \end{aligned}$$

Using $\nabla_Y X - \nabla_X Y + 2\nabla_X Y$, the fact that ∇ is torsion-free, and $[X, Y] = -[Y, X]$, we obtain

$$Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) = -g([X, Y], Z) + 2g(\nabla_X Y, Z) + g([X, Z], Y) + g([Y, Z], X),$$

Hence

$$g(\nabla_X Y, Z) = \frac{1}{2} (Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) + g([X, Y], Z) - g([X, Z], Y) - g([Y, Z], X)).$$

This is the Koszul formula. It defines a connection ∇ by specifying the unique vector $\nabla_X Y$ whose inner product with an arbitrary $Z_p \in T_p M$ is the right-hand side evaluated at p . Since the right-hand side is completely determined by g and arbitrary $X, Y, Z \in \mathfrak{X}(M)$, and since g is nondegenerate, $\nabla_X Y$ is uniquely determined. Thus, if ∇ exists, it is unique.

In a local coordinate frame, write $X = X^i \frac{\partial}{\partial x^i}$ and $Y = Y^j \frac{\partial}{\partial x^j}$. Then $\nabla_X Y$ is uniquely determined:

$$\begin{aligned} \nabla_X Y &= \nabla_{X^i \frac{\partial}{\partial x^i}} \left(Y^j \frac{\partial}{\partial x^j} \right) \\ &= X^i \nabla_{\frac{\partial}{\partial x^i}} \left(Y^j \frac{\partial}{\partial x^j} \right) \\ &= X^i \left(\frac{\partial}{\partial x^i} Y^j \frac{\partial}{\partial x^j} + Y^j \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} \right) \\ &= X^i \left(\frac{\partial}{\partial x^i} Y^j \frac{\partial}{\partial x^j} + Y^j \Gamma_{ij}^k \frac{\partial}{\partial x^k} \right) \\ &= X^i \left(\frac{\partial}{\partial x^i} Y^k + Y^j \Gamma_{ij}^k \right) \frac{\partial}{\partial x^k}, \end{aligned}$$

Thus $\nabla_X Y$ is determined by Γ_{ij}^k .

Now let $g_{ij} = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)$ for each $1 \leq i, j \leq n$. Then

$$\begin{aligned} \frac{\partial}{\partial x^i} g_{jk} &= \frac{\partial}{\partial x^i} g\left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}\right) \\ &= g\left(\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}\right) + g\left(\frac{\partial}{\partial x^i}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}\right) \\ &= g\left(\Gamma_{ij}^l \frac{\partial}{\partial x^l}, \frac{\partial}{\partial x^k}\right) + g\left(\frac{\partial}{\partial x^i}, \Gamma_{ik}^l \frac{\partial}{\partial x^l}\right) \\ &= g_{lk} \Gamma_{ij}^l + g_{il} \Gamma_{ik}^l, \end{aligned}$$

whence

$$\frac{\partial g_{jk}}{\partial x^i} = g_{lk} \Gamma_{ij}^l + g_{jl} \Gamma_{ik}^l, \quad (1.5)$$

$$\frac{\partial g_{ki}}{\partial x^j} = g_{li} \Gamma_{jk}^l + g_{kl} \Gamma_{ji}^l, \quad (1.6)$$

$$\frac{\partial g_{ij}}{\partial x^k} = g_{lj} \Gamma_{ki}^l + g_{il} \Gamma_{kj}^l. \quad (1.7)$$

Adding and subtracting equations (1.5) + (1.6) - (1.7) gives

$$\frac{\partial g_{jk}}{\partial x^i} + \frac{\partial g_{ki}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k} = 2g_{lk}\Gamma_{ij}^l,$$

Since (g^{ij}) is the inverse of the metric matrix (g_{ij}) , we have $g^{sk}g_{kl} = \delta_l^s$ for every $1 \leq s, l \leq n$. Therefore

$$\Gamma_{ij}^k = \frac{1}{2}g^{kl} \left(\frac{\partial g_{li}}{\partial x^j} + \frac{\partial g_{lj}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right) \quad (1.8)$$

□

Lemma 1.5.12. ∇ is torsion-free if and only if, in every coordinate chart,

$$\Gamma_{ij}^k = \Gamma_{ji}^k, \quad \text{for all } 1 \leq i, j, k \leq n.$$

Proof. By definition, ∇ is torsion-free if

$$0 = \nabla_X Y - \nabla_Y X - [X, Y],$$

In a coordinate frame, this becomes

$$\begin{aligned} 0 &= \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial y^j} - \nabla_{\frac{\partial}{\partial y^j}} \frac{\partial}{\partial x^i} - \left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial y^j} \right] \\ &= \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial y^j} - \nabla_{\frac{\partial}{\partial y^j}} \frac{\partial}{\partial x^i} \\ &= (\Gamma_{ij}^k - \Gamma_{ji}^k) \frac{\partial}{\partial x^k}, \end{aligned}$$

Therefore

$$\Gamma_{ij}^k = \Gamma_{ji}^k \quad \text{for all } 1 \leq i, j, k \leq n.$$

□

Lemma 1.5.13. If $X, Y \in \mathfrak{X}(M)$ and $X_p = Y_p$ for some $p \in M$, then $(\nabla_X Z)_p = (\nabla_Y Z)_p$ for every $Z \in \mathfrak{X}(M)$

Proof. Since $X_p = Y_p$, we have $X_p - Y_p = 0$. Set $W_p = X_p - Y_p = 0$. In a coordinate chart around p , we can write $W = f^i \frac{\partial}{\partial x^i}$, with $f^i(p) = 0$ for every i . Then, for any $Z \in \mathfrak{X}(M)$,

$$\begin{aligned} (\nabla_X Z)_p &= (\nabla_{f^i \frac{\partial}{\partial x^i}} Z)_p \\ &= \sum f^i(p) \left(\nabla_{\frac{\partial}{\partial x^i}} Z \right)_p \\ &= 0. \end{aligned}$$

□

Definition 1.5.14. A vector field along a curve $\alpha : I \rightarrow M$ is a map $Z : I \rightarrow TM$ such that $Z_{\alpha(t)} \in T_{\alpha(t)}M$ for every $t \in I$.

Definition 1.5.15. Let $\alpha : I \rightarrow M$ be a curve in M . Its tangent vector field T_α is defined by $(T_\alpha)_{\alpha(t)} = (\alpha_*)_t \left(\frac{d}{dt} \right)$, where t is the natural coordinate on I .

Remark: In local coordinates,

$$T_{\alpha(t)} = \frac{d\alpha^i}{dt}(t) \left(\frac{\partial}{\partial x^i} \right)_{\alpha(t)}.$$

Definition 1.5.16. Let $\alpha : I \rightarrow M$ and $Y \in \mathfrak{X}(M)$. The covariant derivative of Y along α is denoted by $\nabla_T Y$. Given t_0 and $X \in \mathfrak{X}(M)$ such that $X_{\alpha(t_0)} = T_{\alpha(t_0)}$, $\nabla_T Y$ is the vector field along α defined by

$$(\nabla_T Y)_{\alpha(t_0)} = (\nabla_X Y)_{\alpha(t_0)}, \quad \text{for all } x \in \mathfrak{X}(M).$$

Example 1.5.17. Let M be two-dimensional Euclidean space and $\alpha(t) = (\cos t, \sin t)$ for $0 < t < 2\pi$. Let $Y \in \mathfrak{X}(\mathbb{R}^2)$ be given by $Y = ye_1 - xe_2$. With $T_{\alpha(t)} = \frac{d\alpha^i}{dt}(t) \left(\frac{\partial}{\partial x^i} \right)_{\alpha(t)}$, we have $T = -\sin t e_1 + \cos t e_2$. If $X = -ye_1 + xe_2$ satisfies $X_{\alpha(t)} = \frac{d\alpha^i}{dt}(t) \left(\frac{\partial}{\partial x^i} \right)_{\alpha(t)}$, then $T_{\alpha(t)} = X_{\alpha(t)}$, and hence $\nabla_T Y = \nabla_X Y$. Thus

$$\begin{aligned} \nabla_X Y &= \nabla_X (ye_1 - xe_2) \\ &= Xye_1 - Xxe_2 && \text{(by the definition of the connection in Example 1.5.10)} \\ &= (-ye_1 + xe_2)ye_1 - (-ye_1 + xe_2)xe_2 \\ &= \left(-y \frac{\partial y}{\partial x} + x \frac{\partial y}{\partial y}\right)e_1 - \left(-y \frac{\partial x}{\partial x} + x \frac{\partial x}{\partial y}\right)e_2 \\ &= xe_1 + ye_2. \end{aligned}$$

Evaluating along the curve gives $(\nabla_{T_\alpha} Y)_{\alpha(t)} = (\nabla_X Y)_{\alpha(t)}$, therefore $(\nabla_X Y)_{\alpha(t)} = \cos t e_1 + \sin t e_2$.

Lemma 1.5.18. If $f \in \mathfrak{T}(M)$ and $\alpha : I \rightarrow M$ satisfy that $f \circ \alpha$ is constant, then $T_\alpha f = 0$.

Proof. By definition,

$$\begin{aligned} T_\alpha f &= \alpha_* \left(\frac{d}{dt} \right) f \\ &= \left(\frac{d}{dt} \right) (f \circ \alpha) \\ &= 0. \end{aligned}$$

□

Lemma 1.5.19. Let $\alpha : I \rightarrow M$ and $Y, Z \in \mathfrak{X}(M)$ satisfy $Y_{\alpha(t)} = Z_{\alpha(t)}$ for every $t \in I$. Then $\nabla_{T_\alpha} Y = \nabla_{T_\alpha} Z$ along α .

Proof. Set $W = Y - Z$, so $W_{\alpha(t)} = 0$. We must prove that $\nabla_T W = 0$.

In local coordinates near a point on the curve, we can write $W = f^i \left(\frac{\partial}{\partial x^i} \right)$, where $f^i(\alpha(t)) = 0$. By the definition of ∇ , we have

$$\nabla_T W = T_\alpha f^i \frac{\partial}{\partial x^i} + f^i \nabla_T \left(\frac{\partial}{\partial x^i} \right).$$

Since $f^i \circ \alpha$ is identically zero, the preceding lemma gives $T_\alpha f^i = 0$. Also, since f^i vanishes at $\alpha(t)$, we have $f^i \nabla_T \left(\frac{\partial}{\partial x^i} \right) = 0$ when evaluated along $\alpha(t)$. $\square$

If Y is a vector field along a curve α , its derivative Y along α is given by $Y' = \nabla_{T_\alpha} Y$, where T_α is the tangent vector field of α .

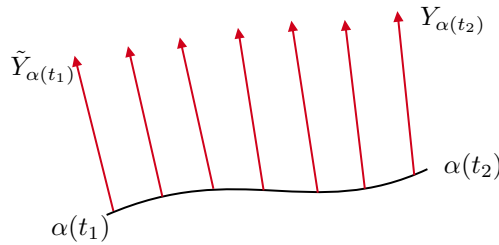
Definition 1.5.20. Let $Y \in \mathfrak{X}(M)$. We say that Y is parallel transported along the curve $\alpha : I \rightarrow M$ if $\nabla_{T_\alpha} Y = 0$.

Covariant differentiation allows us to construct parallel transport along curves.

Definition 1.5.21. The parallel transport map along α is defined by

$$\begin{aligned} P_{\alpha, t_1, t_2} : T_{\alpha(t_1)} M &\longrightarrow T_{\alpha(t_2)} M \\ \bar{Y}_{\alpha(t_1)} &\longmapsto Y_{\alpha(t_2)} \end{aligned}$$

where Y is the unique parallel vector field along α such that $Y_{\alpha(t_2)} = \bar{Y}_{\alpha(t_1)}$.



Definition 1.5.22. A curve α on M is a geodesic with respect to the connection if

$$\nabla_{T_\alpha} T_\alpha = 0.$$

A geodesic is a curve whose tangent vector is parallel transported.

Since $Y' = \nabla_{T_\alpha} Y$, with $Y = y^i \frac{\partial}{\partial x^i}$ and $\alpha = (x^i(t))$, and since $\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \Gamma_{ij}^m \frac{\partial}{\partial x^m}$, we have

$$\begin{aligned} Y' &= \nabla_{\dot{x}^i \frac{\partial}{\partial x^i}} y^j \frac{\partial}{\partial x^j} && \text{with } \dot{x} = \frac{\partial x}{\partial t}, \\ &= \dot{x}^i \nabla_{\frac{\partial}{\partial x^i}} \left(y^j \frac{\partial}{\partial x^j} \right) \\ &= \dot{x}^i \left(\frac{\partial}{\partial x^i} y^j \frac{\partial}{\partial x^j} + y^j \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} \right) \\ &= \dot{x}^i \left(\frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial x^j} + y^j \Gamma_{ij}^m \frac{\partial}{\partial x^m} \right) \\ &= \dot{x}^i \left(\frac{\partial y^j}{\partial x^i} + y^j \Gamma_{ij}^m \right) \frac{\partial}{\partial x^m}. \end{aligned}$$

We conclude that Y is parallel transported along α if and only if

$$\frac{dx^i}{dt} \frac{\partial y^m}{\partial x^i} + y^j \Gamma_{ij}^m \frac{dx^i}{dt} = 0 \quad m = 1, \dots, n,$$

that is, if

$$\frac{dy^m}{dt} + \Gamma_{ij}^m(\alpha(t)) \frac{dx^i}{dt} y^j(t) = 0, \quad m = 1, \dots, n.$$

Given an initial vector

$$v = v^m \left. \frac{\partial}{\partial x^m} \right|_{\alpha(t_1)} \in T_{\alpha(t_1)}M,$$

the initial conditions are

$$y^m(t_1) = v^m, \quad m = 1, \dots, n.$$

This linear system has a unique solution on any interval on which the curve lies in the coordinate chart. The corresponding local solutions agree on overlaps by uniqueness and define the parallel vector field Y along α with $Y(t_1) = v$.

A special case occurs when $\alpha'(t)$ is parallel transported. In this case, $\alpha'(t) = \dot{x}^i \frac{\partial}{\partial x^i}$, and the differential equation for parallel transport becomes

$$\frac{d^2}{dt^2} x^m + \Gamma_{ij}^m \frac{dx^i}{dt} \frac{dx^j}{dt} = 0, \quad \text{with } m=1,2,\dots,n, \quad (1.9)$$

since $\Gamma_{ij}^m(t) = \Gamma_{ji}^m(t)$.

These are the differential equations for geodesics in M . For every $p \in M$ and $v \in T_p M$, the existence and uniqueness theorem for ordinary differential equations gives a unique local geodesic α satisfying

$$\alpha(0) = p, \quad \dot{\alpha}(0) = v.$$

In local coordinates, these initial conditions become

$$x^i(0) = x^i(p), \quad \dot{x}^i(0) = v^i, \quad i = 1, \dots, n.$$

The manifold (M, g) is called geodesically complete if every maximal geodesic is defined for all $t \in \mathbb{R}$. Otherwise, some geodesic cannot be extended to all real parameter values.

Leaving a particular coordinate chart does not by itself imply geodesic incompleteness: the geodesic may continue in another chart.

Definition 1.5.23. Let ∇ be the Levi-Civita connection of (M, g) . The Riemann curvature tensor is defined by

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, \quad X, Y, Z \in \mathfrak{X}(M).$$

This expression is $C^\infty(M)$ -linear in each of X , Y , and Z , and therefore defines a tensor of type $(1, 3)$.

In local coordinates, write $\partial_i = \frac{\partial}{\partial x^i}$. We use the component convention

$$R(\partial_j, \partial_k)\partial_i = R_{ijk}^l \partial_l.$$

Thus R_{ijk}^l are the component functions of R , rather than individual tensors. With this convention, the first lower index corresponds to the vector on which the curvature operator acts, while the second and third lower indices correspond to its two direction arguments.

For $X = X^j \partial_j$, $Y = Y^k \partial_k$, and $Z = Z^i \partial_i$, tensoriality gives

$$R(X, Y)Z = X^j Y^k Z^i R_{ijk}^l \partial_l.$$

Remark. The metric determines a covariant curvature tensor $\mathcal{R}$ of type $(0, 4)$ by

$$\mathcal{R}(X, Y, Z, W) = g(R(X, Y)Z, W).$$

Define its components in the order of these arguments:

$$\mathcal{R}_{ijkl} := \mathcal{R}(\partial_i, \partial_j, \partial_k, \partial_l).$$

Then

$$\mathcal{R} = \mathcal{R}_{ijkl} dx^i \otimes dx^j \otimes dx^k \otimes dx^l.$$

Using our convention for R_{ijk}^l , we obtain

$$\begin{aligned} \mathcal{R}_{ijkl} &= g(R(\partial_i, \partial_j)\partial_k, \partial_l) \\ &= g(R_{kij}^m \partial_m, \partial_l) \\ &= g_{lm} R_{kij}^m. \end{aligned}$$

The permutation of indices is essential: $g_{lm} R_{ijk}^m$ equals $\mathcal{R}_{jkil}$, not $\mathcal{R}_{ijkl}$.

Lemma 1.5.24. *Let (U, ϕ) be a chart, and let $\left\{ \left(\frac{\partial}{\partial x^1} \right)_p, \dots, \left(\frac{\partial}{\partial x^n} \right)_p \right\}$ be a basis of $T_p M$ for $p \in U$.*

Then

$$R\left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}\right) \frac{\partial}{\partial x^i} = \left(\frac{\partial}{\partial x^j} \Gamma_{ik}^l + \Gamma_{ik}^m \Gamma_{jm}^l - \frac{\partial}{\partial x^k} \Gamma_{ij}^l - \Gamma_{ij}^m \Gamma_{km}^l \right) \frac{\partial}{\partial x^l}.$$

Proof. By Definition 1.5.23, in the coordinate basis of $T_p M$ we have

$$R\left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}\right) \frac{\partial}{\partial x^i} = \nabla_{\frac{\partial}{\partial x^j}} \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^i} \right) - \nabla_{\frac{\partial}{\partial x^k}} \left(\nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^i} \right) - \nabla_{\left[\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right]} \frac{\partial}{\partial x^i},$$

where $\left[\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right] f = \frac{\partial}{\partial x^j} \frac{\partial}{\partial x^k} f - \frac{\partial}{\partial x^k} \frac{\partial}{\partial x^j} f$, so $\left[\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right] = 0, \forall f \in \mathfrak{T}(M)$.

Hence

$$\begin{aligned}
R\left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}\right) \frac{\partial}{\partial x^i} &= \nabla_{\frac{\partial}{\partial x^j}} \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^i} \right) - \nabla_{\frac{\partial}{\partial x^k}} \left(\nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^i} \right) \\
&= \nabla_{\frac{\partial}{\partial x^j}} \left(\Gamma_{ki}^l \frac{\partial}{\partial x^l} \right) - \nabla_{\frac{\partial}{\partial x^k}} \left(\Gamma_{ji}^l \frac{\partial}{\partial x^l} \right) \\
&= \frac{\partial}{\partial x^j} \Gamma_{ki}^l \frac{\partial}{\partial x^l} + \Gamma_{ki}^m \left(\nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^m} \right) - \frac{\partial}{\partial x^k} \Gamma_{ji}^l \frac{\partial}{\partial x^l} - \Gamma_{ji}^m \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^m} \right) \\
&= \frac{\partial}{\partial x^j} \Gamma_{ki}^l \frac{\partial}{\partial x^l} + \Gamma_{ki}^m \Gamma_{jm}^l \frac{\partial}{\partial x^l} - \frac{\partial}{\partial x^k} \Gamma_{ji}^l \frac{\partial}{\partial x^l} - \Gamma_{ji}^m \Gamma_{km}^l \frac{\partial}{\partial x^l} \\
&= \left(\frac{\partial}{\partial x^j} \Gamma_{ik}^l + \Gamma_{ik}^m \Gamma_{jm}^l - \frac{\partial}{\partial x^k} \Gamma_{ij}^l - \Gamma_{ij}^m \Gamma_{km}^l \right) \frac{\partial}{\partial x^l}, \quad \text{with } 1 \leq m \leq n.
\end{aligned}$$

□

Example 1.5.25. We compute the curvature of Euclidean three-space in spherical coordinates. On a spherical-coordinate chart, with $r > 0$, $0 < \theta < \pi$, and ϕ restricted to an appropriate open interval, the Euclidean metric is

$$g = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

Thus, in the coordinate order (r, θ, ϕ) ,

$$(g_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

We compute the Christoffel symbols using the equation ; in our notation, we strike out all terms that vanish because of the metric or its inverse. Recall also that $\Gamma_{ij}^k = \Gamma_{ji}^k$.

$$\begin{aligned}
\Gamma_{22}^1 &= \frac{1}{2} \left[g^{11} \left(\cancel{\frac{\partial g_{12}}{\partial \theta}} + \cancel{\frac{\partial g_{12}}{\partial \theta}} - \frac{\partial g_{22}}{\partial r} \right) + \cancel{g^{12}} \left(\frac{\partial g_{22}}{\partial \theta} + \frac{\partial g_{22}}{\partial \theta} - \frac{\partial g_{22}}{\partial \theta} \right) + \cancel{g^{13}} \left(\frac{\partial g_{32}}{\partial \theta} + \frac{\partial g_{32}}{\partial \theta} - \frac{\partial g_{22}}{\partial \phi} \right) \right] \\
&= -\frac{1}{2} \frac{\partial r^2}{\partial r} \\
&= -r,
\end{aligned}$$

$$\begin{aligned}
\Gamma_{33}^1 &= \frac{1}{2} \left[g^{11} \left(\cancel{\frac{\partial g_{13}}{\partial \phi}} + \cancel{\frac{\partial g_{13}}{\partial \phi}} - \frac{\partial g_{33}}{\partial r} \right) + \cancel{g^{12}} \left(\frac{\partial g_{23}}{\partial \phi} + \frac{\partial g_{23}}{\partial \phi} - \frac{\partial g_{33}}{\partial \theta} \right) + \cancel{g^{13}} \left(\frac{\partial g_{33}}{\partial \phi} + \frac{\partial g_{33}}{\partial \phi} - \frac{\partial g_{33}}{\partial \phi} \right) \right] \\
&= -\frac{1}{2} \frac{\partial r^2 \sin^2 \theta}{\partial r} \\
&= -r \sin^2 \theta,
\end{aligned}$$

$$\begin{aligned}
\Gamma_{12}^2 &= \frac{1}{2} \left[g^{21} \left(\frac{\partial g_{11}}{\partial \theta} + \frac{\partial g_{12}}{\partial r} - \frac{\partial g_{12}}{\partial r} \right) + g^{22} \left(\frac{\partial g_{21}}{\partial \theta} + \frac{\partial g_{22}}{\partial r} - \frac{\partial g_{12}}{\partial \theta} \right) + g^{13} \left(\frac{\partial g_{31}}{\partial \theta} + \frac{\partial g_{32}}{\partial r} - \frac{\partial g_{12}}{\partial \phi} \right) \right] \\
&= \frac{1}{2} \frac{1}{r^2} \frac{\partial r^2}{\partial r} \\
&= \frac{1}{r},
\end{aligned}$$

$$\text{Thus } \Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{r}.$$

$$\begin{aligned}
\Gamma_{13}^3 &= \frac{1}{2} \left[g^{31} \left(\frac{\partial g_{11}}{\partial \phi} + \frac{\partial g_{13}}{\partial r} - \frac{\partial g_{13}}{\partial r} \right) + g^{32} \left(\frac{\partial g_{21}}{\partial \phi} + \frac{\partial g_{23}}{\partial r} - \frac{\partial g_{13}}{\partial \theta} \right) + g^{33} \left(\frac{\partial g_{31}}{\partial \phi} + \frac{\partial g_{33}}{\partial r} - \frac{\partial g_{13}}{\partial \phi} \right) \right] \\
&= \frac{1}{2} \frac{1}{r^2 \sin^2 \theta} \frac{\partial r^2 \sin^2 \theta}{\partial r} \\
&= \frac{1}{r},
\end{aligned}$$

$$\text{Thus } \Gamma_{13}^3 = \Gamma_{31}^3 = \frac{1}{r}.$$

$$\begin{aligned}
\Gamma_{33}^2 &= \frac{1}{2} \left[g^{21} \left(\frac{\partial g_{13}}{\partial \phi} + \frac{\partial g_{13}}{\partial \phi} - \frac{\partial g_{33}}{\partial r} \right) + g^{22} \left(\frac{\partial g_{23}}{\partial \phi} + \frac{\partial g_{23}}{\partial \phi} - \frac{\partial g_{33}}{\partial \theta} \right) + g^{23} \left(\frac{\partial g_{33}}{\partial \phi} + \frac{\partial g_{33}}{\partial \phi} - \frac{\partial g_{33}}{\partial \phi} \right) \right] \\
&= -\frac{1}{2} \frac{1}{r^2} \frac{\partial r^2 \sin^2 \theta}{\partial \theta} \\
&= -\sin \theta \cos \theta,
\end{aligned}$$

$$\begin{aligned}
\Gamma_{23}^3 &= \frac{1}{2} \left[g^{31} \left(\frac{\partial g_{12}}{\partial \phi} + \frac{\partial g_{13}}{\partial \theta} - \frac{\partial g_{23}}{\partial r} \right) + g^{32} \left(\frac{\partial g_{22}}{\partial \phi} + \frac{\partial g_{23}}{\partial \theta} - \frac{\partial g_{23}}{\partial \theta} \right) + g^{33} \left(\frac{\partial g_{32}}{\partial \phi} + \frac{\partial g_{33}}{\partial \theta} - \frac{\partial g_{23}}{\partial \phi} \right) \right] \\
&= \frac{1}{2} \frac{1}{r^2 \sin^2 \theta} \frac{\partial r^2 \sin^2 \theta}{\partial \theta} \\
&= \cot \theta,
\end{aligned}$$

$$\text{Thus } \Gamma_{23}^3 = \Gamma_{32}^3 = \cot \theta.$$

The remaining Christoffel symbols vanish.

We can now compute the components of the curvature tensor using Lemma 1.5.24. All curvature components must vanish because this is the Euclidean metric expressed in spherical coordinates: its curvature tensor vanishes in Cartesian coordinates, and the vanishing of a tensor is independent of

coordinates. We check some of them, again striking out the Christoffel symbols that vanish.

$$\begin{aligned}
 R_{221}^1 &= \frac{\partial}{\partial \theta} \cancel{\Gamma_{21}^1} + \cancel{\Gamma_{21}^1} \cancel{\Gamma_{12}^1} + \Gamma_{21}^2 \Gamma_{22}^1 + \cancel{\Gamma_{21}^3} \cancel{\Gamma_{32}^1} - \frac{\partial}{\partial r} \Gamma_{22}^1 - \cancel{\Gamma_{22}^1} \cancel{\Gamma_{11}^1} - \cancel{\Gamma_{22}^2} \cancel{\Gamma_{21}^1} - \cancel{\Gamma_{22}^3} \Gamma_{31}^1 \\
 &= \frac{1}{r}(-r) + \frac{\partial r}{\partial r} \\
 &= 0,
 \end{aligned}$$

$$\begin{aligned}
 R_{111}^2 &= \frac{\partial}{\partial r} \cancel{\Gamma_{11}^2} + \cancel{\Gamma_{11}^1} \cancel{\Gamma_{11}^2} + \cancel{\Gamma_{11}^2} \Gamma_{21}^2 + \cancel{\Gamma_{11}^3} \cancel{\Gamma_{31}^2} - \frac{\partial}{\partial r} \Gamma_{11}^2 - \cancel{\Gamma_{11}^1} \cancel{\Gamma_{11}^2} - \cancel{\Gamma_{11}^2} \Gamma_{21}^2 - \Gamma_{11}^3 \Gamma_{31}^2 \\
 &= 0,
 \end{aligned}$$

$$\begin{aligned}
 R_{121}^3 &= \frac{\partial}{\partial \theta} \cancel{\Gamma_{11}^3} + \cancel{\Gamma_{11}^1} \Gamma_{12}^3 + \cancel{\Gamma_{11}^2} \cancel{\Gamma_{22}^3} + \cancel{\Gamma_{11}^3} \cancel{\Gamma_{32}^3} - \frac{\partial}{\partial r} \Gamma_{12}^3 - \cancel{\Gamma_{12}^1} \cancel{\Gamma_{11}^3} - \Gamma_{12}^2 \Gamma_{21}^3 - \cancel{\Gamma_{12}^3} \cancel{\Gamma_{31}^3} \\
 &= -\frac{\partial}{\partial r} \frac{1}{r} - \frac{1}{r^2} \\
 &= 0,
 \end{aligned}$$

$$\begin{aligned}
 R_{232}^3 &= \frac{\partial}{\partial \phi} \cancel{\Gamma_{22}^3} + \Gamma_{22}^1 \Gamma_{13}^3 + \cancel{\Gamma_{22}^2} \Gamma_{23}^3 + \cancel{\Gamma_{22}^3} \cancel{\Gamma_{33}^3} - \frac{\partial}{\partial \theta} \Gamma_{23}^3 - \cancel{\Gamma_{23}^1} \cancel{\Gamma_{12}^3} - \cancel{\Gamma_{23}^2} \cancel{\Gamma_{22}^3} - \Gamma_{23}^3 \Gamma_{32}^3 \\
 &= -r \frac{1}{r} - \frac{\partial \cot \theta}{\partial \theta} - \cot^2 \theta \\
 &= -1 + (\csc^2 \theta - \cot^2 \theta) \\
 &= 0.
 \end{aligned}$$

These computations illustrate the flatness of Euclidean three-space: $R_{ijk}^l = 0$ for all $i, j, k, l = 1, 2, 3$, despite the presence of nonzero Christoffel symbols.

This must be distinguished from the intrinsic geometry of the sphere $r = R$, whose induced metric is

$$g_{\mathbb{S}_R^2} = R^2 d\theta^2 + R^2 \sin^2 \theta d\phi^2.$$

The sphere has Gaussian curvature $K = 1/R^2$ and is not flat.

Definition 1.5.26. The Ricci tensor $R_{ij} dx^i \otimes dx^j$ is the tensor of type $\binom{0}{2}$ obtained by contracting the curvature tensor:

$$R_{ij} = R_{ikj}^k.$$

Definition 1.5.27. The scalar curvature is denoted by R and is obtained by contracting the Ricci tensor: $g^{ij} R_{ij} = R$.

Remark: Remark: $R_i^i = g^{ij} R_{ilj}^l$ expresses the scalar curvature in terms of the curvature tensor.

Lemma 1.5.28. *The Riemann–Christoffel curvature tensor of type $\binom{1}{3}$ has the following symmetry properties:*

1. $R(X, Y)Z = -R(Y, X)Z$,
2. $g(R(X, Y)Z, W) = -g(R(X, Y)W, Z)$,
3. $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$.

Proof. The first two symmetries follow directly from the definition, so we prove the third.

Let $X, Y, Z \in \mathfrak{X}(M)$ be expressed as coordinate vector fields $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}, Z = \frac{\partial}{\partial x^k}$, so that $[X, Y] = [Y, Z] = [Z, X] = 0$ and hence $\nabla_X Y = \nabla_Y X$. Substituting from Definition 1.5.23 and using the preceding observations gives

$$\begin{aligned} R(X, Y)Z &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z, \\ R(Y, Z)X &= \nabla_Y \nabla_Z X - \nabla_Z \nabla_Y X, \\ R(Z, X)Y &= \nabla_Z \nabla_X Y - \nabla_X \nabla_Z Y, \end{aligned}$$

It follows directly that $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$. □

Remark: In coordinates, these properties become:

1. $R_{ijk}^l = -R_{ikj}^l$,
2. $R_{ijkm} = -R_{jikm}$,
3. $R_{ijk}^l + R_{jki}^l + R_{kij}^l = 0$.

Part 3. of Lemma 1.5.28 is called the first Bianchi identity. The identity satisfied by the covariant derivative of the curvature tensor is called the second Bianchi identity:

Proposition 1.5.29. *Let $X, Y, Z, V \in \mathfrak{X}(M)$. The covariant derivative of the Riemann–Christoffel curvature tensor satisfies*

$$\nabla_X R(Y, Z)V + \nabla_Y R(Z, X)V + \nabla_Z R(X, Y)V = 0.$$

Proof. The covariant derivative of R , applied to the tensor R , has the form

$$\nabla_V R(X, Y)Z = \nabla_V (R(X, Y)Z) - R(\nabla_V X, Y)Z - R(X, \nabla_V Y)Z - R(X, Y)\nabla_V Z,$$

By the definition of $R(X, Y)Z$, we have

$$\begin{aligned} \nabla_X R(Y, Z)V &= \nabla_X (\nabla_Y \nabla_Z V - \nabla_Z \nabla_Y V) - R(\nabla_X Y, Z)V - R(Y, \nabla_X Z)V - (\nabla_Y \nabla_Z - \nabla_Z \nabla_Y)\nabla_X V, \\ \nabla_Y R(Z, X)V &= \nabla_Y (\nabla_Z \nabla_X V - \nabla_X \nabla_Z V) - R(\nabla_Y Z, X)V - R(Z, \nabla_Y X)V - (\nabla_Z \nabla_X - \nabla_X \nabla_Z)\nabla_Y V, \\ \nabla_Z R(X, Y)V &= \nabla_Z (\nabla_X \nabla_Y V - \nabla_Y \nabla_X V) - R(\nabla_Z X, Y)V - R(X, \nabla_Z Y)V - (\nabla_X \nabla_Y - \nabla_Y \nabla_X)\nabla_Z V, \end{aligned}$$

Since $R(\nabla_Y Z, X) = -R(X, \nabla_Y Z)V$, it follows directly that

$$\nabla_X R(Y, Z)V + \nabla_Y R(Z, X)V + \nabla_Z R(X, Y)V = 0.$$

□

Remark: We can write the vector field as $\nabla_{\frac{\partial}{\partial x^m}} R\left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}\right) \frac{\partial}{\partial x^i} := \nabla_m R_{ijk}^l \frac{\partial}{\partial x^l}$. Thus, the Bianchi identity in coordinates is

$$\nabla_m R_{ijk}^l + \nabla_k R_{imj}^l + \nabla_j R_{ikm}^l = 0.$$

This last property is important in determining the Einstein tensor.

Chapter 2

General Relativity

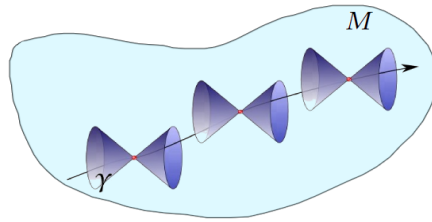
The structure of this chapter follows [10] and [12]. We present the definitions and theorems needed to establish the conditions under which we shall solve the problem.

Let (M, g) be a Lorentzian manifold. We briefly introduce causality and time orientation in order to define a spacetime.

2.1 Introduction to Relativity

A spacetime is a four-dimensional Hausdorff Lorentzian manifold (M, g) . Usually, (M, g) is equipped with additional information modelling electromagnetism, matter, fluids, and other physical phenomena. Thus M describes the entire history of a physical process. In other words, M is interpreted as a space of events rather than merely a set of points, and the metric g encodes information about spacetime and gravity. For example, g allows us to define causality, distance, time, velocity, acceleration, rotation, rigidity, simultaneity, orthogonality, and related notions.

For example, a particle is modelled by a curve $\gamma : \varepsilon \subseteq \mathbb{R} \rightarrow M$ with mass $m \in (0, \infty)$, where ε is an interval.



When studying particles, we require g to equip M with a sense of *future* and *past*. Thus M must be *time-orientable*.

The metric g also defines an arc length that models time along γ , replacing Newtonian time.

For example, consider $\mathbb{R}^2$ with the Lorentzian metric

$$g = du^1 \otimes du^1 - du^2 \otimes du^2,$$

Here $\eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the matrix of the Minkowski metric in the basis $\{e_0, e_1\}$ of $\mathbb{R}^2$. If $V = (x, y)$ is a lightlike vector, then

$$(x, y) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x^2 - y^2 = 0,$$

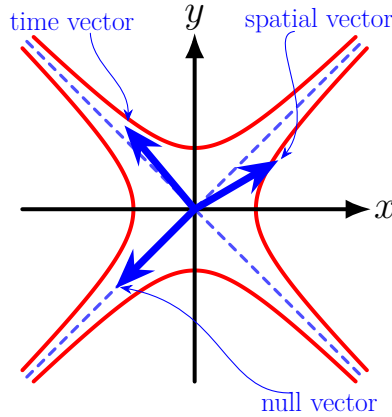
Thus the lightlike vectors lie along the bisectors of the quadrants and divide the space into four regions.

For spacelike vectors, $x^2 - y^2 > 0$. For some $r \in \mathbb{R}^2 - \{0\}$, we have

$x^2 - y^2 = r^2$. Hence $(\frac{x}{r})^2 - (\frac{y}{r})^2 = 1$. Thus spacelike vectors lie inside the lateral hyperbolas.

For timelike vectors, $x^2 - y^2 < 0$, and for some

$r \in \mathbb{R}^2 - \{0\}$, we have $x^2 - y^2 = -r^2$. Hence $(\frac{y}{r})^2 - (\frac{x}{r})^2 = 1$. Thus spacelike vectors lie inside the upper and lower hyperbolas.



2.2 Causal Structure

To examine the physical equivalence of geometric objects, recall that, for each $p \in M$, $g(p)$ is an inner product. It induces an isomorphism between $T_p M$ and its dual $T_p^* M$, sending $V \in T_p M$ to $g(p)(V, \cdot) \in T_p^* M$. In local coordinates, this is $V^i \frac{\partial}{\partial x^i} \mapsto g_{ij}(p) V^i dx^j$. This isomorphism induces isomorphisms between tensor spaces corresponding to raising and lowering indices.

Two tensors are therefore physically equivalent if they are related by such isomorphisms.

Example 2.2.1. Suppose that $X, Y \in \mathfrak{X}(M)$. To find the tensors physically equivalent to $X \otimes Y$, write $X = X^i \frac{\partial}{\partial x^i}$ and $Y = Y^j \frac{\partial}{\partial x^j}$ in a coordinate frame. Then $X \otimes Y = X^i Y^j \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j}$.

$$\begin{aligned} g(X, \cdot) \otimes Y &= g_{ij} X^i dx^j \otimes Y^k \frac{\partial}{\partial x^k} \\ &= g_{ij} X^i Y^k dx^j \otimes \frac{\partial}{\partial x^k}, \end{aligned}$$

On the other hand,

$$\begin{aligned} X \otimes g(Y, \cdot) &= X^i \frac{\partial}{\partial x^i} \otimes g_{jk} Y^j dx^k \\ &= g_{jk} X^i Y^j \frac{\partial}{\partial x^i} \otimes dx^k, \end{aligned}$$

Finally,

$$\begin{aligned} g(X, \cdot) \otimes g(Y, \cdot) &= g_{im} X^i dx^m \otimes g_{jk} Y^j dx^k \\ &= g_{im} g_{jk} X^i Y^j dx^m \otimes dx^k. \end{aligned}$$

We now discuss some aspects of the vector space structure. We refer to the Lorentzian metric as a Lorentzian inner product through the identification $g(v, w) = \langle v, w \rangle$.

Definition 2.2.2. Let (V, g) be a vector space with a Lorentzian inner product, and let W be a subspace of V . We say that:

1. W is spacelike if $g|_W$ is positive definite.
2. W is lightlike if $g|_W$ is positive semidefinite but not positive definite.
3. W is timelike if $g|_W$ is neither spacelike nor lightlike.
4. The causal character of a vector v is the causal character of $\langle v \rangle$.

Example 2.2.3. Let (V, g) be a vector space with a Lorentzian inner product, and let $v \in V$.

1. $\vec{O}$ is spacelike, since $g|_{\langle O \rangle}$ is positive definite, there being no basis vectors.
2. V is spacelike if and only if $g(v, v) > 0, \forall v \in V$.
3. V is lightlike if and only if $g(v, v) = 0$.
4. V is timelike if and only if $g(v, v) < 0$.

Example 2.2.4. A subspace W is spacelike if and only if all its vectors are spacelike. It is lightlike if W contains a lightlike vector but no timelike vector.

Proposition 2.2.5. Let W be a subspace of V . Then W is timelike if and only if $W^\perp$ is spacelike, and W is spacelike if and only if $W^\perp$ is timelike.

W is lightlike if and only if $W \cap W^\perp \neq \{0\}$, which is equivalent to $W^\perp$ being lightlike.

Proof. Suppose that W is timelike. Then there is a timelike vector $w \in W$, which we may assume to be unit. Completing it to an orthonormal basis of V of the form $\{e_1, \dots, e_n, w\}$ gives $W^\perp \subseteq \text{gen}\{e_1, \dots, e_{n-1}\}$. Since $g(w, w) = -1$, the restriction $g|_{\langle e_1, \dots, e_{n-1} \rangle}$ is positive definite. Hence $\text{gen}\{e_1, \dots, e_{n-1}\}$ is spacelike, and so is $W^\perp$.

Now suppose that $W^\perp$ is spacelike, and let $v \in V$ be timelike. Since $V = W \oplus W^\perp$, we can write $v = w + w^\perp$. Thus

$$\begin{aligned} g(v, v) &= g(w, w) + 2g(w, w^\perp) + g(w^\perp, w^\perp) \\ &= g(w, w) + g(w^\perp, w^\perp) \end{aligned}$$

Hence $g(w, w) = g(v, v) - g(w^\perp, w^\perp) < 0$. Thus $w \in W$ is timelike, and consequently W is timelike. $\square$

Theorem 2.2.6. *Two lightlike vectors are orthogonal if and only if they are proportional.*

Proof. $\Rightarrow$ Let $v, w \in V$ be lightlike and $e \in V$ timelike, and suppose that $g(v, w) = 0$. We must have $g(e, v) \neq 0$. Indeed, since e is timelike, the restriction of g to $e^\perp$ is positive definite. If $g(e, v) = 0$, then the nonzero lightlike vector v would belong to $e^\perp$, contradicting $g(v, v) = 0$.

There is therefore $a \in \mathbb{R}$ such that $g(e, w + av) = 0$. Indeed, this requires $g(e, w) + ag(e, v) = 0$, so we take $a = -\frac{g(e, w)}{g(e, v)}$. It follows that $w + av$ is spacelike. Then

$$g(w + av, w + av) = \cancel{g(w, w)} + 2a\cancel{g(w, v)} + a^2\cancel{g(v, v)}.$$

Since $w + av \in e^\perp$ and g is positive definite on $e^\perp$, this implies $w + av = 0$, that is, $w = -av$. Hence v and w are proportional.

$\Leftarrow$ Conversely, if v and w are proportional, then $w = av$ for some $a \in \mathbb{R}$. Thus

$$g(v, w) = ag(v, v)$$

Therefore v and w are orthogonal. $\square$

Definition 2.2.7. *Let (M, g) be a Lorentzian manifold and $\phi : N \rightarrow M$ an immersion. If $\phi_*(T_p N) \subseteq T_{\phi(p)} M$ has the same causal character for every $p \in N$, we call this the causal character of the immersion ϕ and of its image $\phi(N) \subseteq M$.*

This definition applies to curves in M and vector fields on M . In particular, a vector field X defined on $U \subseteq M$ is timelike (respectively lightlike or spacelike) if and only if, for every $p \in U$, X_p is timelike (respectively lightlike or spacelike).

Let ω be a 1-form on $U \subseteq M$, and let X be a vector field physically equivalent to ω . If X has a causal character, we assign it to ω . For example, if X is lightlike, then $g(X, \cdot)$ is a lightlike differential 1-form.

The set $\mathcal{L}_0 \subseteq V$ of all lightlike vectors is called the light cone in V .

Proposition 2.2.8. *The light cone $\mathcal{L}_0$ is a lightlike manifold.*

Proof. Consider the smooth function

$$\begin{aligned} F : V \setminus \{0\} &\longrightarrow \mathbb{R} \\ v &\longmapsto g(v, v) \end{aligned}$$

Then $\mathcal{L}_0 = F^{-1}(0)$. Under the canonical identification $T_v V \simeq V$, its differential is

$$dF_v(w) = 2g(v, w).$$

Since g is nondegenerate and $v \neq 0$, the differential dF_v is nonzero. Thus 0 is a regular value of F , and $\mathcal{L}_0$ is a smooth hypersurface of V .

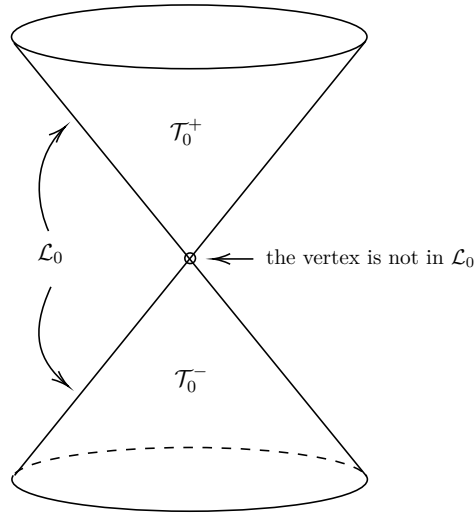
For each $v \in \mathcal{L}_0$, we have

$$T_v \mathcal{L}_0 = \ker dF_v = \{w \in V : g(v, w) = 0\} = v^\perp.$$

Since v is lightlike, the line spanned by v is lightlike. By Proposition 2.2.5, its orthogonal complement $v^\perp$ is also lightlike. Hence $\mathcal{L}_0$ is a lightlike hypersurface. $\square$

Example 2.2.9. Let $N = 3$ and let $\{e_1, e_2, e_3\}$ be an orthonormal basis. Then $v \in \mathcal{L}_0$ if and only if its components satisfy $(v^3)^2 = (v^1)^2 + (v^2)^2 > 0$.

Thus $\mathcal{L}_0$ is represented by a cone with its vertex removed. Timelike vectors satisfy $(v^3)^2 > (v^1)^2 + (v^2)^2$ and are represented by points inside $\mathcal{L}_0$. They form two connected components, $\mathcal{T}_0^+$ and $\mathcal{T}_0^-$.



More generally, in an n -dimensional Lorentzian vector space, the set $\mathcal{T}_0$ of timelike vectors is an open manifold with two connected components, $\mathcal{T}_0^+$ and $\mathcal{T}_0^-$, each diffeomorphic to $\mathbb{R}^n$. For $n \geq 3$, the light cone also has two connected components, $\mathcal{L}_0^+$ and $\mathcal{L}_0^-$, each diffeomorphic to $\mathbb{R} \times S^{n-2}$.

A Lorentzian manifold (M, g) is said to admit a time orientation if one can choose a connected component $\mathcal{T}_0^+$ of $\mathcal{T}_0$ at every $p \in M$ so that, in each chart U of M , there is a differentiable vector field X satisfying $X_p \in \mathcal{T}_0^+$. Such a choice of $\mathcal{T}_0^+$ at each p is called a time orientation of M .

Definition 2.2.10. A spacetime is a triple (M, g, ∇) , where the manifold M is C^∞ , connected, and Hausdorff; ∇ is the Levi-Civita connection; and (M, g) is Lorentzian, orientable, and time-orientable. Thus $\mathcal{T} = \{(p, X_p) \in TM \mid (p, X_p) \text{ is timelike}\}$ has two connected components.

We now have the geometric structures needed to formulate Einstein's general theory. We have seen that the metric-compatible connection determines curvature through the Riemann tensor. We have not yet determined the metric corresponding to a particular physical system.

Recall that the Riemann curvature tensor $\mathfrak{R}$ is given by

$R_{ijk}^l dx^i \otimes dx^j \otimes dx^k \otimes \frac{\partial}{\partial x^l}$, the Ricci tensor by $R_{ik} dx^i \otimes dx^k$, and the scalar curvature S by $R = g^{jk} R_{kj}$. We now introduce another tensor relevant to our purpose.

Definition 2.2.11. The Einstein gravitational tensor of a spacetime M is

$$G = R_{ij} - \frac{1}{2} g_{ij} R.$$

Remark: An Einstein manifold is a pseudo-Riemannian manifold whose Ricci tensor satisfies $\text{Ric} = \lambda g$ for some constant λ . In dimension four, the condition $G = 0$ is equivalent to $\text{Ric} = 0$, corresponding to the special case $\lambda = 0$.

Question: How is the energy tensor T_{ab} related to the curvature tensor G ?

Einstein answered this question by proposing the formula

$$G = k T_{ab}, \quad \text{with } k \text{ constant,}$$

Note that G is a symmetric $(0, 2)$ -tensor. To satisfy energy-momentum conservation, we require $\text{div} T_{ij} = 0$; thus $G = k T_{ab}$ implies $\text{div} G = 0$.

Remark: Some texts write $\text{div} T_{ij} = 0$ as $\nabla_i T^{ij} = 0$, whereas others use $\nabla^i T_{ij} = 0$, because $\nabla_i T^{ij}$ and $\nabla^i T_{ij}$ are physically equivalent; see [3]. We shall use $\nabla^i T_{ij}$. Moreover, for a symmetric $(0, 2)$ -tensor T , $\text{div} T$ is the trace of the covariant derivative of T . It therefore makes sense to show that the covariant derivative of T vanishes, since this implies that its divergence also vanishes.

We require G in Definition 2.2.11 to satisfy this constraint. Consider the following identity.

Proposition 2.2.12.

$$\nabla^a \left(R_{ab} - \frac{1}{2} g_{ab} R \right) = 0. \quad (2.1)$$

Proof. The second Bianchi identity gives

$$\nabla_m R_{ijk}^l + \nabla_k R_{imj}^l + \nabla_j R_{ikm}^l = 0,$$

Contracting $j = l$ yields

$$\begin{aligned} 0 &= \nabla_m R_{ilk}^l + \nabla_k R_{iml}^l + \nabla_l R_{ikm}^l \\ &= \nabla_m R_{ik} - \nabla_k R_{ilm}^l + \nabla_l R_{ikm}^l \\ &= \nabla_m R_{ik} - \nabla_k R_{im} + \nabla_l R_{ikm}^l \end{aligned}$$

Multiplying by g^{im} gives

$$\begin{aligned} 0 &= \nabla_m g^{im} R_{ik} - \nabla_k g^{im} R_{im} + \nabla_l g^{im} R_{ikm}^l \\ &= \nabla_m R_k^m - \nabla_k R + \nabla_l R_k^l, \end{aligned}$$

Contracting $m = l$ gives

$$\begin{aligned} 0 &= \nabla_l R_k^l - \nabla_k R + \nabla_l R_k^l \\ &= \nabla_l R_k^l - \frac{1}{2} \nabla_k R, \end{aligned}$$

Since $\nabla_l R_k^l = \nabla_l g^{sl} g_{ps} R_k^p = \nabla^s R_{sk}$, we obtain

$$\begin{aligned} 0 &= \nabla^s R_{sk} - \frac{1}{2} g_{sk} \nabla^s R \\ &= \nabla^s \left(R_{sk} - \frac{1}{2} g_{sk} R \right), \end{aligned}$$

Relabelling the indices yields the desired result. $\square$

Einstein proposed several possibilities for G , testing them notably against the precession of Mercury's perihelion. This led to

$$R_{ab} = k T_{ab} + \frac{1}{2} g_{ab} R.$$

where R_{ab} is the Ricci tensor, yielding the tensor introduced in Definition 2.2.11.

Remark: Not every spacetime is a model of general relativity. It must satisfy the equations in Definition 2.2.11.

Definition 2.2.13. *If M is a spacetime containing matter with energy-momentum tensor T_{ab} , Einstein's equation is*

$$R_{ab} - \frac{1}{2} g_{ab} R = 8\pi T_{ab}.$$

Here R_{ab} is the Ricci tensor, R the scalar curvature, and the universal constant $k = 8\pi$ is determined by comparison with Newton's theory; see [10], p. 406.

The left-hand side of Einstein's equations is geometric, whereas the right-hand side depends on the physical content of the model. For example, T_{ab} may represent an electromagnetic field or a fluid, or it may be 0 in the absence of material content.

Einstein's equations tell us, among other things, that knowledge of matter in a region of the universe provides information about spacetime curvature, and conversely. If we know g , we can also define T_{ab} using Einstein's equations, but this would not necessarily be meaningful, since T_{ab} need not have a physical interpretation. If $T_{ab} = 0$, then $R_{ab} - \frac{1}{2}g_{ab}R = 0$, giving a system of ten nonlinear second-order equations for the components g_{ab} . This system has nontrivial solutions, which we shall compute under the stated assumptions.

The interaction between matter and curvature expressed by Einstein's equations is not derived from earlier principles. Rather, the equations constitute a proposed model of physical phenomena involving gravitational interactions, whose accuracy must be assessed through experiment and observation. Moreover, despite their difficulty, Einstein's equations admit explicit solutions. Every explicit solution has some form of symmetry.

We can summarize the postulates of general relativity as follows:

- Spacetime is as specified in Definition 2.2.10.
- Light travels on the surface of the cone, while particles travel inside it.
- The energy, momentum, and stress of continuous matter distributions are described by a symmetric tensor T_{ab} satisfying $\nabla^a T_{ab} = 0$, in accordance with energy–momentum conservation.
- Curvature is related to mass and energy through Einstein's equations in Definition 2.2.13.

2.3 Examples of Spacetimes

We briefly introduce Minkowski spacetime, which describes physical phenomena in special relativity. We then introduce Schwarzschild spacetime and the assumptions Karl Schwarzschild used to solve Einstein's equations.

2.3.1 Minkowski Spacetime

Minkowski spacetime is $M = \mathbb{R}^4$ with the metric

$$g = \sum_{i=1}^3 du^i \otimes du^i - du^4 \otimes du^4.$$

The spacetime (M, g) is time-oriented by $\frac{\partial}{\partial u^4}$ and oriented by the ordered basis (e_1, e_2, e_3, e_4) . The connection ∇ is trivial and $R_{ijkl} \equiv 0$, so this spacetime is flat. In this case, $G = 0$ and therefore $kT_{ij} = 0$. Thus Minkowski spacetime represents an empty universe, without material content.

This space is important because small neighbourhoods in a spacetime can locally be identified with Minkowski space.

Expanding the metric $ds^2 = g_{ij}du^i du^j$ and using $g_{ij} = 0$ whenever $i \neq j$ gives

$$ds^2 = g_{11}du^1 du^1 + g_{22}du^2 du^2 + g_{33}du^3 du^3 + g_{44}du^4 du^4.$$

In spherical coordinates (r, θ, ϕ, t) , we have

$$u^1 = r \sin \theta \sin \phi, \quad u^2 = r \sin \theta \cos \phi, \quad u^3 = r \cos \theta, \quad u^4 = t,$$

Hence

$$\begin{aligned} g_{11} &= \sin^2 \theta \sin^2 \phi + \sin^2 \theta \cos^2 \phi + \cos^2 \theta \\ &= 1, \end{aligned}$$

$$\begin{aligned} g_{22} &= r^2 \cos^2 \theta \sin^2 \phi + r^2 \cos^2 \theta \cos^2 \phi + r^2 (-\sin \theta)^2 \\ &= r^2, \end{aligned}$$

$$\begin{aligned} g_{33} &= r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta (-\sin \phi)^2 \\ &= r^2 \sin^2 \theta, \end{aligned}$$

$$g_{44} = -1,$$

Thus the metric g in the new coordinate system is

$$ds^2 = -dt^2 + dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.2)$$

In these coordinates the Christoffel symbols need not vanish, but the Riemann and Ricci curvature tensors satisfy $R_{ijkl} = 0$ and $R_{ij} = 0$, and $R = 0$.

The metric matrix is

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

The coordinate expression becomes singular at $r = 0$ or $\sin \theta = 0$, where spherical coordinates fail to define a chart; the Minkowski metric itself remains nondegenerate. Thus a single coordinate chart cannot cover M . We generally use polar-coordinate charts with domain

$$0 < r < \infty, \quad 0 < \theta < \pi \quad \text{and} \quad 0 < \phi < 2\pi.$$

2.3.2 Schwarzschild Spacetime

We start from the familiar field equations $R_{ij} - \frac{1}{2}g_{ij}R = 8\pi T_{ij}$.

If the energy tensor satisfies $T_{ij} = 0$, there is no matter (energy). Schwarzschild's solutions were obtained using symmetry and model the gravitational field outside a spherically symmetric body, in a region assumed to contain no matter.

Let (S^2, h) be the unit sphere with standard metric h , and let $m \in (0, \infty)$. On $\mathbb{R}^2$, consider coordinates (u^1, u^2) , where $u^i : \mathbb{R}^2 \rightarrow \mathbb{R}$ for $i = 1, 2$, and let h be the usual metric on $\mathbb{R}^2$. Set $A = (u^1)^{-1}[(0, 2m) \cup (2m, \infty)]$. Then $A \subset \mathbb{R}^2$ is open and has two connected components. Define $M = S^2 \times A$, with projections $P : M \rightarrow S^2$ and $Q : M \rightarrow A$. Define $r = u^1 \circ Q : M \rightarrow (0, 2m) \cup (2m, \infty)$ and $t = u^2 \circ Q : M \rightarrow \mathbb{R}$, where $r(p, (u^1, u^2)) = u^1(u^1, u^2) = u^1$ and $t(p, (u^1, u^2)) = u^2$.

The expression $\left(1 - \frac{2m}{r}\right)$ is a function of class C^∞ from M onto $(-\infty, 0) \cup (0, 1)$. In the next section, we shall derive the Schwarzschild metric:

$$g = \left(1 - \frac{2m}{r}\right)^{-1} dr \otimes dr + r^2 P^* h - \left(1 - \frac{2m}{r}\right) dt \otimes dt.$$

The function r is a radius, and t is interpreted as time. For $r > 2m$, the vector fields $\frac{\partial}{\partial r}$ and $\frac{\partial}{\partial t}$ are spacelike and timelike, respectively. For $0 < r < 2m$, their causal characters are reversed: $\frac{\partial}{\partial r}$ is timelike and $\frac{\partial}{\partial t}$ is spacelike. Moreover $P_* \left(\frac{\partial}{\partial t}\right) = 0 = P_* \left(\frac{\partial}{\partial r}\right)$, $Q_* \left(\frac{\partial}{\partial t}\right) = \frac{\partial}{\partial u^2}$ and $Q_* \left(\frac{\partial}{\partial r}\right) = \frac{\partial}{\partial u^1}$.

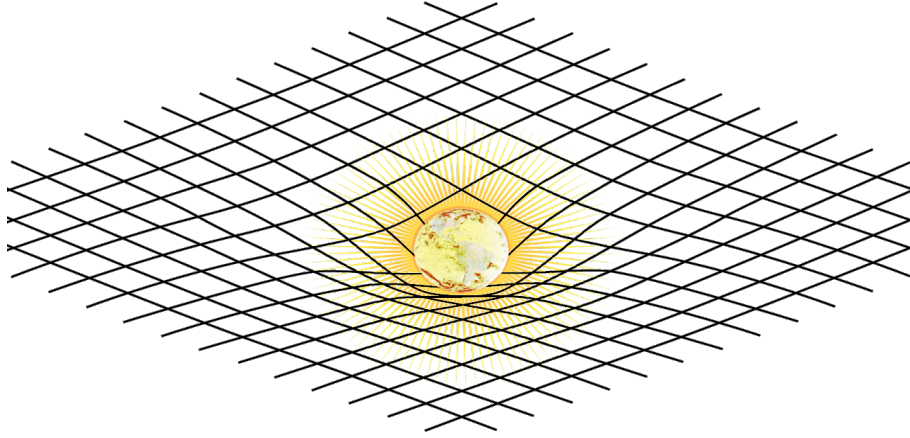
Note that $M = S^2 \times A$ is not connected. Take the connected component $N = r^{-1}(2m, \infty)$. Then $g|_N$ is an orientable Lorentzian metric, time-oriented by $\frac{\partial}{\partial t}$. Thus $(N, g|_N)$ is the Schwarzschild spacetime of mass $8\pi m$.

Physically, consider a spherically symmetric star of radius r_0 and mass $\bar{M} = 8\pi m$, with $r_0 >> \bar{M}$. The submanifold U of N defined by $r > r_0$ models the complete history of the star's exterior. Furthermore, a star is assumed to be nonrotating and electrically uncharged. The more complicated cases are included in the Kerr solution; see [10].

Computing the pullback P_* gives

$$ds^2 = -\left(1 - \frac{2m}{r}\right) dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2).$$

The solution (N, g) is spherically symmetric, that is, invariant under the action of $SO(3)$ on the sphere $S^2(r_0)$. ds^2 is asymptotically flat: $g_{ij} = \eta_{ij} + \Theta\left(\frac{1}{r}\right)$, where η_{ij} are the components of the Minkowski metric. Indeed, as $r \rightarrow \infty$, we have $ds^2 \approx -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$.



More precisely, in Section 2.4 we shall prove the following theorem.

Theorem 2.3.1. *If a solution of the equations $R_{ij} - \frac{1}{2}g_{ij}R = 0$ is spherically symmetric, then it is locally isometric to the Schwarzschild metric*

$$ds^2 = -\left(1 - \frac{2m}{r}\right) dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad t \in \mathbb{R}, r \neq 0 \text{ and } r \neq 2m.$$

To prove that the Schwarzschild metric solves Einstein's equations, in Section 2.4 we shall compute the components of the metric tensor under the assumptions of the problem. This result is known as Birkhoff's theorem.

Thus (M, ds^2) has the following properties:

1. $\frac{\partial}{\partial t}$ is a timelike vector field.
2. (M, ds^2) is time-orientable.
3. $M = S^2 \times A \subset S^2 \times \mathbb{R}^2$
4. $g_{ij} \approx \eta_{ij} + \Theta\left(\frac{1}{r}\right)$, where η_{ij} is the Minkowski metric. Thus ds^2 is asymptotically flat.
5. Every spherically symmetric solution of Einstein's equations is locally Schwarzschild.

In general, we consider ds^2 for $r > r_0 > 2m$ as the solution outside a spherical body of radius r , where $2m$ represents the radius of the star. The metric inside the body is determined by its energy tensor, so an initial restriction is $r > 2m$.

The metric is singular at $\theta = 0$ and $\theta = \pi$, or when $r = 0$ or $r = 2m$. Here $\theta = 0$ and $\theta = \pi$ are trivial singularities due to polar coordinates, whose domain is $0 < \theta < \pi$; the points $r = 0$ and $r = 2m$ must be excluded from the coordinate domain. If $m > 0$, any continuous curve beginning in $r < 2m$ must cross $r = 2m$ before reaching the *outside world*, where $r \rightarrow \infty$ is allowed. We therefore remove the hypersurface $r = 2m$, called the event horizon, from the manifold, turning M into $\mathbb{R} \times \{(0, 2m) \cup (2m, \infty)\} \times S^2$. Since M is connected, we must choose one interval in the domain of r . As $r > 2m$ represents the “outside world” of the star, we take $M = \mathbb{R} \times (2m, \infty) \times S^2$.

Remark: No Lorentzian metric extends the Schwarzschild metric through $r = 0$; see [13].

Question: Can we extend M to another spacetime in which (M, g) is isometrically embedded?

We shall extend M by extending the domain of the coordinate r beyond $r = 2m$, that is, into the star. To do so, define the so-called “tortoise” coordinate

$$\begin{aligned} r^* &= \int \frac{dr}{1 - \frac{2m}{r}} \\ &= r + 2m \ln(r - 2m), \end{aligned}$$

and set $v = t + r^*$. Then $dv = dt + dr^*$, so $dr^* = dv - dt$. Moreover, $dr^* = \frac{1}{1 - \frac{2m}{r}} dr$ and $dt = dv - dr^*$.

We now write ds^2 in terms of (v, r, θ, ϕ) :

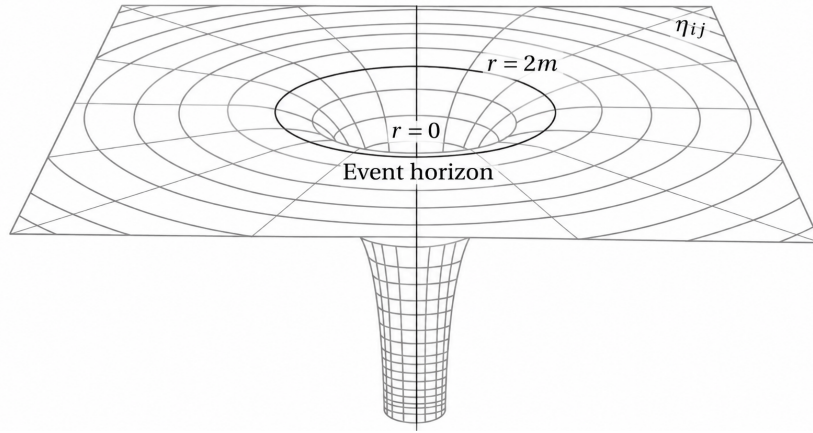
$$\begin{aligned} ds^2 &= -\left(1 - \frac{2m}{r}\right) dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \\ &= -\left(1 - \frac{2m}{r}\right) (dv^2 - 2dvdr^* + d(r^*)^2 + dvdr^* + r^2(d\theta^2 + \sin^2 \theta d\phi^2)) \\ &= -\left(1 - \frac{2m}{r}\right) dv^2 + 2\left(1 - \frac{2m}{r}\right) dvdr^* - \left(1 - \frac{2m}{r}\right) d(r^*)^2 + \left(1 - \frac{2m}{r}\right) d(r^*)^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \\ &= -\left(1 - \frac{2m}{r}\right) dv^2 + 2dvdr + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \end{aligned} \tag{2.3}$$

with $\left(1 - \frac{2m}{r}\right) dr^* = dr$. The metric matrix is therefore

$$\begin{pmatrix} -\left(1 - \frac{2m}{r}\right) & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

Equation (2.3) permits $r = 2m$, since the matrix remains invertible. We can therefore extend the spacetime to $\mathbb{R} \times (0, \infty) \times S^2$, suggesting that the singularity at $r = 2m$ is merely due to an unsuitable choice of coordinates. Intuitively, this is supported by the fact that no scalar polynomial in the curvature tensor diverges as $r \rightarrow 2m$. On the other hand, as $r \rightarrow 0$, we have $R_{ijkm} \rightarrow \infty$, a phenomenon called spaghettification.

(v, r, θ, ϕ) are called Eddington–Finkelstein coordinates. This extension indicates that $r = 0$ is a singularity, meaning that the spacetime cannot be extended to $r = 0$. As $r \rightarrow 0$, we are in the presence of a black hole.



If $r > 2m$, then $\frac{\partial}{\partial t}$ is timelike and $\frac{\partial}{\partial r}$ is spacelike. If $r < 2m$, then $\frac{\partial}{\partial t}$ is spacelike and $\frac{\partial}{\partial r}$ is timelike. This means that an observer crossing $r = 2m$ cannot return; hence the hypersurface $\mathbb{R} \times \{r = 2m\} \times S^2$ is called the event horizon.

Definition 2.3.2. For the metric (2.3), the region $\{r \leq 2m\}$ is a black-hole region in the sense that observers or signals can enter it but can never leave it.

2.4 The Schwarzschild Solution

Consider a spherical body (a star or a black hole) of mass m centred at the coordinate origin in $\mathbb{R}^3$. Its gravitational field is represented by a metric replacing the Minkowski metric, which we must determine. Outside the star, the energy tensor T_{ij} vanishes, so the Ricci tensor of the metric must also vanish. The nature of the body suggests that the spacetime should be spherically symmetric. Thus we can write its metric in spherical coordinates as

$$ds^2 = -Udt^2 + Vdr^2 + r^2(Sd\theta^2 + Q\sin^2\theta d\phi^2)$$

where U, V, S, Q are functions to be determined under the assumptions of the problem:

1. **Spherical symmetry:** The functions do not depend on θ or ϕ . Moreover, the angular part of the metric must satisfy $S = Q$.
2. **Staticity:** In vacuum, spherical symmetry locally implies two symmetries: translations in the coordinate t and reflections of the form $t \mapsto -t$, where $1 - \frac{2m}{r} > 0$. A metric with these two properties is called static. Thus every derivative of a metric component with respect to t vanishes.
3. **Vacuum solution:** There is no matter or energy outside the body producing the curvature. Thus $T_{ij} = 0$.

Without loss of generality, we may assume $S = Q = 1$. The restricted form of the solutions of Einstein's equations is therefore

$$ds^2 = -U dt^2 + V dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

with metric matrix

$$g_{ij} = \begin{pmatrix} -U & 0 & 0 & 0 \\ 0 & V & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{pmatrix}$$

and inverse

$$g^{ij} = \begin{pmatrix} -\frac{1}{U} & 0 & 0 & 0 \\ 0 & \frac{1}{V} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2\theta} \end{pmatrix}.$$

We now use Einstein's equation

$$R_{ij} - \frac{1}{2}g_{ij}R = 8\pi T_{ij}$$

to determine U and V . To do so, we need to compute the Ricci tensor components R_{ij} and the scalar curvature R . We first compute all the Christoffel symbols using equation (1.8), taking into account the symmetries imposed by the assumptions of the problem.

$$\Gamma_{00}^0 = \frac{1}{2}g^{00} \left(\cancel{\frac{\partial g_{00}}{\partial t}} + \cancel{\frac{\partial g_{00}}{\partial t}} - \cancel{\frac{\partial g_{00}}{\partial t}} \right) = 0,$$

$$\begin{aligned} \Gamma_{0j}^0 &= \Gamma_{j0}^0 = \frac{1}{2}g^{00} \left(\frac{\partial g_{00}}{\partial x^j} + \cancel{\frac{\partial g_{0j}}{\partial t}} - \cancel{\frac{\partial g_{0j}}{\partial t}} \right) \\ &= \frac{1}{2}g^{00} \frac{\partial g_{00}}{\partial x^j} \\ &= \frac{1}{2} \frac{1}{U} \frac{\partial U}{\partial x^j}, \end{aligned}$$

Since U depends on r , the expression need not vanish when $j = 1$:

$$\Gamma_{01}^0 = \Gamma_{10}^0 = \frac{1}{2} \frac{1}{U} \frac{\partial U}{\partial r},$$

whereas for $j \neq 1$,

$$\Gamma_{0j}^0 = \Gamma_{j0}^0 = 0.$$

If we now take $i \neq j \neq 0$, we obtain

$$\Gamma_{ij}^0 = \Gamma_{ji}^0 = \frac{1}{2}g^{00} \left(\frac{\partial g_{0i}}{\partial x^j} + \frac{\partial g_{0j}}{\partial x^i} - \cancel{\frac{\partial g_{ij}}{\partial t}} \right) = 0,$$

$$\begin{aligned}
\Gamma_{00}^1 &= \frac{1}{2}g^{11} \left(\cancel{\frac{\partial g_{00}}{\partial t}} + \cancel{\frac{\partial g_{00}}{\partial t}} - \frac{\partial g_{00}}{\partial r} \right) \\
&= -\frac{1}{2}g^{11} \frac{\partial g_{00}}{\partial r} \\
&= \frac{1}{2} \frac{1}{V} \frac{\partial U}{\partial r},
\end{aligned}$$

$$\begin{aligned}
\Gamma_{11}^1 &= \frac{1}{2}g^{11} \left(\frac{\partial g_{11}}{\partial r} + \cancel{\frac{\partial g_{11}}{\partial r}} - \cancel{\frac{\partial g_{11}}{\partial r}} \right) \\
&= \frac{1}{2}g^{11} \frac{\partial g_{11}}{\partial r} \\
&= \frac{1}{2} \frac{1}{V} \frac{\partial V}{\partial r},
\end{aligned}$$

$$\begin{aligned}
\Gamma_{22}^1 &= \frac{1}{2}g^{11} \left(\cancel{\frac{\partial g_{22}}{\partial \theta}} + \cancel{\frac{\partial g_{22}}{\partial \theta}} - \frac{\partial g_{22}}{\partial r} \right) \\
&= -\frac{1}{2}g^{11} \frac{\partial g_{22}}{\partial r} \\
&= -\frac{1}{2} \frac{1}{V} \frac{\partial r^2}{\partial r} \\
&= -\frac{r}{V},
\end{aligned}$$

$$\begin{aligned}
\Gamma_{33}^1 &= \frac{1}{2}g^{11} \left(\cancel{\frac{\partial g_{33}}{\partial \phi}} + \cancel{\frac{\partial g_{33}}{\partial \phi}} - \frac{\partial g_{33}}{\partial r} \right) \\
&= -\frac{1}{2}g^{11} \frac{\partial g_{33}}{\partial r} \\
&= -\frac{1}{2} \frac{1}{V} \frac{\partial r^2 \sin^2 \theta}{\partial r} \\
&= -\frac{r \sin^2 \theta}{V},
\end{aligned}$$

For $j \neq 0$, we have

$$\Gamma_{0j}^1 = \Gamma_{j0}^1 = \frac{1}{2}g^{11} \left(\cancel{\frac{\partial g_{0j}}{\partial x^j}} + \cancel{\frac{\partial g_{0j}}{\partial t}} - \cancel{\frac{\partial g_{0j}}{\partial r}} \right) = 0,$$

For $i \neq j$, since V depends only on r (in particular, $i \neq j \neq 1$), we have

$$\Gamma_{ij}^1 = \Gamma_{ji}^1 = \frac{1}{2}g^{11} \left(\frac{\partial g_{1i}}{\partial x^j} + \frac{\partial g_{1j}}{\partial x^i} - \frac{\partial g_{ij}}{\partial r} \right) = 0,$$

$$\begin{aligned}
\Gamma_{00}^2 &= \frac{1}{2}g^{22} \left(\cancel{\frac{\partial g_{20}}{\partial t}} + \cancel{\frac{\partial g_{20}}{\partial t}} - \frac{\partial g_{00}}{\partial \theta} \right) \\
&= -\frac{1}{2}g^{22} \frac{\partial g_{00}}{\partial \theta} \\
&= -\frac{1}{2} \frac{1}{r^2} \frac{\partial U}{\partial \theta} \\
&= 0,
\end{aligned}$$

Now consider $i=j$:

$$\Gamma_{ii}^2 = \frac{1}{2}g^{22} \left(\frac{\partial g_{2i}}{\partial x^i} + \frac{\partial g_{2i}}{\partial x^i} - \frac{\partial g_{ii}}{\partial \theta} \right),$$

This vanishes for $i = 1$, whereas for $i = 2$,

$$\begin{aligned}
\Gamma_{22}^2 &= \frac{1}{2}g^{22} \left(\frac{\partial g_{22}}{\partial \theta} + \cancel{\frac{\partial g_{22}}{\partial \theta}} - \cancel{\frac{\partial g_{22}}{\partial \theta}} \right) \\
&= \frac{1}{2}g^{22} \frac{\partial g_{22}}{\partial \theta} \\
&= \frac{1}{2} \frac{1}{r^2} \frac{\partial r^2}{\partial \theta} \\
&= 0,
\end{aligned}$$

For $i = 3$,

$$\begin{aligned}
\Gamma_{33}^2 &= \frac{1}{2}g^{22} \left(\cancel{\frac{\partial g_{23}}{\partial \phi}} + \cancel{\frac{\partial g_{23}}{\partial \phi}} - \frac{\partial g_{33}}{\partial \theta} \right) \\
&= -\frac{1}{2}g^{22} \frac{\partial g_{33}}{\partial \theta} \\
&= -\frac{1}{2} \frac{1}{r^2} \frac{\partial r^2 \sin^2 \theta}{\partial \theta} \\
&= -\sin \theta \cos \theta,
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\Gamma_{12}^2 = \Gamma_{21}^2 &= \frac{1}{2}g^{22} \left(\cancel{\frac{\partial g_{21}}{\partial \theta}} + \frac{\partial g_{22}}{\partial r} - \cancel{\frac{\partial g_{12}}{\partial \theta}} \right) \\
&= \frac{1}{2}g^{22} \frac{\partial g_{22}}{\partial r} \\
&= \frac{1}{2} \frac{1}{r^2} \frac{\partial r^2}{\partial r} \\
&= \frac{1}{r},
\end{aligned}$$

$$\begin{aligned}\Gamma_{13}^2 &= \Gamma_{31}^2 = \frac{1}{2}g^{22} \left(\frac{\partial g_{21}}{\partial \theta} + \frac{\partial g_{23}}{\partial r} - \frac{\partial g_{13}}{\partial \theta} \right) \\ &= 0,\end{aligned}$$

$$\begin{aligned}\Gamma_{23}^2 &= \Gamma_{32}^2 = \frac{1}{2}g^{22} \left(\frac{\partial g_{22}}{\partial \phi} + \frac{\partial g_{23}}{\partial \theta} - \frac{\partial g_{23}}{\partial \theta} \right) \\ &= \frac{1}{2}g^{22} \frac{\partial g_{22}}{\partial \phi} \\ &= \frac{1}{2} \frac{1}{r^2} \frac{\partial r^2}{\partial \phi} \\ &= 0,\end{aligned}$$

$$\begin{aligned}\Gamma_{00}^3 &= \frac{1}{2}g^{33} \left(\frac{\partial g_{30}}{\partial t} + \frac{\partial g_{30}}{\partial t} - \frac{\partial g_{00}}{\partial \phi} \right) \\ &= -\frac{1}{2}g^{33} \frac{\partial g_{00}}{\partial \theta} \\ &= -\frac{1}{2} \frac{1}{r^2 \sin^2 \theta} \frac{\partial U}{\partial \phi} \\ &= 0,\end{aligned}$$

For $j \neq 0$,

$$\begin{aligned}\Gamma_{0j}^3 &= \Gamma_{j0}^3 = \frac{1}{2}g^{33} \left(\frac{\partial g_{30}}{\partial x^j} + \frac{\partial g_{3j}}{\partial t} - \frac{\partial g_{0j}}{\partial \phi} \right) \\ &= \frac{1}{2}g^{33} \frac{\partial g_{33}}{\partial t} \\ &= 0,\end{aligned}$$

For $i \neq 0$, using the fact that $g_{ij} \neq 0$ if and only if $i = j$, we obtain

$$\begin{aligned}\Gamma_{ii}^3 &= \frac{1}{2}g^{33} \left(\frac{\partial g_{3i}}{\partial x^i} + \frac{\partial g_{3i}}{\partial x^i} - \frac{\partial g_{ii}}{\partial \phi} \right) \\ &= g^{33} \frac{\partial g_{33}}{\partial \phi} \\ &= \frac{1}{r^2 \sin^2 \theta} \frac{\partial r^2 \sin^2 \theta}{\partial \phi} \\ &= 0,\end{aligned}$$

On the other hand, for $i \neq j$,

$$\begin{aligned}\Gamma_{ij}^3 &= \Gamma_{ji}^3 = \frac{1}{2}g^{33} \left(\frac{\partial g_{3i}}{\partial x^j} + \frac{\partial g_{3j}}{\partial x^i} - \cancel{\frac{\partial g_{ij}}{\partial \phi}} \right) \\ &= \frac{1}{2}g^{33} \left(\frac{\partial g_{3i}}{\partial x^j} + \frac{\partial g_{3j}}{\partial x^i} \right),\end{aligned}$$

It suffices to set $j = 3$ with $i \neq 3$ to obtain the nonzero symbols.

$$\begin{aligned}\Gamma_{13}^3 &= \Gamma_{31}^3 = \frac{1}{2}g^{33} \left(\cancel{\frac{\partial g_{31}}{\partial \phi}} + \frac{\partial g_{33}}{\partial r} \right) \\ &= \frac{1}{2} \frac{1}{r^2 \sin^2 \theta} \frac{\partial r^2 \sin^2 \theta}{\partial r} \\ &= \frac{1}{r},\end{aligned}$$

$$\begin{aligned}\Gamma_{23}^3 &= \Gamma_{32}^3 = \frac{1}{2}g^{33} \left(\cancel{\frac{\partial g_{32}}{\partial \phi}} + \frac{\partial g_{33}}{\partial \theta} \right) \\ &= \frac{1}{2} \frac{1}{r^2 \sin^2 \theta} \frac{\partial r^2 \sin^2 \theta}{\partial \theta} \\ &= \frac{\cos \theta}{\sin \theta} \\ &= \cot \theta.\end{aligned}$$

We now compute the Ricci tensor components in terms of the Christoffel symbols. We know that

$$R_{ij} = R_{ikj}^k = \frac{\partial}{\partial x^k} \Gamma_{ij}^k + \Gamma_{ij}^m \Gamma_{km}^k - \frac{\partial}{\partial x^j} \Gamma_{ik}^k - \Gamma_{ik}^m \Gamma_{jm}^k, \quad (2.4)$$

Expanding the sum gives

$$\begin{aligned}R_{ij} &= \cancel{\frac{\partial}{\partial t} \Gamma_{ij}^0} + \Gamma_{ij}^m \Gamma_{0m}^0 - \frac{\partial}{\partial x^j} \Gamma_{i0}^0 - \Gamma_{i0}^m \Gamma_{jm}^0 + \frac{\partial}{\partial r} \Gamma_{ij}^1 + \Gamma_{ij}^m \Gamma_{1m}^1 - \frac{\partial}{\partial x^j} \Gamma_{i1}^1 - \Gamma_{i1}^m \Gamma_{jm}^1 \\ &\quad + \frac{\partial}{\partial \theta} \Gamma_{ij}^2 + \Gamma_{ij}^m \Gamma_{2m}^2 - \frac{\partial}{\partial x^j} \Gamma_{i2}^2 - \Gamma_{i2}^m \Gamma_{jm}^2 + \frac{\partial}{\partial \phi} \Gamma_{ij}^3 + \Gamma_{ij}^m \Gamma_{3m}^3 - \frac{\partial}{\partial x^j} \Gamma_{i3}^3 - \Gamma_{i3}^m \Gamma_{jm}^3, \quad (2.5)\end{aligned}$$

We first compute the components R_{ij} with $i \neq j$ and $j \neq 0$.

$$\begin{aligned}R_{0j} &= \Gamma_{0j}^m \Gamma_{0m}^0 - \cancel{\frac{\partial}{\partial x^j} \Gamma_{00}^0} - \Gamma_{00}^m \Gamma_{jm}^0 + \cancel{\frac{\partial}{\partial r} \Gamma_{0j}^1} + \Gamma_{0j}^m \Gamma_{1m}^1 - \cancel{\frac{\partial}{\partial x^j} \Gamma_{01}^1} - \Gamma_{01}^m \Gamma_{jm}^1 \\ &\quad + \cancel{\frac{\partial}{\partial \theta} \Gamma_{0j}^2} + \Gamma_{0j}^m \Gamma_{2m}^2 - \cancel{\frac{\partial}{\partial x^j} \Gamma_{02}^2} - \Gamma_{02}^m \Gamma_{jm}^2 + \cancel{\frac{\partial}{\partial \phi} \Gamma_{0j}^3} + \Gamma_{0j}^m \Gamma_{3m}^3 - \cancel{\frac{\partial}{\partial x^j} \Gamma_{03}^3} - \Gamma_{03}^m \Gamma_{jm}^3,\end{aligned}$$

For the terms that remain, we have

$$\Gamma_{0j}^m \Gamma_{0m}^0 = 0 \text{ if } m = 0, 2, 3. \text{ And for } m = 1 \text{ we have } \cancel{\Gamma_{0j}^1 \Gamma_{01}^0} = 0,$$

$\Gamma_{00}^m \Gamma_{jm}^0 = 0$ if $m = 0, 2, 3$. And for $m = 1$ and $j \neq 0$ we have $\Gamma_{00}^1 \Gamma_{j1}^0 = 0$,

Similarly,

$$\begin{aligned}\Gamma_{0j}^m \Gamma_{1m}^1 &= 0, \\ \Gamma_{0j}^m \Gamma_{2m}^2 &= 0, \\ \Gamma_{0j}^m \Gamma_{3m}^3 &= 0,\end{aligned}$$

$\Gamma_{01}^m \Gamma_{jm}^1 = 0$ if $m = 1, 2, 3$. And for $m = 0$ and $j \neq 0$ we have $\Gamma_{01}^0 \Gamma_{j0}^1 = 0$.

Hence

$$R_{0j} = 0, \quad \text{for } j \neq 0.$$

For R_{ij} with $i \neq j$ and neither index zero, consider (2.5). Examining the expression term by term gives

$$\begin{aligned}\Gamma_{ij}^m \Gamma_{0m}^0 &\neq 0 \text{ if } m = 1, \text{ then } \Gamma_{ij}^1 \Gamma_{01}^0 \neq 0 \text{ if } i = j, \\ \Gamma_{i0}^0 &\neq 0 \text{ if } i = 1. \text{ Thus } \frac{\partial}{\partial x^j} \Gamma_{i0}^0 = \frac{\partial}{\partial x^j} \left(\frac{1}{2U} \frac{\partial U}{\partial r} \right) = 0 \text{ if } j \neq 1 = i, \\ \Gamma_{jm}^0 &\neq 0 \text{ if } m = 0 \text{ and } j = 1, \text{ then } \Gamma_{i0}^0 \Gamma_{10}^0 \neq 0 \text{ if } i = j = 1, \\ \frac{\partial}{\partial r} \Gamma_{ij}^1 &= 0, \text{ since } \Gamma_{ij}^1 \neq 0 \text{ if } i = j, \\ \Gamma_{ij}^m \Gamma_{1m}^1 &\neq 0 \text{ if } \Gamma_{1m}^1 = 0 \text{ if } m = 1, \text{ then } \Gamma_{ij}^1 \Gamma_{11}^1 \neq 0 \text{ if } \Gamma_{ij}^1 \neq 0 \text{ if } i = j, \\ \frac{\partial}{\partial x^j} \Gamma_{i1}^1 &\neq 0 \text{ if } i = 1, \text{ but } \frac{\partial}{\partial x^j} \left(\frac{1}{2V} \frac{\partial V}{\partial r} \right) \neq 0 \text{ if } i = j = 1,\end{aligned}$$

Since

$$\Gamma_{i1}^m \Gamma_{jm}^1 = \Gamma_{i1}^0 \Gamma_{j0}^1 + \Gamma_{i1}^1 \Gamma_{j1}^1 + \Gamma_{i1}^2 \Gamma_{j2}^1 + \Gamma_{i1}^3 \Gamma_{j3}^1,$$

and since

$$\begin{aligned}\Gamma_{j0}^1 &\neq 0 \text{ if } j = 0, \text{ then } \Gamma_{i1}^0 \Gamma_{j0}^1 = 0, \\ \Gamma_{i1}^1 &\neq 0 \text{ if } i = 1, \text{ then } \Gamma_{11}^1 \Gamma_{j1}^1 \neq 0 \text{ if } i = j = 1, \text{ then } \Gamma_{i1}^1 \Gamma_{j1}^1 = 0 \text{ for } i \neq j, \\ \Gamma_{j2}^1 &\neq 0 \text{ if } j = 2, \text{ then } \Gamma_{i1}^2 \Gamma_{22}^1 \neq 0 \text{ if } i = j = 2. \text{ Thus } \Gamma_{i1}^2 \Gamma_{j2}^1 = 0 \text{ for } i \neq j, \\ \Gamma_{i1}^3 &\neq 0 \text{ if } i = 3, \text{ then } \Gamma_{31}^3 \Gamma_{j3}^1 \neq 0 \text{ if } i = j = 3, \text{ then } \Gamma_{i1}^3 \Gamma_{j3}^1 = 0 \text{ for } i \neq j,\end{aligned}$$

Therefore

$$\Gamma_{i1}^m \Gamma_{jm}^1 = 0,$$

On the other hand,

$$\Gamma_{ij}^2 \neq 0 \text{ if } i = j = 3, \text{ or if } i = 1 \text{ and } j = 2 \text{ (or conversely). Since } \Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{r},$$

$$\text{then } \frac{\partial}{\partial \theta} \frac{1}{r} = 0 \text{ and thus } \frac{\partial}{\partial \theta} \Gamma_{ij}^2 = 0,$$

$$\Gamma_{ij}^m \Gamma_{2m}^2 \neq 0 \text{ if } m = 1, \text{ then } \Gamma_{ij}^1 \Gamma_{21}^2 \neq 0 \text{ if } i = j, \text{ then } \Gamma_{ij}^m \Gamma_{2m}^2 = 0 \text{ for } i \neq j,$$

$$\Gamma_{i2}^2 \neq 0 \text{ if } i = 1, \text{ thus } \frac{\partial}{\partial x^j} \Gamma_{12}^2 = \frac{\partial}{\partial x^j} \left(\frac{1}{r} \right) \neq 0 \text{ if } i = j = 1, \text{ then } \frac{\partial}{\partial x^j} \Gamma_{i2}^2 = 0 \text{ for } i \neq j,$$

Moreover, since

$$\Gamma_{i2}^m \Gamma_{jm}^2 = \cancel{\Gamma_{i2}^0 \Gamma_{j0}^2} + \Gamma_{i2}^1 \Gamma_{j1}^2 + \Gamma_{i2}^2 \Gamma_{j2}^2 + \Gamma_{i2}^3 \Gamma_{j3}^2,$$

we have

$$\Gamma_{i2}^1 \Gamma_{j1}^2 \neq 0 \text{ if } i = j = 2,$$

$$\Gamma_{i2}^2 \Gamma_{j2}^2 \neq 0 \text{ if } i = j = 1,$$

and

$$\Gamma_{i2}^3 \Gamma_{j3}^2 \neq 0 \text{ if } i = j = 3,$$

Thus

$$\Gamma_{i2}^m \Gamma_{jm}^2 = 0.$$

On the other hand,

$$\Gamma_{ij}^3 \neq 0 \text{ if } i = 2 \text{ and } j = 3 \text{ (or conversely), or if } i = 1 \text{ and } j = 3, \text{ where } \frac{\partial}{\partial \phi} \cot = 0 \text{ and } \frac{\partial}{\partial \phi} \frac{1}{r} = 0,$$

Thus

$$\Gamma_{ij}^3 = 0.$$

Expanding the terms gives

$$\Gamma_{ij}^m \Gamma_{3m}^3 = \cancel{\Gamma_{ij}^0 \Gamma_{30}^3} + \Gamma_{ij}^1 \Gamma_{31}^3 + \Gamma_{ij}^2 \Gamma_{32}^3 + \cancel{\Gamma_{ij}^3 \Gamma_{33}^3},$$

Since

$$\Gamma_{ij}^1 \neq 0 \text{ if } i = j = 1, \text{ then } \Gamma_{ij}^1 \Gamma_{31}^3 = 0, \text{ for } i \neq j,$$

and expanding the nonzero terms, we obtain

$$\begin{aligned} \Gamma_{ij}^2 \Gamma_{32}^3 &= \Gamma_{12}^2 \Gamma_{32}^3 + \Gamma_{21}^2 \Gamma_{32}^3 \\ &= \frac{1}{r} \cot \theta + \frac{1}{r} \cot \theta, \end{aligned}$$

Therefore

$$\Gamma_{ij}^m \Gamma_{3m}^3 = \frac{2}{r} \cot \theta.$$

Furthermore, note that

$$\Gamma_{i3}^3 \neq 0 \text{ if } i = 1 \text{ or if } i = 2. \text{ Since } \frac{\partial}{\partial x^j} \frac{1}{r} \neq 0 \text{ if } i = j = 1, \text{ and } \frac{\partial}{\partial x^j} \cot \theta \neq 0 \text{ if } i = j = 2, \\ \text{then } \frac{\partial}{\partial x^j} \Gamma_{i3}^3 = 0 \text{ for } i \neq j.$$

Finally, expanding gives

$$\Gamma_{i3}^m \Gamma_{jm}^3 = \cancel{\Gamma_{i3}^0 \Gamma_{j0}^3} + \Gamma_{i3}^1 \Gamma_{j1}^3 + \Gamma_{i3}^2 \Gamma_{j2}^3 + \Gamma_{i3}^3 \Gamma_{j3}^3,$$

where

$$\Gamma_{i3}^1 \Gamma_{j1}^3 \neq 0 \text{ if } i = j = 3,$$

$$\Gamma_{i3}^2 \Gamma_{j2}^3 \neq 0 \text{ if } i = j = 3,$$

$$\Gamma_{i3}^3 \Gamma_{j3}^3 \neq 0 \text{ if } i = 1 \text{ and } j = 2 \text{ (or conversely),}$$

Thus

$$\Gamma_{i3}^m \Gamma_{jm}^3 = \Gamma_{13}^3 \Gamma_{23}^3 + \Gamma_{23}^3 \Gamma_{13}^3 \\ = \frac{1}{r} \cot \theta + \frac{1}{r} \cot \theta,$$

Therefore, for $i \neq j$, we have

$$R_{ij} = \frac{2}{r} \cot \theta - \frac{2}{r} \cot \theta \\ = 0.$$

We now examine the components R_{ii} , with $i = 0, 1, 2, 3$.

$$R_{00} = \cancel{\frac{\partial}{\partial t} \Gamma_{00}^0} + \cancel{\Gamma_{00}^m \Gamma_{0m}^0} - \cancel{\frac{\partial}{\partial t} \Gamma_{00}^0} - \cancel{\Gamma_{00}^m \Gamma_{0m}^0} + \frac{\partial}{\partial r} \Gamma_{00}^1 + \Gamma_{00}^m \Gamma_{1m}^1 - \cancel{\frac{\partial}{\partial t} \Gamma_{01}^1} - \Gamma_{01}^m \Gamma_{0m}^1 \\ \frac{\partial}{\partial \theta} \Gamma_{00}^2 + \Gamma_{00}^m \Gamma_{2m}^2 - \cancel{\frac{\partial}{\partial t} \Gamma_{02}^2} - \cancel{\Gamma_{02}^m \Gamma_{0m}^2} + \cancel{\frac{\partial}{\partial \phi} \Gamma_{00}^3} + \Gamma_{00}^m \Gamma_{3m}^3 - \cancel{\frac{\partial}{\partial t} \Gamma_{03}^3} - \cancel{\Gamma_{03}^m \Gamma_{0m}^3},$$

Taking $m = 0, 1, 2, 3$ and retaining the nonzero terms ($m = 1$), we obtain

$$R_{00} = \frac{\partial}{\partial r} \Gamma_{00}^1 + \Gamma_{00}^1 \Gamma_{11}^1 - \Gamma_{10}^0 \Gamma_{00}^1 + \Gamma_{00}^1 \Gamma_{21}^2 + \Gamma_{00}^1 \Gamma_{31}^3,$$

Substituting the corresponding Christoffel symbols gives

$$R_{00} = \frac{\partial}{\partial r} \left(\frac{1}{2V} \frac{\partial U}{\partial r} \right) + \frac{1}{2V} \frac{\partial U}{\partial r} \frac{1}{2V} \frac{\partial V}{\partial r} - \frac{1}{2U} \frac{\partial U}{\partial r} \frac{1}{2V} \frac{\partial U}{\partial r} + \frac{1}{2V} \frac{\partial U}{\partial r} \frac{1}{r} + \frac{1}{2V} \frac{\partial U}{\partial r} \frac{1}{r}.$$

Since

$$\begin{aligned}\frac{\partial}{\partial r} \left(\frac{1}{2V} \frac{\partial U}{\partial r} \right) &= \frac{1}{2} \left(\frac{\frac{\partial^2 U}{\partial r^2} V - \frac{\partial U}{\partial r} \frac{\partial V}{\partial r}}{V^2} \right) \\ &= \frac{1}{2V} \frac{\partial^2 U}{\partial r^2} - \frac{1}{2V^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r},\end{aligned}$$

we obtain

$$R_{00} = \frac{1}{2V} \frac{\partial^2 U}{\partial r^2} - \frac{1}{4V^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{4UV} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial U}{\partial r}. \quad (2.6)$$

On the other hand, (2.5) gives

$$R_{11} = \Gamma_{11}^1 \Gamma_{01}^0 - \frac{\partial}{\partial r} \Gamma_{10}^0 - \Gamma_{10}^0 \Gamma_{10}^0 + \Gamma_{11}^1 \Gamma_{21}^2 - \frac{\partial}{\partial r} \Gamma_{12}^2 - \Gamma_{12}^2 \Gamma_{12}^2 + \Gamma_{11}^1 \Gamma_{31}^3 - \frac{\partial}{\partial r} \Gamma_{13}^3 - \Gamma_{13}^3 \Gamma_{13}^3, \quad ,$$

Substituting and simplifying, we obtain

$$\begin{aligned}R_{11} &= \frac{1}{4UV} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \left(\frac{2U \left(\frac{\partial^2 U}{\partial r^2} \right) - 2 \left(\frac{\partial U}{\partial r} \right)^2}{4U^2} \right) - \frac{1}{4U^2} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial V}{\partial r} + \frac{2}{r^2} - \frac{2}{r^2} \\ &= \frac{1}{4UV} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{2U} \frac{\partial^2 U}{\partial r^2} + \frac{1}{4U^2} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial V}{\partial r}.\end{aligned} \quad (2.7)$$

Equation (2.5) also gives the nonzero terms

$$R_{22} = \Gamma_{22}^1 \Gamma_{01}^0 + \frac{\partial}{\partial \theta} \Gamma_{22}^1 + \Gamma_{22}^1 \Gamma_{11}^1 - \Gamma_{21}^2 \Gamma_{22}^1 + \Gamma_{22}^1 \Gamma_{31}^3 - \frac{\partial}{\partial \theta} \Gamma_{23}^3 - \Gamma_{23}^3 \Gamma_{23}^3, \quad ,$$

Substituting and using $\frac{\partial \cot \theta}{\partial \theta} = -(\csc^2 \theta)$, we obtain

$$\begin{aligned}R_{22} &= \frac{-r}{2UV} \frac{\partial U}{\partial r} + \left(\frac{-V + r \left(\frac{\partial V}{\partial r} \right)}{V^2} \right) - \frac{r}{2V^2} \left(\frac{\partial V}{\partial r} \right) + 1 + \cot^2 \theta - \cot^2 \theta \\ &= \frac{-r}{2UV} \frac{\partial U}{\partial r} - \frac{1}{V} + \frac{r}{2V^2} \frac{\partial V}{\partial r} + 1.\end{aligned} \quad (2.8)$$

Similarly, from (2.5),

$$R_{33} = \Gamma_{33}^1 \Gamma_{01}^0 + \frac{\partial}{\partial r} \Gamma_{33}^1 + \Gamma_{33}^1 \Gamma_{11}^1 - \Gamma_{31}^3 \Gamma_{33}^1 + \frac{\partial}{\partial \theta} \Gamma_{33}^2 + \Gamma_{33}^1 \Gamma_{21}^2 - \Gamma_{32}^3 \Gamma_{33}^2, \quad ,$$

Substituting and simplifying gives

$$\begin{aligned}R_{33} &= \frac{-r}{2UV} \sin^2 \theta \frac{\partial}{\partial r} - \frac{\sin^2 \theta}{V} + \frac{r \sin^2 \theta}{V^2} \left(\frac{\partial V}{\partial r} \right) - \frac{r \sin^2 \theta}{2V^2} \left(\frac{\partial V}{\partial r} \right) + \sin^2 \theta \\ &= \left(\frac{-r}{2UV} \frac{\partial U}{\partial r} - \frac{1}{V} + \frac{r}{2V^2} \frac{\partial V}{\partial r} + 1 \right) \sin^2 \theta,\end{aligned}$$

Therefore

$$R_{33} = R_{22} \sin^2 \theta. \quad (2.9)$$

We now compute the Ricci scalar $R = g^{ij} R_{ij}$. Substitution gives

$$\begin{aligned} R &= g^{00} R_{00} + g^{11} R_{11} + g^{22} R_{22} + g^{33} R_{33} \\ &= g^{00} R_{00} + g^{11} R_{11} + g^{22} R_{22} + g^{33} R_{22} \sin^2 \theta \\ &= -\frac{1}{U} R_{00} + \frac{1}{V} R_{11} + \frac{1}{r^2} R_{22} + \frac{1}{r^2 \sin^2 \theta} R_{22} \sin^2 \theta \\ &= -\frac{1}{U} R_{00} + \frac{1}{V} R_{11} + \frac{2}{r^2} R_{22}, \end{aligned}$$

Substituting (2.6), (2.7), and (2.8), we obtain

$$\begin{aligned} R &= -\frac{1}{2UV} \frac{\partial^2 U}{\partial r^2} + \frac{1}{4UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} + \frac{1}{4U^2 V} \left(\frac{\partial U}{\partial r} \right)^2 - \frac{1}{rUV} \frac{\partial U}{\partial r} + \frac{1}{4UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} \\ &\quad - \frac{1}{2UV} \frac{\partial^2 U}{\partial r^2} + \frac{1}{4U^2 V} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV^2} \frac{\partial V}{\partial r} - \frac{1}{rUV} \frac{\partial U}{\partial r} - \frac{2}{r^2 V} + \frac{1}{rV^2} \frac{\partial V}{\partial r} + \frac{2}{r^2}, \end{aligned}$$

Therefore

$$R = -\frac{1}{UV} \frac{\partial^2 U}{\partial r^2} + \frac{1}{2UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} + \frac{1}{2U^2 V} \left(\frac{\partial U}{\partial r} \right)^2 - \frac{2}{rUV} \frac{\partial U}{\partial r} + \frac{2}{rV^2} \frac{\partial V}{\partial r} + \frac{2}{r^2} \left(1 - \frac{1}{V} \right). \quad (2.10)$$

The Schwarzschild solution describes spacetime outside a spherically symmetric star, where there is no mass. We therefore take $T_{ij} = 0$, and Einstein's equations become

$$R_{ij} - \frac{1}{2} g_{ij} R = 0.$$

This gives four tensor equations of the form

- $R_{00} - \frac{1}{2} g_{00} R = 0,$
- $R_{11} - \frac{1}{2} g_{11} R = 0,$
- $R_{22} - \frac{1}{2} g_{22} R = 0,$
- $R_{33} - \frac{1}{2} g_{33} R = 0.$

Substituting the expressions for g_{ij} and R_{ij} , we solve

$$R_{00} - \frac{1}{2} g_{00} R = R_{00} + \frac{U}{2} R,$$

Substituting (2.6) and (2.10) gives

$$\begin{aligned}
0 &= \frac{1}{2V} \frac{\partial^2 U}{\partial r^2} - \frac{1}{4V^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{4UV} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial U}{\partial r} \\
&\quad + \frac{U}{2} \left(-\frac{1}{UV} \frac{\partial^2 U}{\partial r^2} + \frac{1}{2UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} + \frac{1}{2U^2V} \left(\frac{\partial U}{\partial r} \right)^2 - \frac{2}{rUV} \frac{\partial U}{\partial r} + \frac{2}{rV^2} \frac{\partial V}{\partial r} + \frac{2}{r^2} \left(1 - \frac{1}{V} \right) \right) \\
&= \frac{1}{2V} \frac{\partial^2 U}{\partial r^2} - \frac{1}{4V^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{4UV} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial U}{\partial r} \\
&\quad - \frac{1}{2V} \frac{\partial^2 U}{\partial r^2} + \frac{1}{4V^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} + \frac{1}{4UV} \left(\frac{\partial U}{\partial r} \right)^2 - \frac{1}{rV} \frac{\partial U}{\partial r} + \frac{U}{rV^2} \frac{\partial V}{\partial r} + \frac{U}{r^2} \left(1 - \frac{1}{V} \right) \\
&= \frac{U}{r} \left(\frac{1}{V^2} \frac{\partial V}{\partial r} + \frac{1}{r} \left(1 - \frac{1}{V} \right) \right).
\end{aligned}$$

Thus the first tensor equation yields

$$0 = \frac{1}{V^2} \frac{\partial V}{\partial r} + \frac{1}{r} \left(1 - \frac{1}{V} \right). \quad (2.11)$$

The second tensor equation gives

$$R_{11} - \frac{1}{2} g_{11} R = R_{11} - \frac{V}{2} R,$$

Substituting (2.7) and (2.10),

$$\begin{aligned}
0 &= \frac{1}{4UV} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{2U} \frac{\partial^2 U}{\partial r^2} + \frac{1}{4U^2} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial V}{\partial r} \\
&\quad - \frac{V}{2} \left(-\frac{1}{UV} \frac{\partial^2 U}{\partial r^2} + \frac{1}{2UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} + \frac{1}{2U^2V} \left(\frac{\partial U}{\partial r} \right)^2 - \frac{2}{rUV} \frac{\partial U}{\partial r} + \frac{2}{rV^2} \frac{\partial V}{\partial r} + \frac{2}{r^2} \left(1 - \frac{1}{V} \right) \right) \\
&= \frac{1}{4UV} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{2U} \frac{\partial^2 U}{\partial r^2} + \frac{1}{4U^2} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rV} \frac{\partial V}{\partial r} \\
&\quad + \frac{1}{2U} \frac{\partial^2 U}{\partial r^2} - \frac{1}{4UV} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{1}{4U^2} \left(\frac{\partial U}{\partial r} \right)^2 + \frac{1}{rU} \frac{\partial U}{\partial r} - \frac{1}{rV} \frac{\partial V}{\partial r} - \frac{V}{r^2} \left(1 - \frac{1}{V} \right),
\end{aligned}$$

that is,

$$0 = \frac{1}{rU} \frac{\partial U}{\partial r} - \frac{V}{r^2} \left(1 - \frac{1}{V} \right),$$

Multiplying by $\frac{-r}{V}$ gives

$$0 = -\frac{1}{UV} \frac{\partial U}{\partial r} + \frac{1}{r} \left(1 - \frac{1}{V} \right). \quad (2.12)$$

The third tensor equation gives

$$R_{22} - \frac{1}{2} g_{22} R = R_{22} - \frac{r^2}{2} R,$$

Substituting (2.8) and (2.10),

$$\begin{aligned}
0 &= \frac{-r}{2UV} \frac{\partial U}{\partial r} - \frac{1}{V} + \frac{r}{2V^2} \frac{\partial V}{\partial r} + 1 \\
&\quad - \frac{r^2}{2} \left(-\frac{1}{UV} \frac{\partial^2 U}{\partial r^2} + \frac{1}{2UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} + \frac{1}{2U^2V} \left(\frac{\partial U}{\partial r} \right)^2 - \frac{2}{rUV} \frac{\partial U}{\partial r} + \frac{2}{rV^2} \frac{\partial V}{\partial r} + \frac{2}{r^2} \left(1 - \frac{1}{V} \right) \right) \\
&= \frac{r}{2UV} \frac{\partial U}{\partial r} - \frac{r}{2V^2} \frac{\partial V}{\partial r} + \frac{r^2}{2UV} \frac{\partial^2 U}{\partial r^2} - \frac{r^2}{4UV^2} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} - \frac{r^2}{4U^2V} \left(\frac{\partial U}{\partial r} \right)^2,
\end{aligned}$$

and multiplying by $\frac{-2V}{r}$, we obtain

$$0 = -\frac{1}{U} \frac{\partial U}{\partial r} + \frac{1}{V} \frac{\partial V}{\partial r} - \frac{r}{U} \frac{\partial^2 U}{\partial r^2} + \frac{r}{2UV} \frac{\partial U}{\partial r} \frac{\partial V}{\partial r} V + \frac{r}{2U^2} \left(\frac{\partial U}{\partial r} \right)^2. \quad (2.13)$$

The fourth tensor equation gives

$$\begin{aligned}
0 &= R_{33} - \frac{1}{2} g_{33} R \\
&= R_{22} \sin^2 \theta - \frac{r^2 \sin^2 \theta}{2} R \\
&= \sin^2 \theta (R_{22} - \frac{r^2}{2} R),
\end{aligned}$$

Finally, we obtain

$$0 = R_{22} - \frac{r^2}{2} R. \quad (2.14)$$

It remains to determine U and V explicitly. We use the preceding tensor equations, noting that the first equation, (2.11), depends only on V :

$$\begin{aligned}
0 &= \frac{1}{V^2} \frac{\partial V}{\partial r} + \frac{1}{r} \left(\frac{V-1}{V} \right) \\
-\frac{1}{r} &= \frac{1}{V(V-1)} \frac{\partial V}{\partial r}.
\end{aligned}$$

This is a separable differential equation:

$$-\frac{\partial r}{r} = \frac{\partial V}{V(V-1)},$$

Integrating and using $\frac{1}{V(V-1)} = \frac{1}{V-1} - \frac{1}{V}$, we obtain

$$\begin{aligned}
\int \frac{dV}{V(V-1)} &= \int \frac{dV}{V-1} - \int \frac{dV}{V} \\
&= \ln |V-1| - \ln |V| + C_1 \\
&= \ln \left| \frac{V-1}{V} \right| + C_1
\end{aligned}$$

where C_1 is a constant. Then

$$\begin{aligned} -\int \frac{dr}{r} &= \int \frac{dV}{V(V-1)} \\ -\ln|r| + C_2 &= \ln\left|\frac{V-1}{V}\right| + C_1 \end{aligned}$$

where C_2 is a constant. Equivalently,

$$\ln\left|\frac{1}{r}\right| + C_3 = \ln\left|\frac{V-1}{V}\right|,$$

Exponentiating and setting $e^{C_3} = C$, with C constant, gives

$$\frac{1}{r} \cdot C = \frac{V-1}{V},$$

Therefore

$$V = \frac{1}{1 - \frac{C}{r}}. \quad (2.15)$$

To determine U , we now substitute the expression for V into equation (2.12) and use $\frac{1}{V} = 1 - \frac{C}{r}$:

$$\begin{aligned} 0 &= -\frac{1}{U} \frac{\partial U}{\partial r} \left(1 - \frac{C}{r}\right) + \frac{1}{r} \left(1 - \left(1 - \frac{C}{r}\right)\right) \\ 0 &= -\frac{1}{U} \frac{\partial U}{\partial r} \left(1 - \frac{C}{r}\right) + \frac{C}{r^2} \end{aligned}$$

This implies

$$\frac{1}{U} \frac{\partial U}{\partial r} = \frac{C}{r^2 - rC}.$$

Again, we have a separable differential equation:

$$\frac{dU}{U} = \frac{Cdr}{r^2 - rC},$$

Using $\frac{1}{C(r-C)} - \frac{1}{Cr} = \frac{1}{r(r-C)}$, we obtain

$$\begin{aligned} \int \frac{dr}{r(r-C)} &= \frac{1}{C} \int \frac{dr}{r-C} - \frac{1}{C} \int \frac{dr}{r} \\ &= \frac{1}{C} \ln|r-C| - \frac{1}{C} \ln|r| + C_3 \\ &= \frac{1}{C} \ln\left|1 - \frac{C}{r}\right| + C_3, \end{aligned}$$

Solving the differential equation gives

$$\begin{aligned}\int \frac{dU}{U} &= C \int \frac{dr}{r(r-C)} \\ \ln |U| + C_4 &= C \left(\frac{1}{C} \ln \left| 1 - \frac{C}{r} \right| + C_3 \right) \\ \ln |U| &= \ln \left| 1 - \frac{C}{r} \right| + C_5, \quad \text{with } C_5 = C \cdot C_3,\end{aligned}$$

Exponentiating, we obtain

$$U = \left(1 - \frac{C}{r} \right) C_6, \quad \text{with } C_6 \text{ constant.} \quad (2.16)$$

Using the functions V and U obtained in (2.15) and (2.16), respectively, and rescaling time by $\sqrt{C_6}t \rightarrow t$, our metric becomes

$$ds^2 = - \left(1 - \frac{C}{r} \right) dt^2 + \left(1 - \frac{C}{r} \right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad t \in \mathbb{R}, r \neq C \text{ and } r \neq 0. \quad (2.17)$$

Setting $C = 2m$, the so-called **Schwarzschild radius**, gives the metric in Theorem (2.3.1). Moreover, as $m \rightarrow 0$ ($r \rightarrow \infty$), the metric reduces to the Minkowski metric in spherical coordinates given in equation (2.2).

2.5 Euler–Lagrange Equations

Euler first coined the term calculus of variations in 1756 to describe the method introduced the previous year by Joseph Louis Lagrange for finding minima or maxima of functionals on spaces of functions. The method was subsequently developed and studied by Euler, Hamilton, Legendre, Weierstrass, and others.

In this section, consider the C^∞ map

$$\begin{aligned}u : \quad R \subset \mathbb{R}^n &\longrightarrow \mathbb{R}^m \\ (x^i) &\longmapsto u^\alpha(x^i)\end{aligned}$$

where R is open and bounded.

Define the functional

$$F(u) = \int_R L \left(x^i, u^\alpha(x^i), \frac{\partial u^\alpha}{\partial x^i}(x^i) \right) dx^1 \wedge dx^2 \wedge \dots \wedge dx^n,$$

where F is a smooth real-valued function of $m + n + mn$ variables. The fundamental problem of the calculus of variations is to find functions u^α minimizing $F(u)$ and taking prescribed values on ∂R .

Notation: We denote u_{x_I} by u_I , where $x_I = (x_1, x_2, \dots, x_n)$.

Theorem 2.5.1. *If $u^\alpha = u^\alpha(x^i)$ minimizes $F(u)$, then $u^\alpha = u^\alpha(x^i)$ solves the m Euler–Lagrange equations $E_\alpha(L) = 0$, where*

$$E_\alpha(L) = \frac{\partial L}{\partial u^\alpha} - D_I \left(\frac{\partial L}{\partial u_I^\alpha} \right), \quad \text{for } \alpha = 1, 2, \dots, m. \quad I \text{ is a multi-index,} \quad (2.18)$$

and D_I is the total derivative operator defined by

$$D_I = \frac{\partial}{\partial x^i} + u_i^\alpha \frac{\partial}{\partial u^\alpha} + u_{ji}^\alpha \frac{\partial}{\partial u_j^\alpha} + \dots + u_{kji}^\alpha \frac{\partial}{\partial u_{kj}^\alpha} + \dots$$

The quantities u_I^α are regarded as “coordinates” and must therefore be treated as independent variables when computing total derivatives.

Example 2.5.2. We compute $D_x(u_x^2 + x + u_{yy})$ for $u = u(x, y)$. By the definition of the total derivative operator,

$$D_x = \frac{\partial}{\partial x^i} + u_x \frac{\partial}{\partial u} + u_{xx} \frac{\partial}{\partial u_x} + u_{xy} \frac{\partial}{\partial u_y} + u_{xxx} \frac{\partial}{\partial u_{xx}} + u_{xxy} \frac{\partial}{\partial u_{xy}} + \dots + u_{xyy} \frac{\partial}{\partial u_{yy}} + \dots$$

Retaining the nonzero terms gives

$$D_x(u_x^2 + x + u_{yy}) = u_{xx} \frac{\partial}{\partial u_x}(u_x^2) + \frac{\partial}{\partial x}x + u_{xyy} \frac{\partial}{\partial u_{yy}}(u_{yy}),$$

Hence

$$D_x(u_x^2 + x + u_{yy}) = 2u_x u_{xx} + 1 + u_{xyy}$$

Example 2.5.3. Suppose that $L = (1 + u_x^2 + u_y^2)^{\frac{1}{2}}$. To find u^α minimizing the area of the surface $z = u(x, y)$, we must minimize the functional

$$F(u) = \int_R (1 + u_x^2 + u_y^2)^{\frac{1}{2}} dx \wedge dy$$

using the Euler–Lagrange equations 2.18.

We have

$$\begin{aligned} E(L) &= \frac{\partial L}{\partial u} - D_I \left(\frac{\partial L}{\partial u_I} \right) \\ &= -D_x \left(\frac{\partial L}{\partial u_x} \right) - D_y \left(\frac{\partial L}{\partial u_y} \right), \end{aligned}$$

and note that

$$D_x = \frac{\partial}{\partial x} + u_x \left(\frac{\partial}{\partial u} \right) + u_{xx} \left(\frac{\partial}{\partial u_x} \right) + u_{xy} \left(\frac{\partial}{\partial u_y} \right) + \dots$$

$$D_y = \frac{\partial}{\partial y} + u_y \left(\frac{\partial}{\partial u} \right) + u_{yx} \left(\frac{\partial}{\partial u_x} \right) + u_{yy} \left(\frac{\partial}{\partial u_y} \right) + \dots$$

On the other hand,

$$\begin{aligned} \frac{\partial L}{\partial u_x} &= \frac{\partial}{\partial u_x} (1 + u_x^2 + u_y^2)^{\frac{1}{2}} \\ &= \frac{u_x}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}}, \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial u_y} &= \frac{\partial}{\partial u_y} (1 + u_x^2 + u_y^2)^{\frac{1}{2}} \\ &= \frac{u_y}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}}, \end{aligned}$$

Moreover,

$$\frac{\partial}{\partial u_x} \left(\frac{u_x}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) = \frac{1 + u_y^2}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}}$$

and

$$\frac{\partial}{\partial u_y} \left(\frac{u_x}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) = \frac{-u_x u_y}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}}.$$

Similarly,

$$\frac{\partial}{\partial u_x} \left(\frac{u_y}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) = \frac{-u_x u_y}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}}$$

and

$$\frac{\partial}{\partial u_y} \left(\frac{u_y}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) = \frac{1 + u_x^2}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}}.$$

Hence

$$\begin{aligned} D_x \left(\frac{\partial L}{\partial u_x} \right) &= u_{xx} \left(\frac{\partial}{\partial u_x} \left(\frac{u_x}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) \right) + u_{xy} \left(\frac{\partial}{\partial u_y} \left(\frac{u_x}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) \right) \\ &= u_{xx} \left(\frac{1 + u_y^2}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right) - u_{xy} \left(\frac{u_x u_y}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right), \end{aligned}$$

$$\begin{aligned}
D_y \left(\frac{\partial L}{\partial u_x} \right) &= u_{yx} \left(\frac{\partial}{\partial u_x} \left(\frac{u_y}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) \right) + u_{yy} \left(\frac{\partial}{\partial u_y} \left(\frac{u_y}{(1 + u_x^2 + u_y^2)^{\frac{1}{2}}} \right) \right) \\
&= -u_{yx} \left(\frac{u_x u_y}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right) + u_{yy} \left(\frac{1 + u_x^2}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right),
\end{aligned}$$

Thus

$$\begin{aligned}
E(L) &= -D_x \left(\frac{\partial L}{\partial u_x} \right) - D_y \left(\frac{\partial L}{\partial u_y} \right) \\
&= -u_{xx} \left(\frac{1 + u_y^2}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right) + u_{xy} \left(\frac{u_x u_y}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right) \\
&\quad + u_{yx} \left(\frac{u_x u_y}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right) - u_{yy} \left(\frac{1 + u_x^2}{(1 + u_x^2 + u_y^2)^{\frac{3}{2}}} \right)
\end{aligned}$$

The surface therefore has minimal area if and only if

$$u_{xx}(1 + u_y^2) - 2u_{xy}u_{yx} + u_{yy}(1 + u_x^2) = 0.$$

This is one of the Monge–Ampere equations, which locally determine surfaces of minimal area.

Example 2.5.4. Let (M, g) be a Riemannian manifold and $\sigma(t) = (u^1(t), \dots, u^n(t))$ a curve on M . We seek the curve minimizing the energy of a particle.

The total energy of a particle moving along this curve is $\int_{\sigma} \frac{1}{2} \|\dot{\sigma}(t)\|^2 dt$, assuming mass $m = 1$. If $g = (g_{\alpha\beta})_n$, then

$$\begin{aligned}
L(u_t) &= \frac{1}{2} \|\dot{\sigma}(t)\|^2 \\
&= \frac{1}{2} g_{\alpha\beta} u_t^\alpha u_t^\beta,
\end{aligned}$$

Thus finding a curve of minimum energy amounts to minimizing the functional

$$\begin{aligned}
F(u) &= \int_{\sigma} L(u_t) dt \\
&= \int_{\sigma} \frac{1}{2} g_{\alpha\beta} u_t^\alpha u_t^\beta dt.
\end{aligned}$$

If σ minimizes $F(u)$, the Euler–Lagrange equations give

$$E_{\gamma}(L) = \frac{\partial L}{\partial u^{\gamma}} - D_T \left(\frac{\partial}{\partial u_t^{\gamma}} \right) = 0,$$

where

$$\begin{aligned}
\frac{\partial L}{\partial u^\gamma} - D_T \left(\frac{\partial}{\partial u_t^\gamma} \right) &= \frac{1}{2} \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} u_t^\alpha u_t^\beta - D_T \left(\frac{1}{2} g_{\alpha\beta} \delta_\gamma^\alpha u_t^\beta + \frac{1}{2} g_{\alpha\beta} u_t^\alpha \delta_\gamma^\beta \right) \\
&= \frac{1}{2} \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} u_t^\alpha u_t^\beta - D_T \left(\frac{1}{2} g_{\gamma\beta} u_t^\beta + \frac{1}{2} g_{\alpha\gamma} u_t^\alpha \right) \\
&= \frac{1}{2} \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} u_t^\alpha u_t^\beta - \underbrace{\frac{1}{2} \frac{\partial g_{\gamma\beta}}{\partial u^\phi} u_t^\phi u_t^\beta}_{\text{if } \phi=\alpha} - \underbrace{\frac{1}{2} g_{\gamma\beta} u_{tt}^\beta}_{\text{if } \beta=\alpha} - \underbrace{\frac{1}{2} \frac{\partial g_{\alpha\gamma}}{\partial u^\phi} u_t^\phi u_t^\alpha}_{\text{if } \phi=\beta} - \frac{1}{2} g_{\alpha\gamma} u_{tt}^\alpha \\
&= -g_{\alpha\gamma} u_{tt}^\alpha + \frac{1}{2} \left(\frac{\partial g_{\alpha\beta}}{\partial u^\gamma} - \frac{\partial g_{\gamma\beta}}{\partial u^\alpha} - \frac{\partial g_{\alpha\gamma}}{\partial u^\beta} \right) u_t^\alpha u_t^\beta.
\end{aligned}$$

Thus $E_\gamma(L) = 0$ if and only if

$$g_{\alpha\gamma} u_{tt}^\alpha + \frac{1}{2} \left(\frac{\partial g_{\gamma\beta}}{\partial u^\alpha} + \frac{\partial g_{\alpha\gamma}}{\partial u^\beta} - \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} \right) u_t^\alpha u_t^\beta = 0,$$

Multiplying by $g^{v\gamma}$ gives

$$\begin{aligned}
g^{v\gamma} g_{\alpha\gamma} u_{tt}^\alpha + \frac{1}{2} g^{v\gamma} \left(\frac{\partial g_{\gamma\beta}}{\partial u^\alpha} + \frac{\partial g_{\alpha\gamma}}{\partial u^\beta} - \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} \right) u_t^\alpha u_t^\beta &= 0 \\
\delta_\alpha^v u_{tt}^\alpha + \frac{1}{2} g^{v\gamma} \left(\frac{\partial g_{\gamma\beta}}{\partial u^\alpha} + \frac{\partial g_{\alpha\gamma}}{\partial u^\beta} - \frac{\partial g_{\alpha\beta}}{\partial u^\gamma} \right) u_t^\alpha u_t^\beta &= 0, \quad \text{for } v = 0, 1, \dots, n-1,
\end{aligned}$$

Hence

$$u_{tt}^v + \Gamma_{\alpha\beta}^v u_t^\alpha u_t^\beta = 0, \quad \text{for } v = 0, 1, \dots, n-1. \quad (2.19)$$

We conclude that energy-minimizing curves are geodesics of (M, g) .

2.6 Schwarzschild Geodesics

There are two ways to study Schwarzschild geodesics: computing the symbols Γ_{ij}^k and substituting them into (2.19), which is rather cumbersome, or using a variational principle. More precisely, we use the Euler–Lagrange equations $E_\alpha(L) = 0$ for $\alpha = 0, 1, 2, 3$, namely

$$\frac{\partial L}{\partial u^\alpha} = D_T \left(\frac{\partial L}{\partial u_t^\alpha} \right). \quad (2.20)$$

Let $\gamma(\tau) = (t(\tau), r(\tau), \theta(\tau), \phi(\tau))$ be a curve in Schwarzschild spacetime. The Lagrangian L we shall study is

$$\begin{aligned} L(t, r, \theta, \phi, \dot{t}, \dot{r}, \dot{\theta}, \dot{\phi}) &= \begin{pmatrix} \dot{t} & \dot{r} & \dot{\theta} & \dot{\phi} \end{pmatrix} \begin{pmatrix} -\left(1 - \frac{2m}{r}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2m}{r}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix} \begin{pmatrix} \dot{t} \\ \dot{r} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} \\ &= -\left(1 - \frac{2m}{r}\right) \dot{t}^2 + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2, \end{aligned}$$

where $\dot{}$ denotes differentiation with respect to the independent variable τ . If $u^0(\tau) \equiv t(\tau)$, then, since $u_t^0(\tau) = \frac{d}{d\tau}(u^0(\tau)) = \frac{dt(\tau)}{d\tau} = \dot{t}$, equation (2.20) gives, for $\alpha = 0$,

$$\frac{\partial L}{\partial t} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{t}} \right) = 0, \quad (2.21)$$

Equation (2.21) further implies that

$$\begin{aligned} 0 &= \frac{d}{d\tau} \left(-\left(1 - \frac{2m}{r}\right) 2\dot{t} \right) \\ &= \frac{d}{d\tau} \left(-2\dot{t} + \frac{4m}{r} \dot{t} \right) \\ &= -2\ddot{t} + \left(\frac{r\ddot{t} - \dot{r}\dot{t}}{r^2} \right) 4m, \quad \text{where } \ddot{t} := \frac{d^2 t}{d\tau^2} \\ &= -2\ddot{t} + \frac{4m\ddot{t}}{r} - \frac{4m\dot{r}\dot{t}}{r^2} \\ &= 2 \left(-1 + \frac{2m}{r} \right) \ddot{t} - \frac{4m\dot{r}\dot{t}}{r^2} \\ &= -2 \left(1 - \frac{2m}{r} \right) \ddot{t} - \frac{4m\dot{r}\dot{t}}{r^2} \\ &= \ddot{t} + \frac{2m}{r^2 \left(1 - \frac{2m}{r} \right)} \dot{r}\dot{t}. \end{aligned} \quad (2.22)$$

This last equation shows that, on the Schwarzschild manifold,

$$\Gamma_{01}^0 = \Gamma_{10}^0 = \frac{m}{r^2 \left(1 - \frac{2m}{r} \right)}.$$

On the other hand, equation (2.20) with $u^1(\tau) = r(\tau)$ gives

$$\frac{\partial L}{\partial r} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{r}} \right) = 0,$$

Since

$$\frac{\partial L}{\partial r} = -\frac{2m}{r} \dot{t}^2 - \left(1 - \frac{2m}{r} \right)^{-2} \left(\frac{2m}{r} \right) \dot{r}^2 + 2r \dot{\theta}^2 + 2r \sin^2 \theta \dot{\phi}^2$$

we obtain

$$0 = -\frac{2m}{r}\dot{t}^2 - \left(1 - \frac{2m}{r}\right)^{-2} \left(\frac{2m}{r}\right) \dot{r}^2 + 2r\dot{\theta}^2 + 2r\sin^2\theta\dot{\phi}^2 - \frac{d}{d\tau} \left[\left(1 - \frac{2m}{r}\right)^{-1} 2\dot{r} \right] \quad (2.23)$$

Now equation (2.20) with $u^2(\tau) = \phi(\tau)$ gives

$$\frac{\partial L}{\partial \theta} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 0,$$

whence

$$0 = 2r^2 \sin\theta \cos\theta \dot{\phi}^2 - \frac{d}{d\tau} (2r\dot{\theta}). \quad (2.24)$$

Finally, for $u^3(\tau) = \phi(\tau)$, we have

$$\frac{\partial L}{\partial \phi} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0,$$

Thus

$$0 = \frac{d}{d\tau} (2r^2 \sin^2\theta \dot{\phi}). \quad (2.25)$$

The Euler–Lagrange operator simplifies the search for solutions, since equations (2.22), (2.23), (2.24), and (2.25) imply

$$\frac{d}{d\tau} \left(\left(1 - \frac{2m}{r}\right) \dot{t} \right) = 0, \quad (2.26)$$

$$\frac{d}{d\tau} \left(\left(1 - \frac{2m}{r}\right)^{-1} \dot{r} \right) = \frac{m}{r^2} \dot{t}^2 - \left(1 - \frac{2m}{r}\right)^{-2} \frac{m}{r^2} \dot{r}^2 + r\dot{\theta}^2 + r\sin^2\theta\dot{\phi}^2, \quad (2.27)$$

$$\frac{d}{d\tau} (r^2 \dot{\theta}) = r^2 \sin\theta \cos\theta \dot{\phi}^2, \quad (2.28)$$

$$\frac{d}{d\tau} (r^2 \sin^2\theta \dot{\phi}) = 0. \quad (2.29)$$

From (2.26) and (2.29),

$$\left(1 - \frac{2m}{r}\right) \dot{t} = E \quad (2.30)$$

and

$$r^2 \sin^2\theta \dot{\phi} = J, \quad (2.31)$$

where E and J are constants of motion. Another constant of motion is

$$L(\tau) = -\left(1 - \frac{2m}{r}\right) \dot{t}^2 + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + r^2(\dot{\theta}^2 + \sin^2\theta \dot{\phi}^2), \quad (2.32)$$

With an arc-length parametrization, $||L(\tau)|| = 1$.

To simplify the analysis, we show that the motion is planar. Consider a geodesic and spherical coordinates (r, θ, ϕ) on $\mathbb{R}^3$. Its initial position is not at the origin, since the metric is singular at $r = 0$. Note that $\theta(\tau) = 0$ and $\theta(\tau) = \frac{\pi}{2}$ satisfy (2.28), since $\cos \theta \sin \theta = 0$ if and only if $\theta = 0 \vee \theta = \frac{\pi}{2}$. Moreover, $\theta(\tau) = \frac{\pi}{2}$ means that if a geodesic orbit passes through the equatorial plane $\theta = \frac{\pi}{2}$, it remains in that plane for every τ .

Without loss of generality, we may assume $\theta(\tau) = \frac{\pi}{2}$. Equations (2.30), (2.31), and (2.32) then give

$$\begin{aligned}\frac{dt}{d\tau} &= \frac{E}{1 - \frac{2m}{r}}, \\ \frac{d\phi}{d\tau} &= \frac{J}{r^2}, \\ L &= -\left(1 - \frac{2m}{r}\right) \dot{t}^2 + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2,\end{aligned}\tag{2.33}$$

where $L = 1$ for timelike geodesics and $L = 0$ for lightlike geodesics (photons).

Since

$$\dot{t} \left(1 - \frac{2m}{r}\right) = E$$

and

$$\dot{\phi} r^2 = J,$$

substituting into (2.33) gives

$$L = -E\dot{t} + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + \dot{\phi} J.$$

Now consider a timelike geodesic in the equatorial plane. In this case, $L = -1$, so

$$\begin{aligned}-1 &= -E\dot{t} + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + \dot{\phi} J \\ \dot{r}^2 &= \left(-1 + E\dot{t} - J\dot{\phi}\right) \left(1 - \frac{2m}{r}\right) \\ \dot{r}^2 &= -1 + \frac{E^2}{\left(1 - \frac{2m}{r}\right)} - \frac{J^2}{r^2} \\ \dot{r}^2 &= E^2 - \frac{J^2}{r^2} \left(1 - \frac{2m}{r}\right) - \left(1 - \frac{2m}{r}\right) \\ \dot{r}^2 &= E^2 - \left(1 - \frac{2m}{r}\right) \left(\frac{J^2}{r^2} + 1\right).\end{aligned}\tag{2.34}$$

For radial motion (a freely falling particle), ϕ is constant, so $\frac{d\phi}{d\tau} = 0$ and hence $J = 0$. Therefore

$$\dot{r}^2 = E^2 - 1 + \frac{2m}{r}.\tag{2.35}$$

If a particle is released from rest at $r = R$, with $\dot{r} = 0$, equation (2.35) gives

$$E^2 = 1 - \frac{2m}{R}, \quad (2.36)$$

where

$$R = \frac{2m}{1 - E^2}$$

is the radius at which the particle has zero velocity. Equation (2.35) becomes

$$\begin{aligned} \dot{r}^2 &= \left(1 - \frac{2m}{r}\right) - \left(1 - \frac{2m}{R}\right) \\ &= \left(\frac{2m}{r} - \frac{2m}{R}\right), \end{aligned}$$

Thus

$$\begin{aligned} \frac{dr}{d\tau} &= \sqrt{\left(\frac{2m}{r} - \frac{2m}{R}\right)} \\ &= \sqrt{\frac{2m(R-r)}{rR}}, \end{aligned}$$

Hence

$$\begin{aligned} d\tau &= \frac{1}{\sqrt{\frac{2m(R-r)}{rR}}} dr \\ &= \sqrt{\frac{rR}{2m(R-r)}} dr, \end{aligned}$$

Therefore

$$\tau = \sqrt{\frac{R}{2m}} \int \sqrt{\frac{r}{R-r}} dr. \quad (2.37)$$

The motion follows a cycloidal law; see [8]. To evaluate this integral, set $r = \frac{R}{2}(1 + \cos \eta)$ and $dr = -\frac{R}{2} \sin \eta d\eta$. Substitution gives

$$\begin{aligned}
\int \sqrt{\frac{r}{R-r}} dr &= \int \sqrt{\frac{\frac{R}{2}(1+\cos\eta)}{R-\frac{R}{2}(1+\cos\eta)}} \left(-\frac{R}{2} \sin\eta d\eta\right) \\
&= -\frac{R}{2} \int \sqrt{\frac{1+\cos\eta}{1-\cos\eta}} \sin\eta d\eta \\
&= -\frac{R}{2} \int \sqrt{\frac{(1+\cos\eta)^2}{1-\cos^2\eta}} \sin\eta d\eta \\
&= -\frac{R}{2} \int \sqrt{\frac{(1+\cos\eta)^2}{\sin^2\eta}} \sin\eta d\eta \\
&= -\frac{R}{2} \int \frac{|1+\cos\eta|}{|\sin\eta|} \sin\eta d\eta.
\end{aligned}$$

Without loss of generality, we choose the fourth quadrant, where $\cos\eta > 0$ and $\sin\eta < 0$. Then

$$\begin{aligned}
\int \sqrt{\frac{r}{R-r}} dr &= -\frac{R}{2} \int \frac{1+\cos\eta}{-\sin\eta} \sin\eta d\eta \\
&= \frac{R}{2} \int (1+\cos\eta) d\eta \\
&= \frac{R}{2} (\eta + \sin\eta),
\end{aligned}$$

Equation (2.37) therefore yields

$$\tau = \frac{R}{2} \sqrt{\frac{R}{2m}} (\eta + \sin\eta). \quad (2.38)$$

To compute the total proper time of a fall from rest at $r = R$ to $r = 0$, note that

- $r = R$ if $\cos\eta = 1$. Thus $\eta = 0 + 2k\pi$, $k \in \mathbb{Z}_0^+$
- $r = 0$ if $\cos\eta = -1$. Thus $\eta = \pi + 2k\pi$, $k \in \mathbb{Z}_0^+$

These give the limits of integration for τ from $r = R$ to $r = 0$. The total proper time is therefore

$$\begin{aligned}
\tau &= \frac{R}{2} \sqrt{\frac{R}{2m}} (\eta + \sin\eta) \Big|_0^\pi \\
&= \frac{\pi R}{2} \sqrt{\frac{R}{2m}}.
\end{aligned}$$

Question: How does the time coordinate t evolve along this motion??

Using (2.30), we obtain

$$\begin{aligned}
 \frac{dr}{d\tau} &= \frac{dr}{dt} \frac{dt}{d\tau} \\
 &= \frac{dr}{dt} \left(\frac{E}{1 - \frac{2m}{r}} \right) \\
 &= E \frac{dr^*}{dt}, \quad \text{with } dr^* = \frac{dr}{1 - \frac{2m}{r}}, \quad (2.39)
 \end{aligned}$$

where

$$\begin{aligned}
 r^* &= \int dr^* \\
 &= \int \frac{dr}{1 - \frac{2m}{r}}
 \end{aligned}$$

is a new tortoise coordinate proposed by Wheeler. Moreover, since

$$\frac{1}{1 - \frac{2m}{r}} = \frac{r}{r - 2m} \text{ and } \frac{2m}{r - 2m} + \frac{r - 2m}{r - 2m} = \frac{r}{r - 2m}, \text{ we have}$$

$$\begin{aligned}
 r^* &= \int \frac{dr}{1 - \frac{2m}{r}} \\
 &= \int \frac{r dr}{r - 2m} \\
 &= \int \left(\frac{2m}{r - 2m} + 1 \right) dr \\
 &= \int \frac{2m}{r - 2m} dr + \int dr \\
 &= 2m \ln |r - 2m| + r + K, \quad \text{with } K \text{ constant.}
 \end{aligned}$$

Substituting (2.39) into (2.34) gives

$$\left(E \frac{dr^*}{dt} \right)^2 = E^2 - \left(1 - \frac{2m}{r} \right) \left(\frac{J^2}{r^2} + 1 \right),$$

whence

$$dt = \frac{E}{\sqrt{E^2 - \left(1 - \frac{2m}{r} \right) \left(\frac{J^2}{r^2} + 1 \right)}} dr^*, \quad \text{with } dr^* = \frac{dr}{1 - \frac{2m}{r}},$$

Substituting (2.36) and integrating this expression yields

$$t = \int \frac{\sqrt{1 - \frac{2m}{R}}}{\sqrt{\left(\frac{2m}{r} - \frac{2m}{R} \right) \left(1 - \frac{2m}{r} \right)}} \frac{dr}{1 - \frac{2m}{r}}, \quad (2.40)$$

We again use $r = \frac{R}{2}(1 + \cos \eta)$ and $dr = -\frac{R}{2} \sin \eta d\eta$. Since

$$\sqrt{\left(\frac{2m}{r} - \frac{2m}{R}\right)} = \sqrt{\frac{2m}{R} \frac{\sin \eta}{1 + \cos \eta}},$$

we obtain

$$\begin{aligned} \int \frac{\sqrt{1 - \frac{2m}{R}}}{\left(\frac{2m}{r} - \frac{2m}{R}\right)^{\frac{1}{2}}} \frac{1}{\left(1 - \frac{2m}{r}\right)} dr &= \frac{R}{2} \sqrt{\frac{R}{2m} - 1} \int \frac{\cos \eta + 1}{\left(1 - 4 \frac{m}{R(\cos \eta + 1)}\right)} d\eta \\ &= \frac{R^2}{2} \sqrt{\frac{R}{2m} - 1} \int \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} d\eta \\ &= \frac{R^2}{2} \sqrt{\frac{R}{2m} - 1} \int \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} d\eta. \end{aligned}$$

We compute

$$\int \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} d\eta,$$

Since

$$\begin{aligned} \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} &= \frac{(\cos \eta + 1)^2 - \frac{16m^2}{R^2} + \frac{16m^2}{R^2}}{R(\cos \eta + 1) - 4m} \\ &= \frac{\frac{1}{R^2} (R^2 (\cos \eta + 1)^2 - 16m^2)}{R(\cos \eta + 1) - 4m} + \frac{\frac{16m^2}{R^2}}{R(\cos \eta + 1) - 4m} \end{aligned}$$

we have

$$\begin{aligned} &\int \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} d\eta \tag{2.41} \\ &= \int \frac{1}{R^2} (R(\cos \eta + 1) + 4m) d\eta + \frac{16m^2}{R^2} \int \frac{1}{R(\cos \eta + 1) - 4m} d\eta \\ &= \frac{1}{R^2} (R \sin \eta + R\eta + 4m\eta) + \frac{16m^2}{R^2} \int \frac{1}{R(\cos \eta + 1) - 4m} d\eta, \end{aligned}$$

We now compute

$$\int \frac{1}{R(\cos \eta + 1) - 4m} d\eta$$

$$\begin{aligned}
& \int \frac{1}{R(\cos \eta + 1) - 4m} d\eta \\
&= \int \frac{1}{R \cos \eta + R - 4m} d\eta \\
&= \frac{1}{R - 4m} \int \frac{1}{1 + \frac{R}{R-4m} \cos \eta} d\eta \\
&= \frac{1}{R - 4m} \int \frac{1}{1 + A \cos \eta} d\eta, \quad \text{with } A = \frac{R}{R - 4m}. \tag{2.42}
\end{aligned}$$

We evaluate

$$\int \frac{1}{1 + A \cos \eta} d\eta,$$

Using the Weierstrass substitution $u = \tan\left(\frac{\eta}{2}\right)$, we have $\eta = 2 \arctan(u)$, hence $d\eta = \frac{2}{u^2+1} du$ and $\cos(\eta) = \frac{1-u^2}{u^2+1}$. Thus

$$\int \frac{1}{1 + A \cos \eta} d\eta = \int \frac{2du}{(u^2 + 1) \left(1 + A \left(\frac{1-u^2}{u^2+1}\right)\right)} = \int \frac{2du}{(1-A)u^2 + (1+A)}$$

If $A > 1$, set $1 + A = p^2$ and $1 - A = -q^2$. Then

$$\int \frac{2du}{(1-A)u^2 + (1+A)} = 2 \int \frac{du}{p^2 - q^2 u^2}$$

Now let $u = \frac{p}{q}v$ and $du = \frac{p}{q}dv$, so that

$$2 \int \frac{du}{p^2 - q^2 u^2} = 2 \int \frac{\frac{p}{q} dv}{p^2(1 - v^2)} = \frac{2}{pq} \int \frac{dv}{(1 - v^2)}$$

Since $\frac{1}{1-v} + \frac{1}{1+v} = \frac{1}{1-v^2}$, we obtain

$$\begin{aligned}
\frac{2}{pq} \int \frac{dv}{(1 - v^2)} &= \frac{2}{pq} \left[\frac{1}{2} \int \frac{dv}{(1 - v)} + \frac{1}{2} \int \frac{dv}{(1 + v)} \right] \\
&= \frac{1}{pq} \left[- \int \frac{dv}{(1 - v)} + \int \frac{dv}{(1 + v)} \right] \\
&= \frac{1}{pq} [-\ln |1 - v| + \ln |1 + v|] \\
&= \frac{1}{pq} \left[\ln \left| \frac{1 + v}{1 - v} \right| \right] \\
&= \frac{1}{pq} \left[\ln \left| \frac{p + qu}{p - qu} \right| \right] \quad \text{with } v = \frac{q}{p}u \\
&= \frac{1}{\sqrt{A^2 - 1}} \left[\ln \left| \frac{\sqrt{1+A} + (\sqrt{A-1}) \tan\left(\frac{\eta}{2}\right)}{\sqrt{1+A} - (\sqrt{A-1}) \tan\left(\frac{\eta}{2}\right)} \right| \right]
\end{aligned}$$

where $p = \sqrt{1+A}$, $q = \sqrt{A-1}$, and $u = \tan\left(\frac{\eta}{2}\right)$, respectively.

Since $A = \frac{R}{R-4m}$, we obtain

$$\begin{aligned}
 &= \frac{1}{\sqrt{\left(\frac{R}{R-4m}\right)^2 - 1}} \left[\ln \left| \frac{\sqrt{1 + \left(\frac{R}{R-4m}\right)} + \left(\sqrt{\left(\frac{R}{R-4m}\right)} - 1\right) \tan\left(\frac{\eta}{2}\right)}{\sqrt{1 + \left(\frac{R}{R-4m}\right)} - \left(\sqrt{\left(\frac{R}{R-4m}\right)} - 1\right) \tan\left(\frac{\eta}{2}\right)} \right| \right] \\
 &= \frac{1}{\sqrt{\left(\frac{R}{R-4m}\right)^2 - 1}} \left[\ln \left| \frac{\frac{\sqrt{1 + \left(\frac{R}{R-4m}\right)}}{\sqrt{\left(\frac{R}{R-4m}\right)} - 1} + \tan\left(\frac{\eta}{2}\right)}{\frac{\sqrt{1 + \left(\frac{R}{R-4m}\right)}}{\sqrt{\left(\frac{R}{R-4m}\right)} - 1} - \tan\left(\frac{\eta}{2}\right)} \right| \right],
 \end{aligned}$$

Note that

$$\frac{1 + \left(\frac{R}{R-4m}\right)}{\left(\frac{R}{R-4m}\right) - 1} = \frac{\frac{R-4m+R}{R-4m}}{\frac{R-(R-4m)}{R-4m}} = \frac{(2R-4m)(R-4m)}{(R-4m)(4m)} = \frac{R-2m}{2m} = \frac{R}{2m} - 1$$

Thus expression (2.42) becomes

$$\begin{aligned}
 \frac{1}{R-4m} \int \frac{1}{1 + A \cos \eta} d\eta &= \frac{1}{(R-4m)\sqrt{\left(\frac{R}{R-4m}\right)^2 - 1}} \left[\ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right| \right] \\
 &= \frac{1}{\sqrt{R^2 - (R-4m)^2}} \left[\ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right| \right] \\
 &= \frac{1}{\sqrt{8m(R-2m)}} \left[\ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right| \right].
 \end{aligned}$$

Substituting into (2.41) gives

$$\begin{aligned}
 &\int \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} d\eta \\
 &= \frac{1}{R^2} (R \sin \eta + R\eta + 4m\eta) + \frac{16m^2}{R^2} \int \frac{1}{R(\cos \eta + 1) - 4m} d\eta \\
 &= \frac{1}{R^2} (R \sin \eta + R\eta + 4m\eta) + \frac{16m^2}{R^2} \frac{1}{\sqrt{8m(R-2m)}} \left[\ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right| \right].
 \end{aligned}$$

Finally, substituting into (2.40),

$$\begin{aligned}
& \int \frac{\sqrt{1 - \frac{R}{2M}}}{\left(\frac{2M}{r} - \frac{2M}{R}\right)^{\frac{1}{2}} \left(1 - \frac{2M}{r}\right)} dr \\
&= \frac{R^2}{2} \sqrt{\frac{R}{2m} - 1} \int \frac{(\cos \eta + 1)^2}{R(\cos \eta + 1) - 4m} d\eta \\
&= \frac{R^2}{2} \sqrt{\frac{R}{2m} - 1} \left(\frac{1}{R^2} (R \sin \eta + R\eta + 4m\eta) + \frac{16m^2}{R^2} \frac{1}{\sqrt{8m(R-2m)}} \left[\ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right| \right] \right) \\
&= \left(\frac{R}{2} \sqrt{\frac{R}{2m} - 1} \right) \sin \eta + \left(\left(\frac{R}{2} + 2m \right) \sqrt{\frac{R}{2m} - 1} \right) \eta \\
&\quad + \left(\frac{R^2}{2} \left(\sqrt{\frac{R}{2m} - 1} \right) \frac{16m^2}{R^2} \frac{1}{\sqrt{8m(R-2m)}} \right) \ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right|,
\end{aligned}$$

where

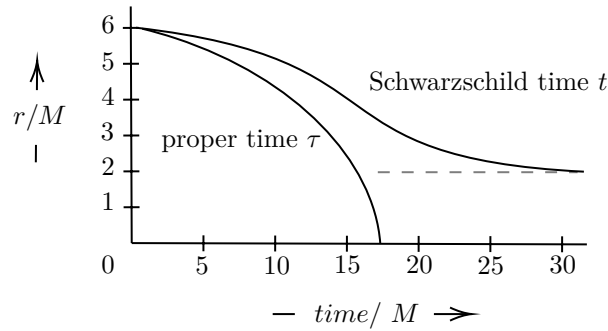
$$\left(\frac{R^2}{2} \left(\sqrt{\frac{R}{2m} - 1} \right) \frac{16m^2}{R^2} \frac{1}{\sqrt{8m(R-2m)}} \right) = \frac{8m^2 \sqrt{R-2m}}{\sqrt{2m} \sqrt{8m(R-2m)}} = 2m.$$

The integral therefore yields

$$t = \left(\frac{R}{2} \sqrt{\frac{R}{2m} - 1} \right) \sin \eta + \left(\left(\frac{R}{2} + 2m \right) \sqrt{\frac{R}{2m} - 1} \right) \eta + 2m \ln \left| \frac{\sqrt{\frac{R}{2m} - 1} + \tan\left(\frac{\eta}{2}\right)}{\sqrt{\frac{R}{2m} - 1} - \tan\left(\frac{\eta}{2}\right)} \right|. \quad (2.43)$$

This result relates the Schwarzschild time coordinate to the radius R at which the particle has zero velocity; see [8].

Thus, for an observer stationary at *infinity*, a particle takes an infinite coordinate time t to reach the horizon, whereas in proper time τ it reaches the horizon in finite time and even reaches the singularity in finite time.



The figure shows an observer freely falling towards a Schwarzschild black hole, described using proper time τ , and a distant observer, described using Schwarzschild coordinate time t . Equation

(2.38) shows that $r = 0$ is reached quickly. In the description given by equation (2.43), however, $r = 0$ is never reached, and $r = 2m$ is approached asymptotically.

We have seen that the Schwarzschild metric describes the gravitational field outside spherically symmetric stars. In deriving it, we required spacetime to be spherically symmetric and free of matter only at the points where we calculated the solution. On the other hand, the metric inside such a star will differ from the Schwarzschild metric; that is, we must examine what happens at $r = 0$ and $r = 2m$, where the metric is singular.

On physical grounds, it is reasonable to expect the light cones to point towards $r = 0$, since we expect the gravitational field to *attract* objects towards the centre. However, we have no mathematical proof of this.

Our calculations with the Schwarzschild metric revealed interesting information, but we encountered considerable difficulties at $r = 2m$. The key to making further progress is to introduce a new tortoise coordinate system. Using ingoing Eddington–Finkelstein coordinates at $r = 2m$, however, one can prove that all future-pointing light cones tilt inwards within $\{r \leq 2m\}$, and therefore that all null and timelike geodesics fall inwards. On the other hand, using outgoing Eddington–Finkelstein coordinates at $r = 2m$, one can prove that all future-pointing light cones within $\{r \leq 2m\}$ point outwards. Now, if Schwarzschild spacetime is static by construction, and we should therefore expect time reflections of the form $t \mapsto -t$, what is happening, and which result is correct?

Apparently, both are correct. To understand this issue, one must combine the ingoing and outgoing Eddington–Finkelstein coordinates in a new coordinate transformation: Kruskal–Szekeres coordinates.

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